GE Healthcare

LightSpeed Series & HiSpeed QX/i

Options Installation

OPERATING DOCUMENTATION

2211224-100
Rev 07

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

このサービスマニュアルには英語版しかありません。

ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

Check for damage to property that may have occurred at the site during delivery, such as damage to floors, door frames or walls. If damage is found, notify the install specialist.

All packages should be closely examined at time of delivery. If damage is apparent, have notation “damage in shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm
- Contact the local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

IMPORTANT...RADIOACTIVE MATERIAL HANDLING

Only employees formally trained in radioactive materials handling and this equipment are authorized by the GE Healthcare Radiation Safety Officer to use radioactive materials to service this equipment.

GE Healthcare Services is required to notify the applicable U.S. state agency PRIOR to any source service event involving pin source handling. See NUC/PET Radioactive material guides for specific instruction or contact your EHS Specialist.

A radiation survey must be performed when a pin source has been removed and replaced. See Radiation Survey Form Instructions or contact your EHS Specialist.

Rev 2 (July 21, 2005)

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION Risk of Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION Danger d'Explosion

Il y a danger d'explosion s'il y a replacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GEMS PQR Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	06/02/00	Initial Release
1	03/07/01	CQA 1010817: Added System (SW) Requirements for each option; Performed formatting maintenance; added DASM/Camera Installation (Chapter 10)
2	11/05/01	CQA 1016595: Description of setup on PPS screen for ConnectPro option added to installation procedure. (Chapter 6, Section 3.1)
3	06/14/02	Updated to include HiSpeed QX/i
4	05/28/03	CQA 1031554: Added check for LightSpeed QX/i Systems to test using Study UID's from hospital HIS/RIS systems. (Chapter 6, Section 3.3)
5	12/08/04	Updated Chapter 3 - Dentascan Option Installation.
6	10/6/05	• In Chapter 2, for the “Install LCD, Power and Video Cables“ section of “Exam Suite Monitor with LCD Installation“, deleted installation instructions and referred reader to 2390072-100 (Rev 3 or later) - CRT & LCD Remote Monitor & Monitor-in-Room Boom Option. • Added Chapter 11 - FX/Xtream Workflow Options. • Added Chapter 12 - Ivy Model 3150 ECG Monitor.
7	9/20/06	Updated Chapter 2, Section 3.0.

This page intentionally blank.

Table of Contents

This page intentionally blank.

Preface
Publication Conventions

Purpose: Standardize conventions for representing information is a uniform way of communicating information to a reader in a consistent manner. Conventions are used so that the reader can easily recognize the actions or decisions that must be made. There are a number of character and paragraph styles used in this publication to accomplish this task. Please become familiar with them before proceeding forward.

Section 1.0
Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

It's important that you read and understand hazard statements, and not just ignore them.

DANGER SEVERE PERSONAL INJURY

DANGER IS USED WHEN A HAZARD EXISTS WHICH <u>WILL CAUSE SEVERE PERSONAL INJURY</u> OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

WARNING SERIOUS PERSONAL INJURY

WARNING IS USED WHEN A HAZARD EXISTS WHICH <u>COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH</u> IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **POTENTIAL FOR SHOCK**
- **EXPOSED WIRES**
- **FAILURE TO TAG AND LOCKOUT SYSTEM POWER COULD ALLOW FOR UN-COMMANDED MOTION.**

CAUTION Minor Personal Injury	**Caution is used when a hazard exists which <u>can or could cause minor injury</u> to self or others if instructions are ignored. They include for example:** • **Loss of critical patient data** • **Crush or pinch points** • **Sharp objects**
NOTICE Property Damage Only	**Notice is used when a hazard is present that can cause property damage but has <u>absolutely no personal injury risk</u>.They can include:** • **Disk drive will be destroyed** • **Internal mechanical damage, such as to the x-ray tube** • **Coasting the rotor through resonance.**

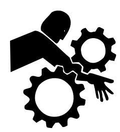
1.2 Graphical Representation

Important information will always be preceded by the exclamation point contained within a triangle, as seen throughout the Preface chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

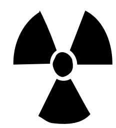
HAZARDS

ELECTRICAL

MECHANICAL



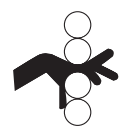
RADIATION



LASER

HEAT

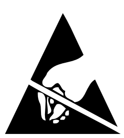
PINCH



.

PROCEDURAL

AVOID STATIC ELECTRICITY

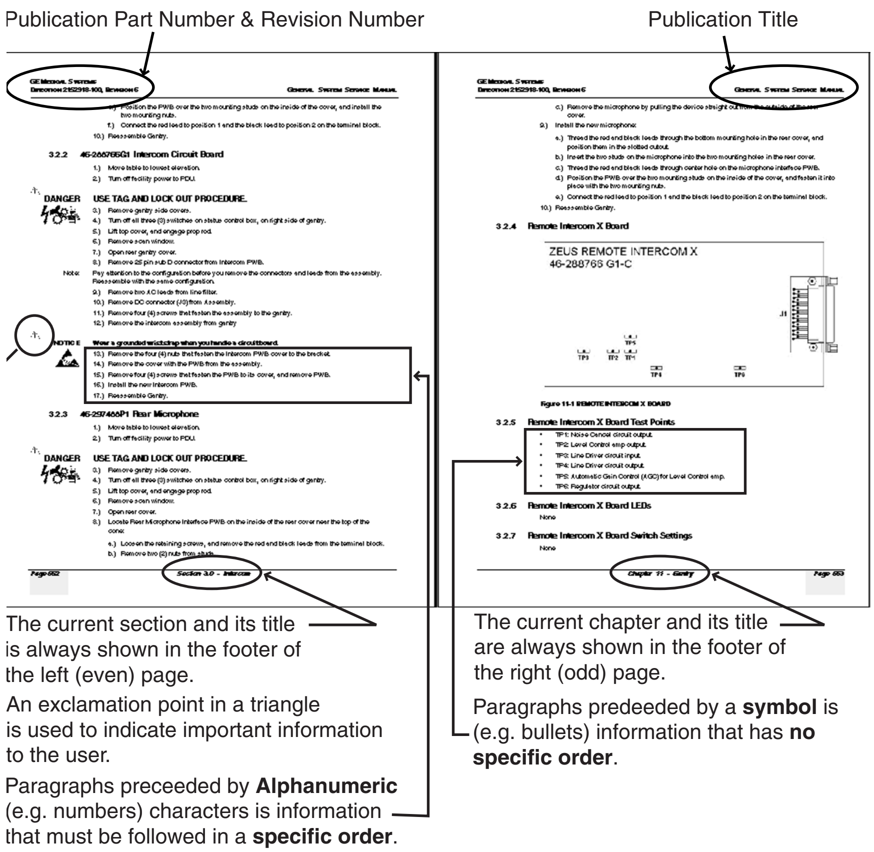


TAG AND LOCK OUT

WEAR EYE PROTECTION

Section 2.0 Publication Conventions

2.1 Page Layout



Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.2 Computer Screen Output/Input Text Character Styles

Within this publication, mono-spaced character styles (fonts) are used to indicate computer text that's either screen input and output. Mono spaced fonts, such courier, are used to indicated text direction. When you type at your keyboard, your are generating computer input. Occasionally you will see the math operators "greater than" and "less than" symbols used to indicate the start and finish of variable output. When reading text generated by the computer, you are reading it as computer generated output. In addition to direction, characters are italicized (e.g. *italics*) to indicate information specific to your system or site.

Example:
Fixed Output

This paragraph's fonts represents computer generated screen "fixed" output. It's output is fixed from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths and text that does not change form system to system. The character style used is a fixed width such as courier.

Example: Variable Output

This paragraph's font represents computer screen output that is "variable". Its used to represent output that varies from application to application or system to system. Variable output is sometimes found placed between greater than and lesser than operators for clarification. For example: <variable_ouput> or <3.45.120.3>. In both cases, the < and > operators are not part of the actual input.

Example: Fixed Input

This paragraph's font represents fixed input. It's computer input that is typed-in via the keyboard. Typed input that does not vary from application to application or system to system. Fixed text the user is required to supply as input. For example: cd /usr/3p

Example: Variable Input

This paragraph's font represents computer input that can vary from application to application or system to system. With variable text, the user is required to supply system dependent input or information. Variable input sometimes is placed between greater-than and less-than operators. For example: <variable_input>. In these cases, the (<>) operators would be dropped prior to input. For example: **ypcat hosts | grep** ***<3.45.120.3>*** **would be typed into the computer as ypcat hosts | grep** ***3.45.120.3 without the greater-than and less-than operators.***

2.3 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys

A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Soft Keys

Whereas the computer MENU button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

Chapter 1
Exam Suite Monitor with CRT Option

Section 1.0
System Requirement

None. Works with all software versions.

Section 2.0
Preinstallation for Exam Suite Monitor with CRT

2.1 Site Preparation Before Equipment Delivery

Determine whether the existing cables run through conduit, wire trough or under raised floor, and, if a sufficient opening exists to run the monitor video cable from the operator console to the gantry base.

- Video Cable: 24.4 m (80 ft.) long and about 5/8 in. in diameter
  - Minimum 45mm (1.75 in.) hole to route cables between Console and Gantry
  - 1.5 m (5 ft.) needed at the back of the gantry base
  - 1.5 m (5 ft.) needed at the console (from center of back side, near floor)
- Gantry to Monitor Cart Cable
  - 120VAC grounded cable, 5 meters (16 ft.) long
  - Video cable, 5 meters (16 ft.) long
- Power source: STC side of Gantry base (left side of Gantry)

2.2 Room Planning

Figure 1-1 contains a typical room layout for the CT components. Overall suite size equals 6.19 m (20.3 ft) x 7.07 m (23.2 ft) deep, with a 2.75 m (9 ft) ceiling. The area consists of the following three rooms:

- Scan Room: 3.81 m x 6.10 m (19.4 ft x 12.9 ft)
- Control Area: 3.81 m x 2.67 m (12.3 ft x 9.0 ft)
- Equipment Room: 8.38 m x 3.20 m (6.7 ft x 9.0 ft)

  Room arrangements vary from facility to facility. Use the cable lengths shown in Table 1-2 to plan the maximum distance between components. For example:
- The control and equipment rooms may be interchanged with the scanner room.
- Options, such as Advantage Windows, the laser camera or line printer, may be located in a separate room.

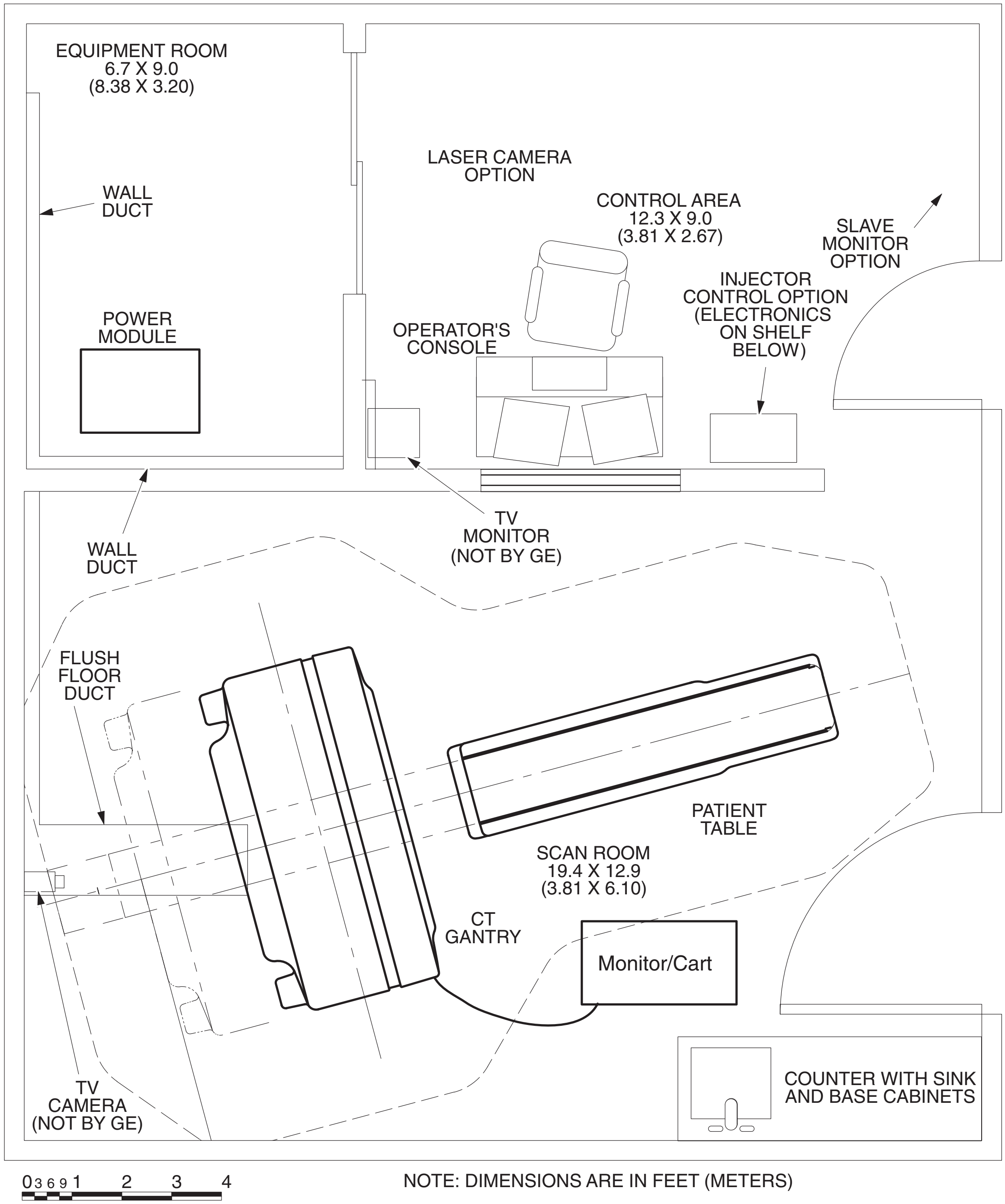


Figure 1-1 Recommended Suite Layout Exam Suite Monitor with CRT

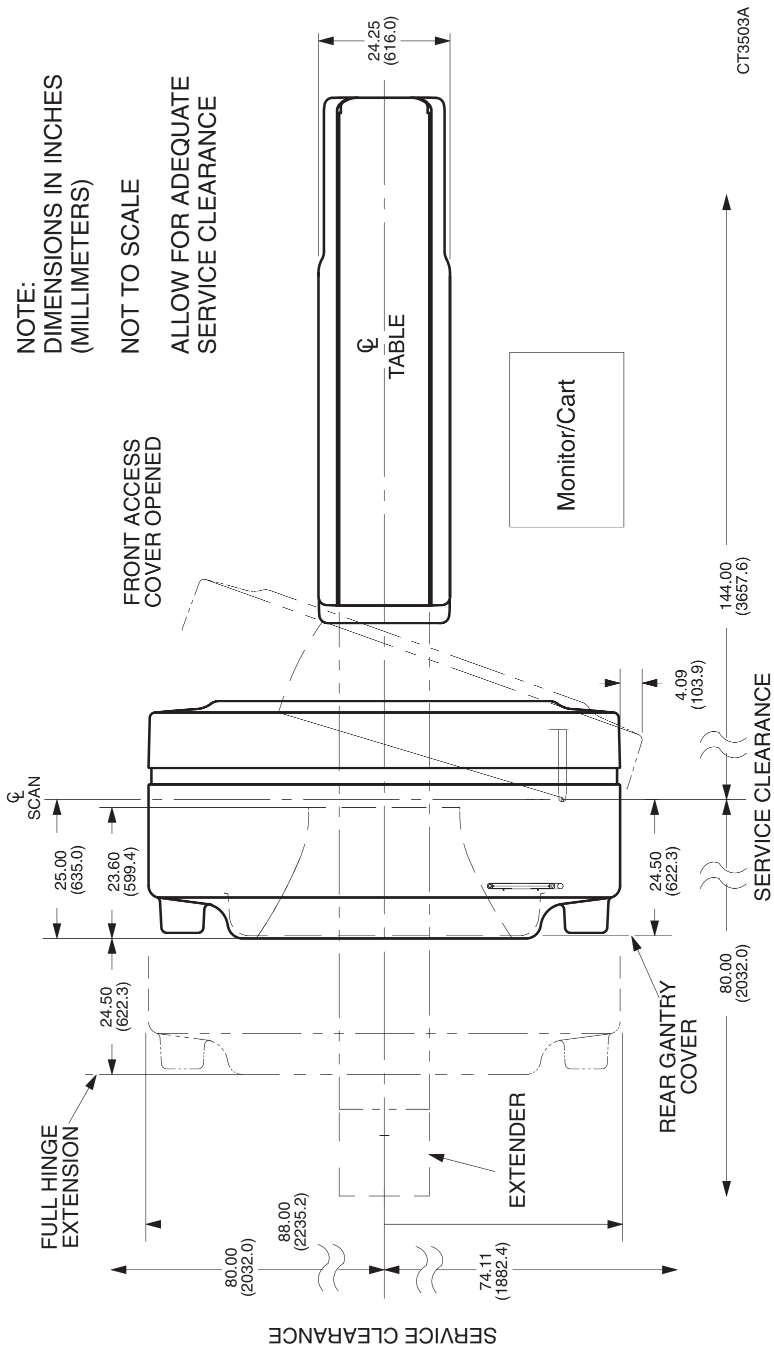


Figure 1-2 Table and Gantry (Top View) Exam Suite Monitor with CRT

Description	Model Number/ Part Number	Width		Depth		Height		Weight	
		Inch	mm	Inch	mm	Inch	mm	Lbs	Kg
Monitor with cart	2200601 (Monitor) 2200602 (Cart)	30	762	25	635	60	1524	195	88.5

Table 1-1 Dimensions of Components: Exam Suite Monitor with CRT

2.3 Environmental Conditions

The environmental condition requirements do not change with this upgrade.

2.3.1 Temperature and Humidity Specifications

- Ambient Temperature:
  - Maintain a temperature of 70^{o} – 75^{o} F (21^{o} – 24^{o} C) in scan room for patient comfort. When scan room is unoccupied, table and gantry temperature limitations equal 60^{o} – 75^{o} F (15^{o} – 24^{o} C).
  - Console/Computer, PDU: 60^{o} – 84^{o} F (15^{o} – 29^{o} C).
- Maintain relative humidity of 30% – 60% (non-condensing) during operation (all areas).
- The maximum temperature rate of change is 5^{o} F/hr (3^{o} C/hr).
- The maximum room temperature gradient is 5^{o} F (3^{o} C).
- The maximum relative humidity rate of change is 5% RH/hr.

2.3.2 Cooling Requirement

The new monitor creates an additional load of 300 BTU (80 Watts) per hour of use.

2.3.3 Electro-Magnetic Interference (EMI)

Color Monitor. Locate color monitors in ambient static magnetic fields of less than 5 x 10^{-5} tesla to guarantee color purity and display geometry.

If fields of excessive EMI are known or suspected to be present, consult GE Medical Systems for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from source of magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- Steel reinforcing in building structure can be a conductor of EMI.
- High-powered radio signals are a source of EMI.
- Maintain good screening of cables and cabinets.

2.3.4 Seismic Data

Follow the specified seismic reinforcement requirements for your area.

2.4 Delivery Data

WARNING: POTENTIAL FOR INJURY **SOME ASSEMBLIES ARE TOP-HEAVY. DO NOT TIP.**

2.4.1 Van Delivery

The Exam Suite Monitor option upgrade kit consists of 2 shipping containers.

2.4.2 Crated Delivery

When shipped overseas, the Exam Suite Monitor option upgrade kit consists of 2 boxes, which may be crated. Total weight of the upgrade kit without crate(s) about 125kg. The total weight of the upgrade kit with crates depends upon the wood used for crating.

2.4.3 Storage Requirements

If you need to store the Exam Suite Monitor option upgrade kit before installation, store in it a warehouse. Protect the kit from weather, dirt and dust. Storage temperature should not exceed -20^{o} to +120^{o} F (-30^{o} to +50^{o} C). Maintain relative humidity (non-condensing) between 0 and 80%. Do not store for more than 90 days.

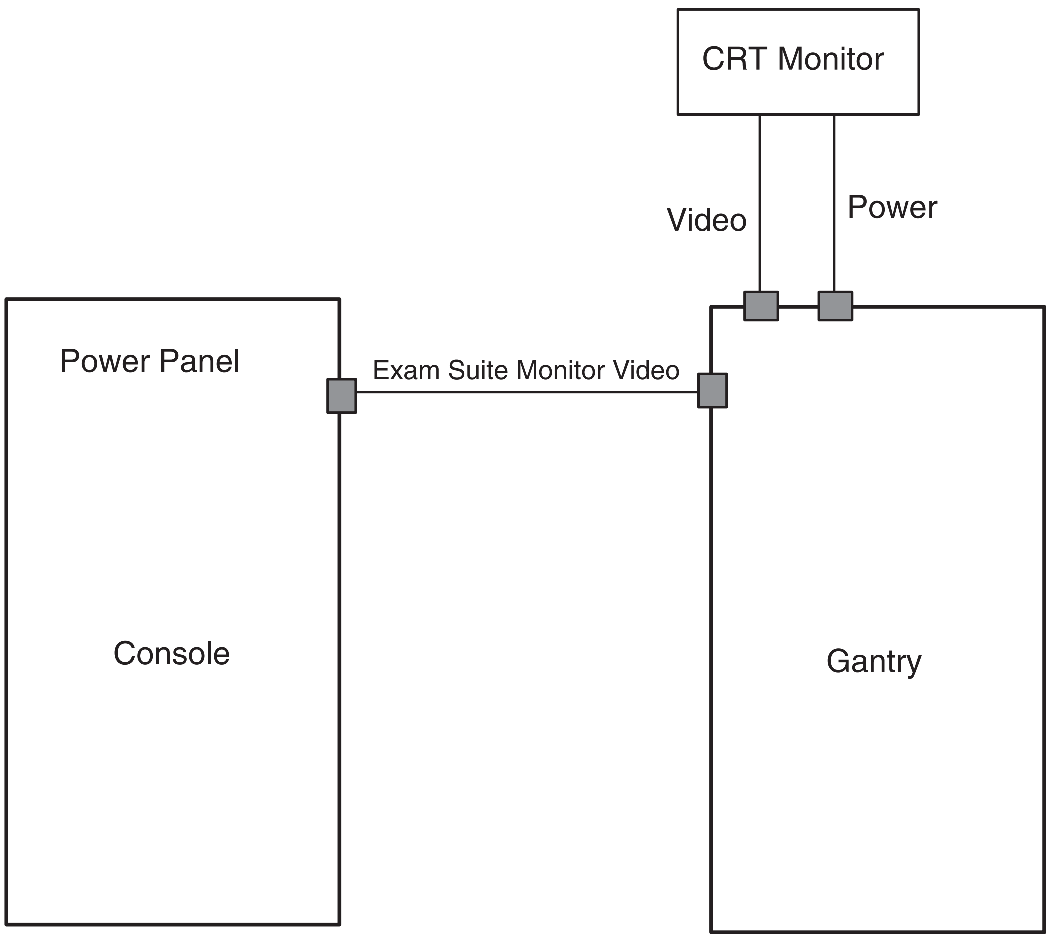


Figure 1-3 Interconnect Diagram Exam Suite Monitor with CRT

2.4.4 Exam Suite Monitor with CRT Cables

The following table contains the list of Exam Suite Monitor Cables and lengths.

Cable Run Number	Cable Part Number	From	To	Cable Length meter/feet
18		Operator Console	Gantry	24.4m/80 ft.
18a	2199957	Left Gantry modified base cover	CRT Monitor	5m/16 ft.
19	2199954	Left Gantry modified base cover	CRT Monitor	5m/16 ft.

Table 1-2 Exam Suite Monitor with CRT Option Cables

2.5 Power Requirements

The monitor receives its power from the Gantry base.

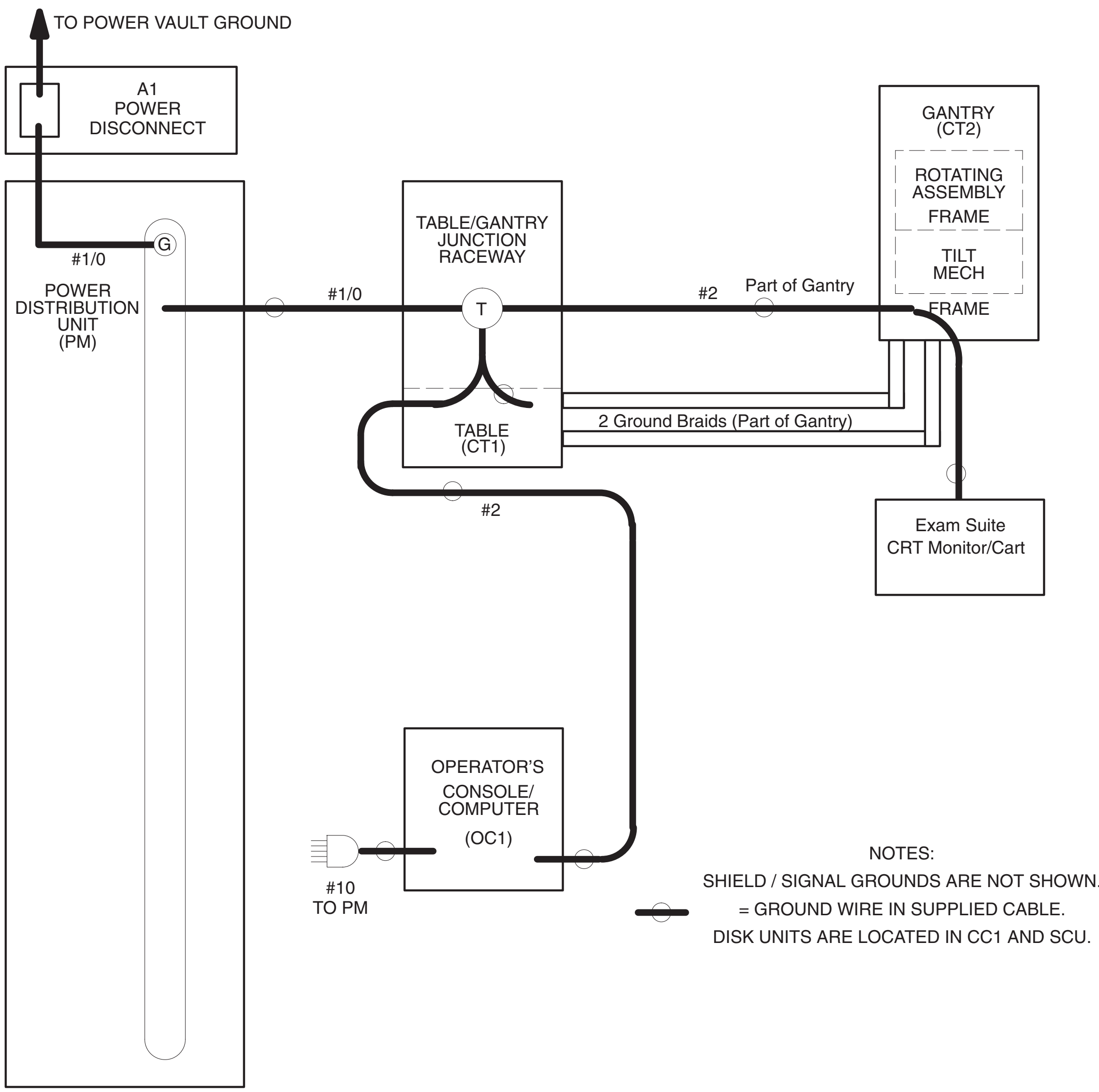


Figure 1-4 Monitor Power and Ground Map for Exam Suite Monitor with CRT

2.6 Preinstallation Checklist for Exam Suite Monitor with CRT

Scheduled
Delivery Date Salesperson

Customer FDO # Room #

Equipment: ______________________________

YES	NO	
		1.) Detailed plans of existing room layout available for the installation crew.
		2.) On site (secure) storage available for upgrade kit.
		3.) Existing cables between Control Area and Scan Room are in (circle one): ***Raceway Conduit Other*** (describe)
		a.) Existing raceway/conduit can accommodate additional cable consisting of 3 video coax cables that require a minimum 1.75 in./45 mm diameter clearance.
		b.) Existing raceway/conduit separates power and signal cables to minimize noise?
		c.) Maximum run length from the back of the console to the rear of the gantry 74 ft. *or less*.
		4.) A least 5 ft. (1.5m) of clearance exists between back of gantry and nearest wall.
		5.) Patients normally transferred from gurney to ***right left*** side of table. (circle one)
		6.) Position CRT Monitor/Cart on (***right left***) side of gantry. (circle one)
		Will monitor location interfere with any non-electrical lines (air, oxygen, vacuum, water)?
		7.) Site conditions:
		a.) All preliminary work completed per plan.
		b.) All work which creates dust in adjacent areas completed.
		c.) All seismic code requirements met.
		d.) Radiation physicist consulted.
		8.) Inspection Dates:

Comments:

Installation Specialist’s Signature Date

Salesperson’s Signature Date

Section 3.0
Exam Suite Monitor with CRT Installation

3.1 Install CRT, Power and Video Cables

1.) Remove the CRT from the corresponding upgrade kit box.
   - Remove the Monitor/Bracket assembly from the box.
   - Remove the cart and hardware mounting kit from the box.
2.) Connect all grounds. Check the Ground/Leakage Current between the CRT and Gantry before using the equipment. Do this immediately after applying power to the system.
3.) Wrap the Monitor power harness (2199954) and video cable (2199957) in the vinyl cable cover (2212717-2).
4.) Refer to Figure 1-5 and Figure 1-6.
   a.) Route the covered cables through the clamp at the base of the Monitor Cart.
   b.) Feed the covered cables through the hole in the base of the cart column. (If you have trouble feeding the cables up the column, tip the base forward, onto the floor, and slide the cables down the column.)
   c.) Follow the enclosed OEM instructions to assemble the yoke to the cart column.

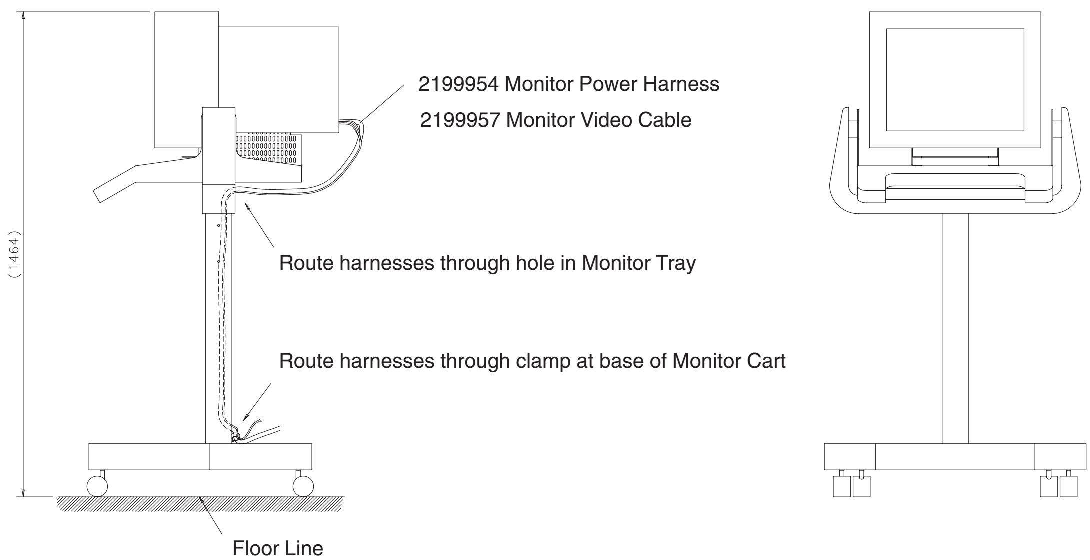


Figure 1-5 Route CRT Harnesses Through Monitor Cart

5.) Route the harnesses through the hole under the Monitor Tray, and out of the hole at the bottom of the Monitor Cart base.

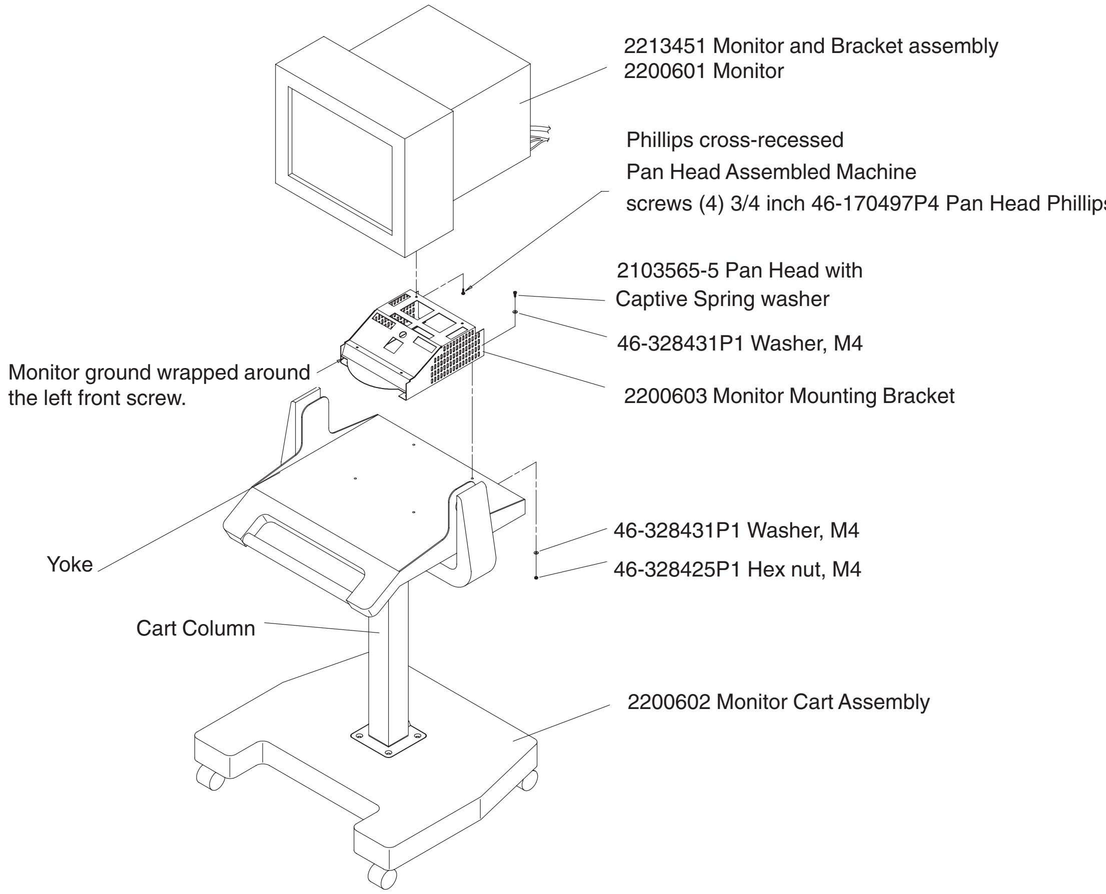


Figure 1-6 CRT Monitor Cart Assembly

6.) Use 4 pan head screws with captive spring washers to attach the Monitor/Bracket assembly to the Monitor Cart assembly.

3.2 Modify the Operator Console

3.2.1 Install the 4-Way Video Splitter

Follow this procedure to place the 4-Way Video Splitter to sit on the console monitor shelf between the two monitors. The splitter provides the video signal to the CRT scan room monitor.

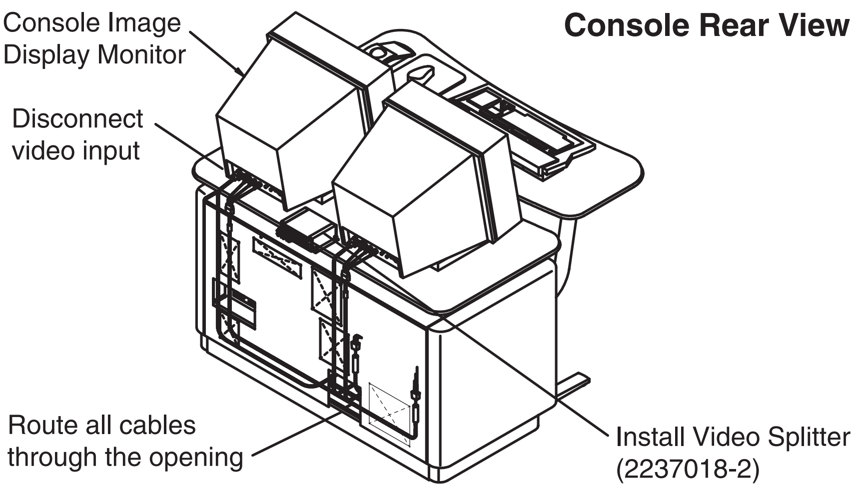


Figure 1-7 Rear View of Console with 4-Way Video Splitter Installed

1.) Remove the front cover from the console.
2.) Refer to Figure 1-7.
3.) Disconnect the video input to the Console Image Display monitor (right monitor).
   If your Image Display Monitor uses a HD15 connector instead of BNC connectors, remove the video cable and replace it with the supplied Monitor Video Cable (2154427). Discard the removed cable.
4.) Remove the 4-Way Video Splitter from the upgrade kit.
5.) Refer to Figure 1-8.

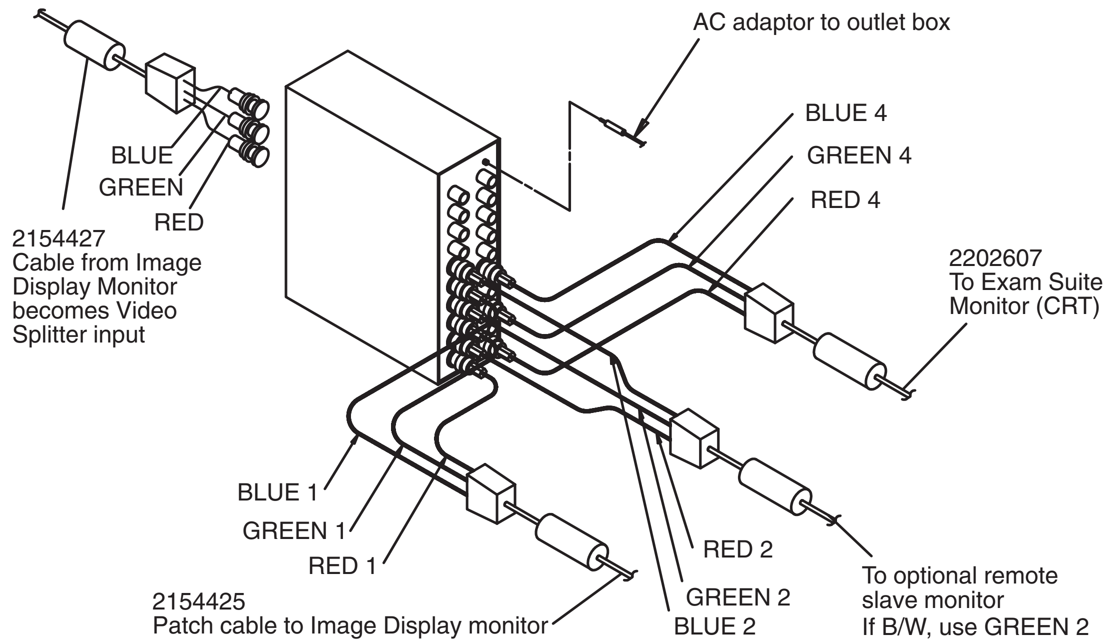


Figure 1-8 Attach the Cables to the Video Splitter

6.) Attach the video cables you removed from the Image Monitor to the video input side of the 4-way splitter.

7.) Remove the video cables from the upgrade kit.

   a.) Install a patch cable from the output of the video splitter to the video input on the Operator Console Image Display Monitor.

   If your Image Display Monitor uses a HD15 connector instead of BNC connectors, attach the supplied BNC-to-HD15 adapter (2256485) between the Image Display Monitor and the video patch cable.

   b.) If present, attach the video cable for the remote slave monitor option to the RED 2/ GREEN 2/BLUE 2 video connectors. If the system has a black and white slave monitor, attach the BNC connectors to the GREEN 2 video output connector.

8.) Attach the AC adaptor to the jack in the upper right corner of the output side of the video splitter.

9.) Plug the video splitter AC adapter into one of the console's power outlets. Refer to Figure 1-9.

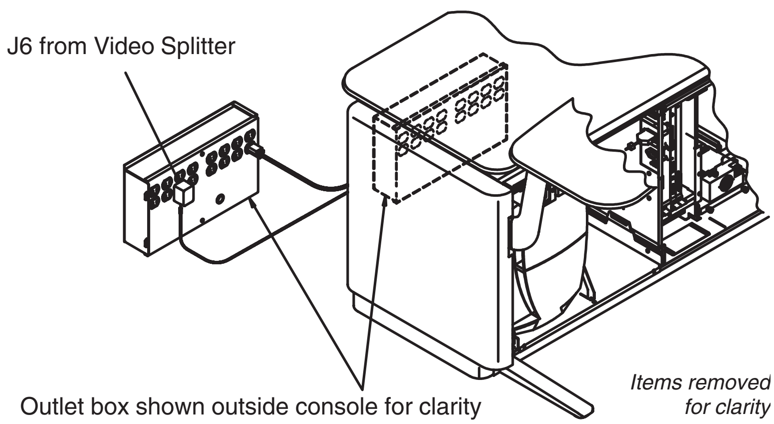


Figure 1-9 Operator Console Outlet Box

3.2.2 Attach Console Modification Plate

1.) Remove the modification plate form the upgrade kit.

2.) Refer to Figure 1-10.

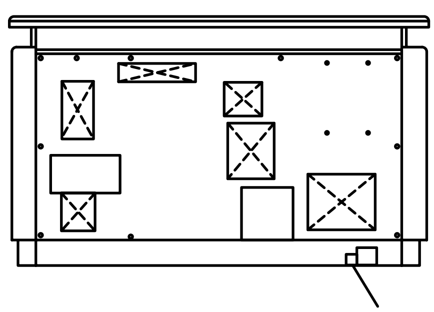


Figure 1-10 Attach Modification Plate to Operator Console

3.) Attach the new plate to the left of the existing rating plate(s).

4.) Take care not to cover any part of the existing plate(s).

3.3 Modify Gantry

3.3.1 Remove Covers from Gantry and Table

1.) Turn off the Mains disconnect power, A1. Tag and lock out.
2.) Remove both side covers from the Gantry.
3.) Turn off the following Gantry Service switches:
   - 550VDC
   - 120VAC
   - Gantry Rotate (Axial Drive Enable)
4.) Open the Front and Rear Gantry covers.
5.) Remove, and set aside, both side covers from the Gantry base.
6.) Refer to Figure 1-7 and Figure 1-11.
7.) Attach CRT Video cable 2202607 to the video output jacks on the 4-Way video splitter you located on the monitor shelf between the two monitors on the Operator Console.
   a.) Thread or lay the cable along the designated route between the operator console and the rear of the gantry, at floor level.
   b.) Route the video cable to the STC side of the gantry base. Final connections at the Gantry will be completed later in this procedure.

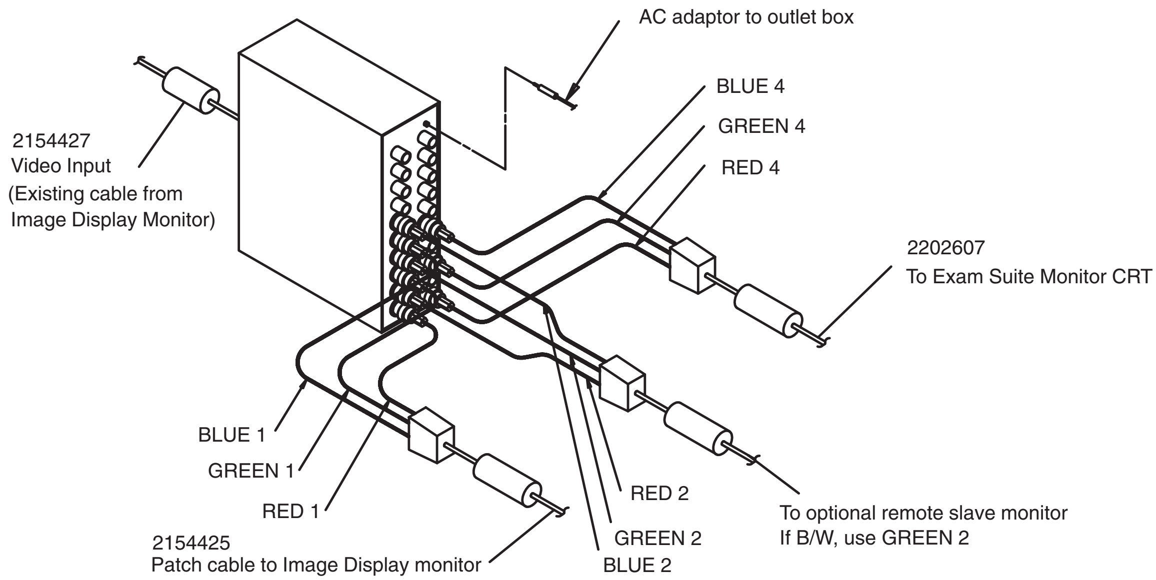


Figure 1-11 Attach CRT Video Cable to 4-Way Splitter

3.3.2 Install the Power Outlet Harness:

1.) Remove the outlet plate from the table side of the left (2215053) modified base cover, to receive the CRT Power and Video cables.

2.) Remove the 2200606 CRT Power Outlet Harness from the upgrade kit. Refer to Figure 1-12.

3.) Thread the ring terminated ends of the power cable through the square hole in the base cover, and fasten the outlet into place with the ground receptacle facing upwards.

4.) Attach the Power outlet harness (2200606) to the 120VAC power strip, located on the STC side of the gantry, on the tilt frame.

- Attach the green/yellow (ground) wire to terminal 1.
- Attach the blue (neutral) wire to terminal 2.
- Attach the black/brown (hot) wire to terminal 3. Check the in-line fuse (10A, 250V) before you install all the gantry baseplate covers.

5.) Refer to Figure 1-12.

a.) Remove the plates nearest the rear of the gantry from both base covers.

b.) Continue to route the video cables along the existing gantry harnesses, toward the STC side of the gantry, but do NOT ty-wrap them into place at this time.

c.) Finish off the cable with the connectors on the Table side of the gantry.

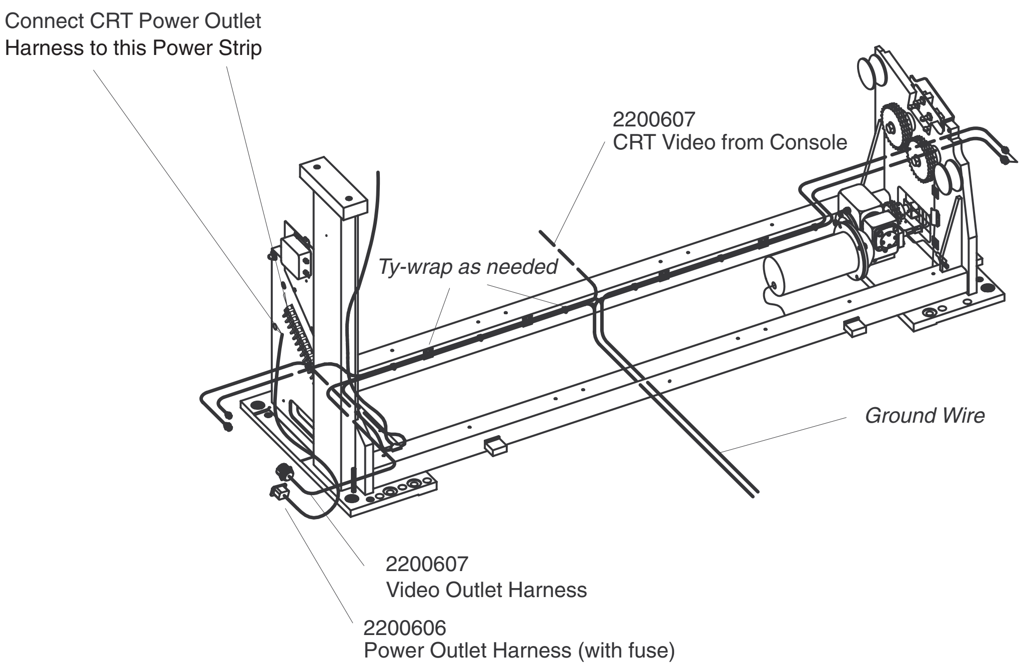


Figure 1-12 Install CRT Power and Video Cables

Section 4.0
Install Modified Gantry Base Side Covers

1.) Refer to Figure 1-13.
2.) Discard the existing side base covers, and remove the modified right (2215054) and left (2215053) base covers from the kit.

When correctly attached to the gantry base, the CRT Power and Video outlets exit the table side of the left (STC side) modified base cover.
3.) Remove the plates nearest the rear of the gantry from both base covers.

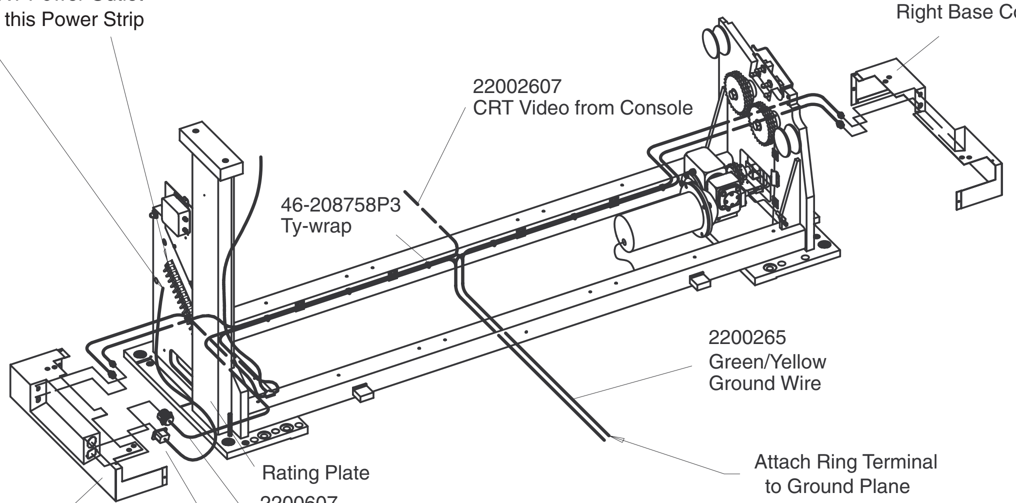


Figure 1-13 Install Modified Gantry Base Covers

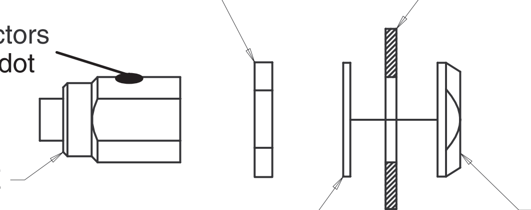


Figure 1-14 Attach Cables to Gantry Side Covers

4.) Attach the video cable connector to the round hole next to the power outlet on the left modified base cover.
5.) Orient the cable with the red dot on the connector facing UP. Check the in-line fuse (10A, 250V) before you install all the gantry baseplate covers.
6.) Attach the new covers to each side of the gantry base. Take care not to pinch any cables when you attach the covers.

Section 5.0
Restore Power to the System

5.1 Restore System Power

1.) Remove the tag, and turn on A1.
2.) Turn on the following Gantry Service switches:
    - 550V
    - 120V
    - Gantry Rotate (Axial Drive Enable)
3.) Use a multimeter to check the CRT outlet in the left Gantry base cover for 120VAC. If you don't get a voltage reading, check the fuse in series with the brown/black hot receptacle, located directly behind the left modified base cover.
4.) Replace both Gantry side covers.
5.) Install and connect the CRT.
    a.) Move the Monitor cart into position.
    b.) Connect the video cables between the connector in the modified Gantry base cover and the back of the CRT monitor
    c.) Plug the CRT monitor power cord into the outlet in the modified Gantry base cover.
    d.) Turn on the CRT monitor.

WARNING POTENTIAL FOR SHOCK — **CAREFULLY CHECK ALL GROUND CONNECTIONS, THEN PERFORM THE LEAKAGE/GROUND CURRENT TEST PROCEDURE IN THE FOLLOWING SECTION.**

5.2 Leakage/Ground Current Test for Exam Suite Monitor with CRT

1.) Turn both the microammeter and CRT off.
2.) Plug the CRT into the test meter, then plug the meter into a tested outlet.
3.) Locate the metal touchpoint on the rear of the CRT monitor, near the bottom, and touch it with one lead.
4.) Touch the other lead to the metal base of the gantry, preferably to a ground stud.
5.) With power on, the meter should read less than 100 microamps. If using a meter, resistance should read less than 0.50 ohms.
6.) Keep one lead on the ground stud in the gantry base.
7.) Touch any metal surface on the CRT Monitor Cart with the other lead.
8.) With power on, the meter should read less than 100 microamps. If using a meter, resistance should read less than 0.50 ohms.

Section 6.0
B7710WR Exam Suite CRT Monitor & Cart Option Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	2199954	Monitor Power Harness	1	1	Monitor to Gantry Base
2	2199957	Monitor Video Cable	1	1	Monitor to Gantry Base
3	2212717-2	Cable Cover for CRT	2	1	Vinyl Cable Cover – 6.5m
4	2213451	Phillips Monitor & Bracket Assembly	1	1	Monitor and Bracket
5	2237018-2	4-Way Video Splitter	2	1	
6	2200602	Cart for Monitor	2	1	
7	2200602-5	4 in. Locking Caster	2	2	Front Casters
8	2200602-6	4 in. Non-Locking Caster	2	2	Rear Casters
9	2200607	Video Outlet	2		
10	2200606	Power Outlet	2		
11	2154427	Monitor Video Cable (Console)	2	1	
12	2154425	Splitter to Console Monitor Cable	1	1	Use to connect new style monitor
13	2256485	Adapter, 3BNC to HD15	1	1	

Table 1-3 B7710WR Exam Suite CRT Monitor and Cart Option Parts List

Chapter 2
Exam Suite Monitor with LCD Option

Section 1.0
System Requirement

None. Works with all software versions.

Section 2.0
Preinstallation for Exam Suite Monitor with LCD

2.1 Site Preparation Before Equipment Delivery

- Determine the location of the LCD ceiling mount, off the right or left side of the Gantry. (Consult the Radiologist for his or her preference.)
- Determine whether existing cables run through conduit, wire trough or under raised floor, and, if a sufficient opening exists to run the video coax and power cables from the operator console to the LCD monitor ceiling plate.

Video Cable

- 21.34 m (70 ft.) video cable from console to ceiling plate
  - About 1 foot needed at the ceiling plate
  - 5 ft. needed at the console (from center of back side, near floor)
- 4.6 m (15 ft.) video cable from ceiling plate to LCD monitor
- Each cable is about 5/8 in. in diameter
- Minimum 1.75 in. (45mm) hole to route cables between console and LCD
- 21.34 m (70 ft.) 120VAC grounded cable between ceiling plate and console
- 4.6 m (15 ft.) 120VAC grounded cable between LCD and ceiling plate

Power source: Operator console outlet box

2.2 Room Planning

Figure 2-1 contains a typical room layout for the CT components. Overall suite size equals 6.19 m (20.3 ft.) x 7.07 m (23.2 ft.) deep, with a 2.75 m (9 ft.) ceiling. The area consists of the following three rooms:

- Scan Room: 3.81 m x 6.10 m (19.4 ft. x 12.9 ft.)
- Control Area: 3.81 m x 2.67 m (12.3 ft. x 9.0 ft.)
- Equipment Room: 8.38 m x 3.20 m (6.7 ft. x 9.0 ft.)

Note: Other room arrangements are possible. Use cable lengths shown in Table 2-2 to plan the maximum distance between components. For example:

- The control and equipment rooms may be interchanged with the scanner room.
- The preferred location of options such as Advantage Windows, laser camera or line printer is in a separate room.

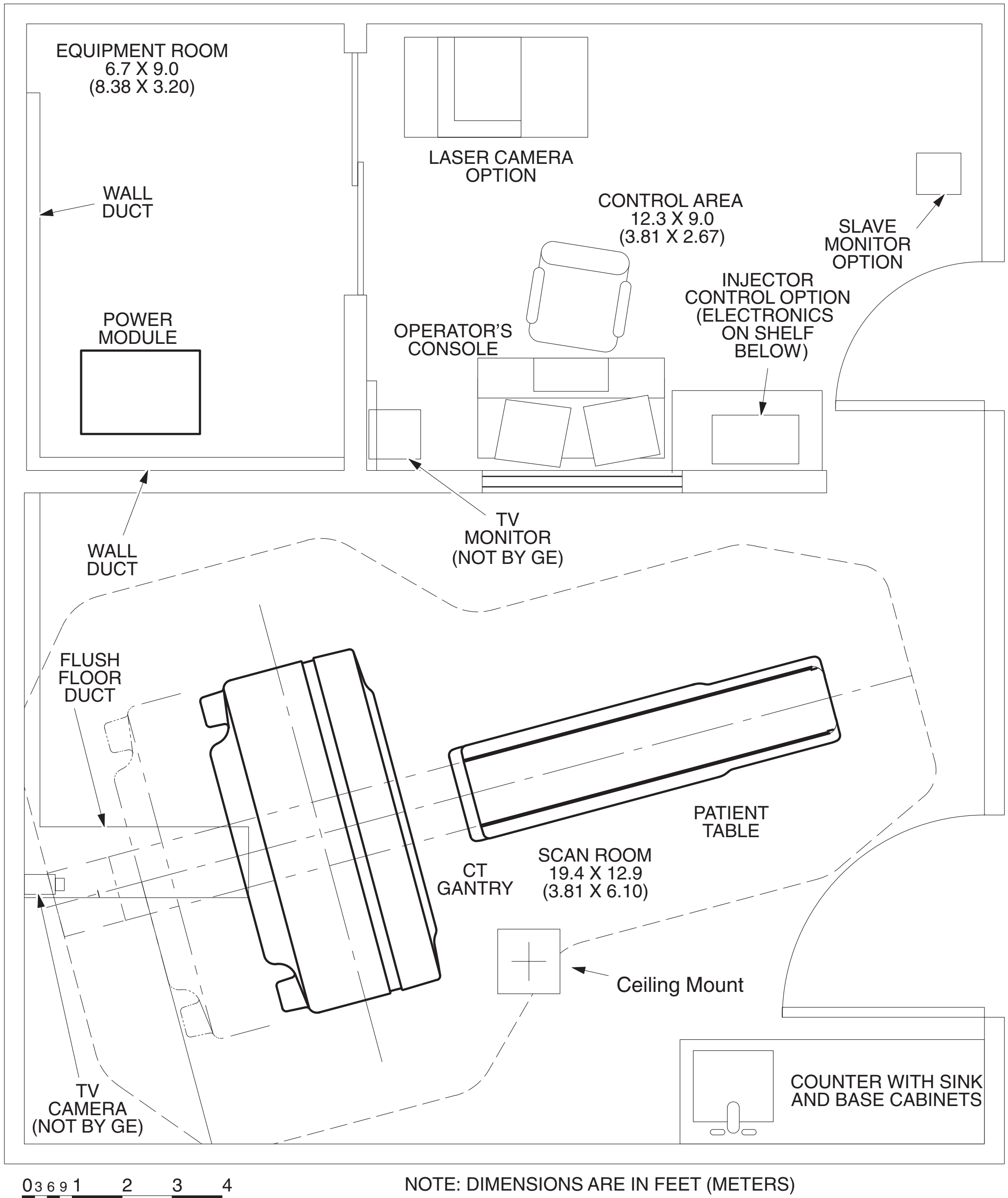


Figure 2-1 Recommended Suite Layout Exam Suite Monitor with Ceiling Mounted LCD

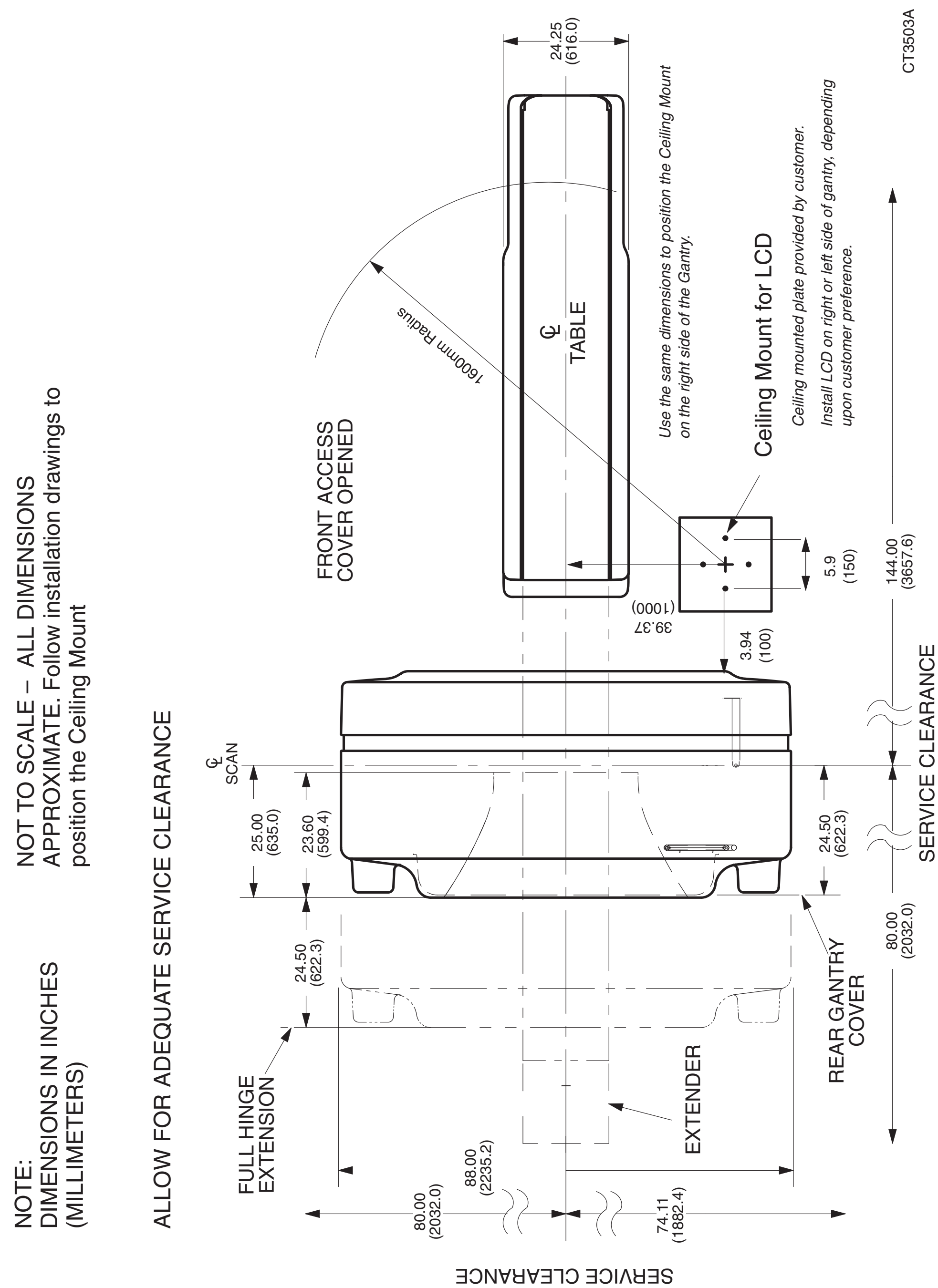


Figure 2-2 Table and Gantry (Top View) Exam Suite Monitor with Ceiling Mounted LCD

DESCRIPTION	MODEL/ PART NUM.	WIDTH		DEPTH		HEIGHT/ LENGTH		WEIGHT	
		Inch	mm	Inch	mm	Inch	mm	Lbs	Kg
LCD Monitor	2202532	20.4	517	3.1	80	17.3	440	16	7.3
LCD Ceiling Column	2214293	N/A	N/A	N/A	N/A	23	585	17	7.7
Extension/Spring Arm	2214292	N/A	N/A	N/A	N/A	31.5	800	16	7.3
Fully Extended		N/A	N/A	N/A	N/A	63	160		

Table 2-1 Dimensions of Components: Exam Suite Monitor with Ceiling Mounted LCD

2.3 Environmental Conditions

The environmental condition requirements do not change with this upgrade.

2.3.1 Temperature and Humidity Specifications

- Ambient Temperature:
  - Maintain a temperature of 70^o – 75^o F (21^o – 24^o C) in scan room for patient comfort. When scan room is unoccupied, table and gantry temperature limitations equal 60^o – 75^o F (15^o – 24^o C).
  - Console/Computer, PDU: 60^o – 84^o F (15^o – 29^o C).
- Maintain relative humidity of 30% – 60% (non-condensing) during operation (all areas).
- The maximum temperature rate of change is 5^o F/hr (3^o C/hr).
- The maximum room temperature gradient is 5^o F (3^o C).
- The maximum relative humidity rate of change is 5% RH/hr.

2.3.2 Cooling Requirement

The new monitor creates an additional load of 300 BTU (80 Watts) per hour of use.

2.3.3 Electro-Magnetic Interference (EMI)

Color Monitor. Locate color monitors in ambient static magnetic fields of less than 5×10^{-5} tesla to guarantee color purity and display geometry.

If fields of excessive EMI are known or suspected to be present, consult GE Medical Systems for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from source of magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- Steel reinforcing in building structure can be a conductor of EMI.
- High-powered radio signals are a source of EMI.
- Maintain good screening of cables and cabinets.

2.3.4 Seismic Data

Follow the specified seismic reinforcement requirements for your area.

2.4 Delivery Data

WARNING SOME ASSEMBLIES ARE TOP-HEAVY. DO NOT TIP.

2.4.1 Van Delivery

The Exam Suite Monitor option upgrade kit consists of approximately 2 shipping containers.

2.4.2 Crated Delivery

When shipped overseas, the Exam Suite Monitor option upgrade kit consists of 2 shipping containers which may be crated. Total weight of the upgrade kit without crate(s): 80 kg. The total weight of the upgrade kit with crates depends upon the wood used for crating.

2.4.3 Storage Requirements

If you need to store the Exam Suite Monitor option upgrade kit before installation, store in it a warehouse. Protect from weather, dirt and dust. Storage temperature should not exceed -20^{o} to +120^{o} F (-30^{o} to +50^{o} C). Maintain relative humidity (non-condensing) between 0 and 80%. Do not store for more than 90 days.

2.5 Exam Suite Monitor with Ceiling Mounted LCD Cables

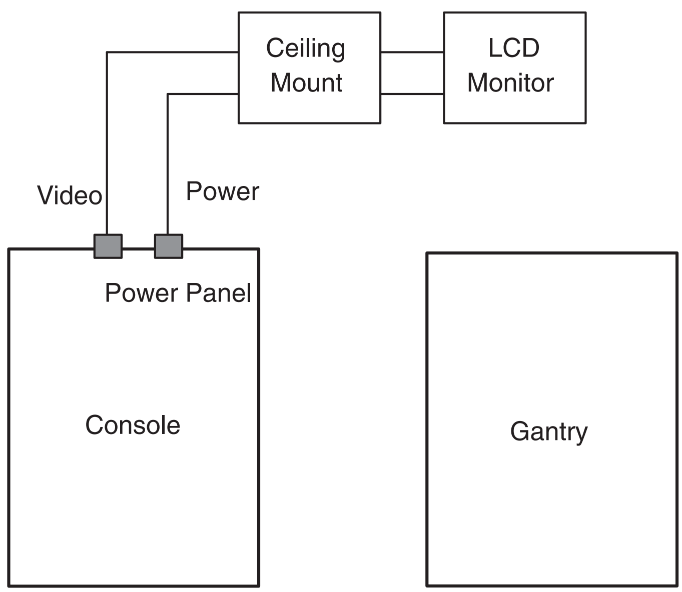


Figure 2-3 Interconnect Diagram Exam Suite Monitor with LCD

The following table contains the list of Exam Suite Monitor Cables and lengths.

RUN NUM	PART NUM	FROM	TO	LENGTH (M/FT)
18	2213203	Operator Console	Ceiling mounting plate	21.34m/70 ft.
19	2213218	Operator Console	Ceiling mounting plate	21.34m/70 ft.

Table 2-2 Exam Suite Monitor with LCD Option Cables

2.6 Power Requirements

The monitor receives its power from the Operator Console.

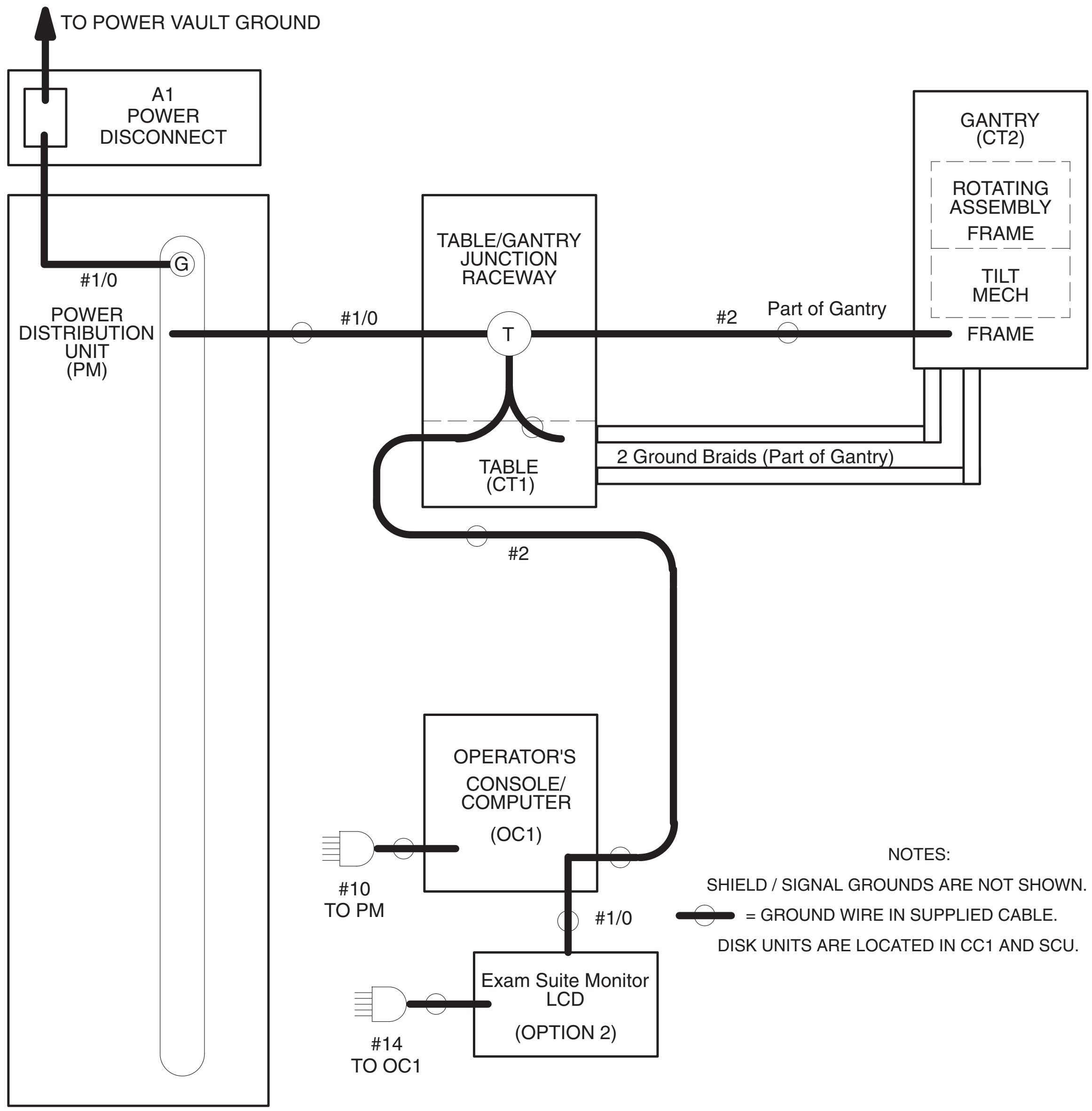


Figure 2-4 Monitor Power and Ground Map

2.7 Preinstallation Checklist for Exam Suite Monitor with LCD

Scheduled
Delivery Date Salesperson

Customer FDO # Room #

Equipment: ____

YES	NO	
		1.) Detailed plans of existing room layout available for the installation crew.
		2.) On site (secure) storage available for upgrade kit.
		3.) Existing conduit between console and LCD ceiling plate?
		a.) Separate conduit for video cable with minimum 1.75 in./45mm diameter clearance.
		b.) Separate conduit for power cable/ground with minimum 1 in./25mm diameter clearance.
		4.) Suspend LCD on the (***right left***) side of gantry. (circle one)
		a.) Will suspension interfere with any non-electrical lines (air, oxygen, vacuum, water)?
		b.) Will the conduit between the LCD monitor and Operator Console equal 64 ft. (1951cm) or less?
		c.) Ceiling mounting plate for LCD present and in correct location, relative to the gantry?
		d.) Electrical box in place in ceiling next LCD mounting plate?
		5.) At least 5 ft. (1.5m) of clearance exists between back of gantry and nearest wall.
		6.) Site conditions:
		a.) All preliminary work completed per plan.
		b.) All work which creates dust in adjacent areas completed.
		c.) All seismic code requirements met.
		d.) Radiation physicist consulted.
		7.) Inspection Dates:

Comments:

Installation Specialist’s Signature Date

Salesperson’s Signature Date

Section 3.0
Exam Suite Monitor with LCD Installation

3.1 Install LCD, Power and Video Cables

Install the recommended, GE supplied LCD monitior. Follow the instructions in 2390072-100 (Rev 3 or later) - CRT & LCD Remote Monitor & Monitor-in-Room Boom Option.

3.2 Modify the Operator Console

3.2.1 Safety Considerations

CAUTION **Do not use a multiple circuit outlet on the Operator Console.**

NOTICE **Do not allow cables behind the Operator Console to drag on the floor. In order to prevent cable damage, route all cables so as to avoide pinch points, casters, etc.**

3.2.2 Install the 4-Way Video Splitter

Follow this procedure to place the 4-Way Video Splitter to sit on the console monitor shelf between the two monitors. The splitter provides the video signal to either the LCD or CRT scan room monitor.

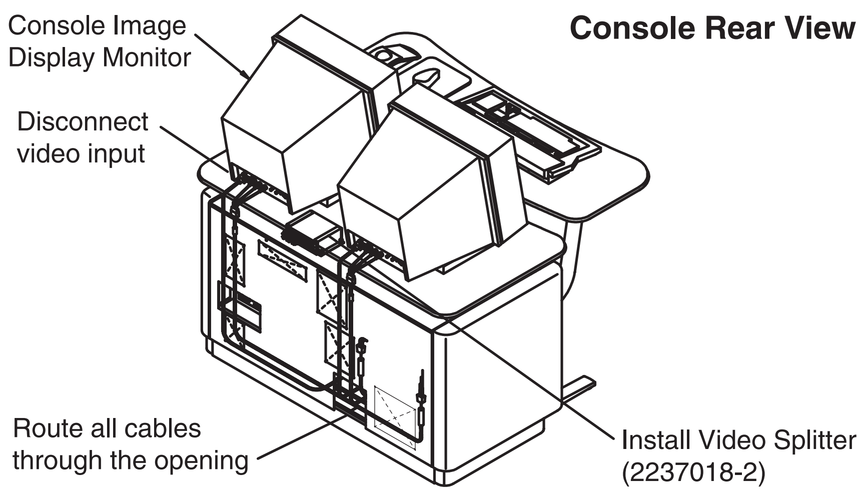


Figure 2-5 Rear View of Console with 4-Way Video Splitter Installed

1.) Remove the front cover from the console.
2.) Refer to Figure 2-5.
3.) Disconnect the video input to the Console Image Display monitor (right monitor).

If your Image Display Monitor uses a HD15 connector instead of BNC connectors, remove the video cable and replace it with the supplied Monitor Video Cable (2154427). Discard the removed cable.

4.) Attach the video cables you removed from the Image Monitor to the video input side of the 4-way splitter. Refer to Figure 2-6.

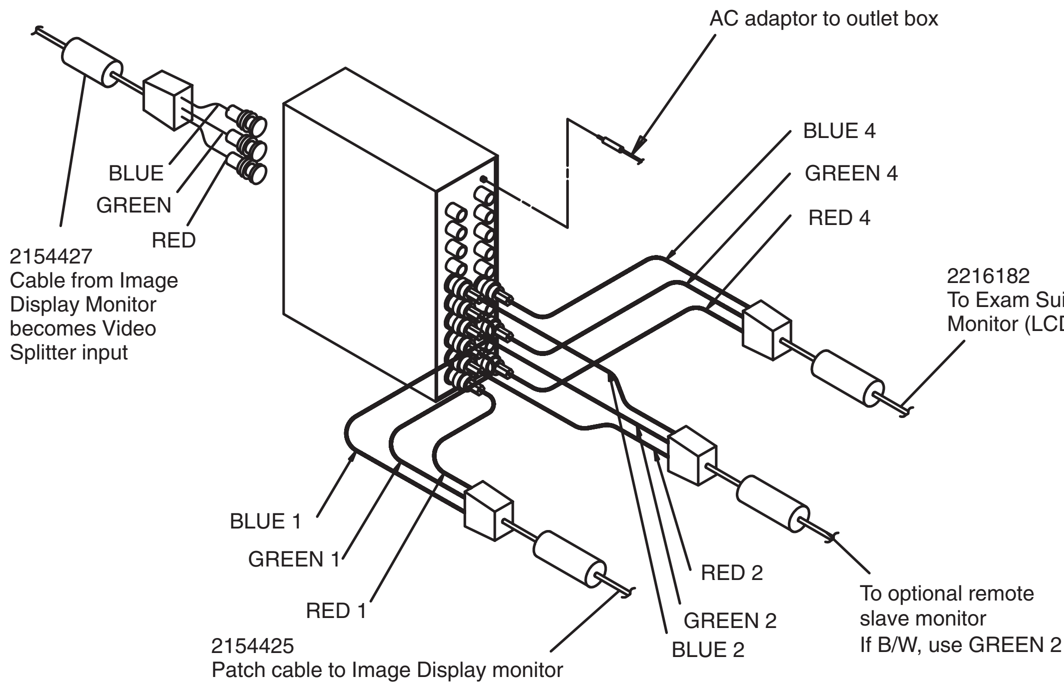


Figure 2-6 Attach the Cables to the Video Splitter

5.) Remove the video cables from the upgrade kit.

a.) Install a patch cable from the output of the video splitter to the video input on the Operator Console Image Display Monitor.

If your Image Display Monitor uses a HD15 connector instead of BNC connectors, attach the supplied BNC-to-HD15 adapter (2256485) between the Image Display Monitor and the video patch cable.

b.) If present, attach the video cable for the remote slave monitor option to the RED 2/ GREEN 2/BLUE 2 video connectors. If the system has a black and white slave monitor, attach the BNC connectors to the GREEN 2 video output connector.

6.) Attach the AC adaptor to the jack in the upper right corner of the output side of the video splitter.

7.) Refer to Figure 2-7.

8.) Plug the video splitter AC adapter into one of the console's power outlets.

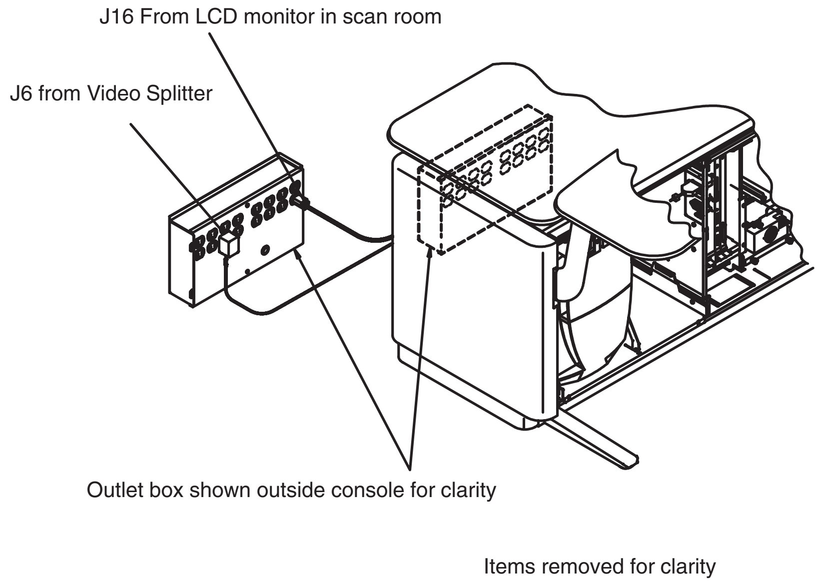


Figure 2-7 Operator Console Outlet Box

9.) Refer to Figure 2-7, Figure 2-8 and Figure 2-9.

a.) Route the LCD power cable into the opening in the rear cover, and plug it into the console's outlet box.

b.) Route the ground cable to the ground stud in the corner of the console, beneath the outlet box, and fasten securely into place.

c.) Route the LCD video cables to the Operator Console. Connect them to the video splitter, as shown in Figure 2-9.

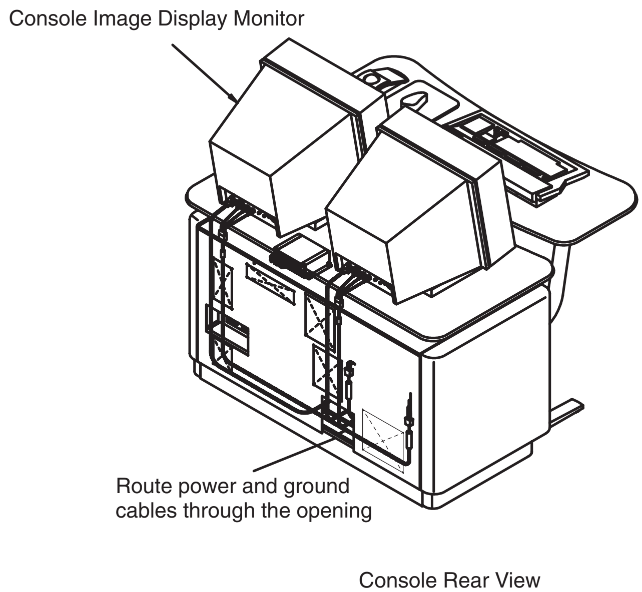


Figure 2-8 Rear View of Console

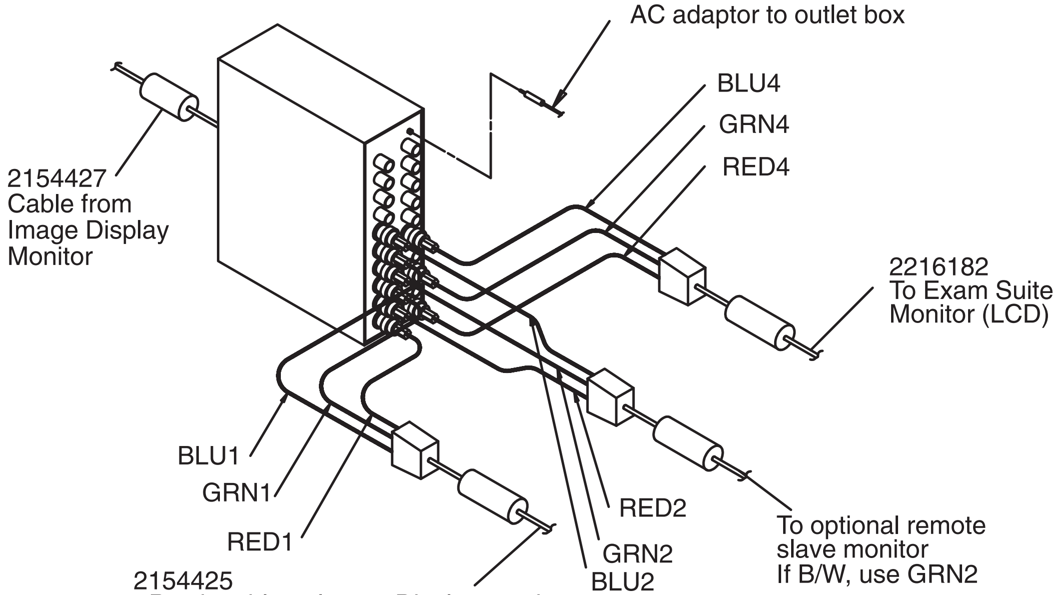


Figure 2-9 Connect LCD Video Cable to the 4-Way Video Splitter

10.) Use ty-wraps, as needed, to secure the cables.

3.2.3 Attach Console Modification Plate

1.) Remove the modification plate from the upgrade kit.

2.) Refer to Figure 2-10.

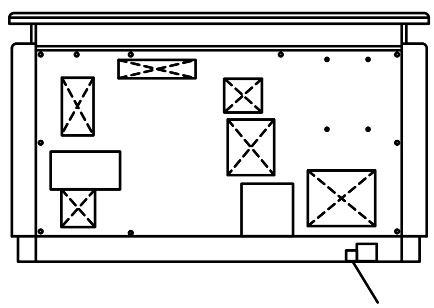


Figure 2-10 Attach Modification Plate to Operator Console

3.) Attach the new plate to the left of the rating plate(s).

4.) Take care not to cover any part of the existing plate(s).

Section 4.0
Balance LCD Extension/Spring Arm Assembly

When it arrives, the Extension/Spring Arm assembly is adjusted to support a heavier load than the LCD monitor and bracket. When out-of-balance, the LCD monitor tends to drift from its rest position. Follow this procedure to adjust the arm to balance the LCD monitor.

1.) Refer to Figure 2-11. Remove the plastic end cover on the adjustable arm by pulling outward on each side while lifting the plastic cover upwards.

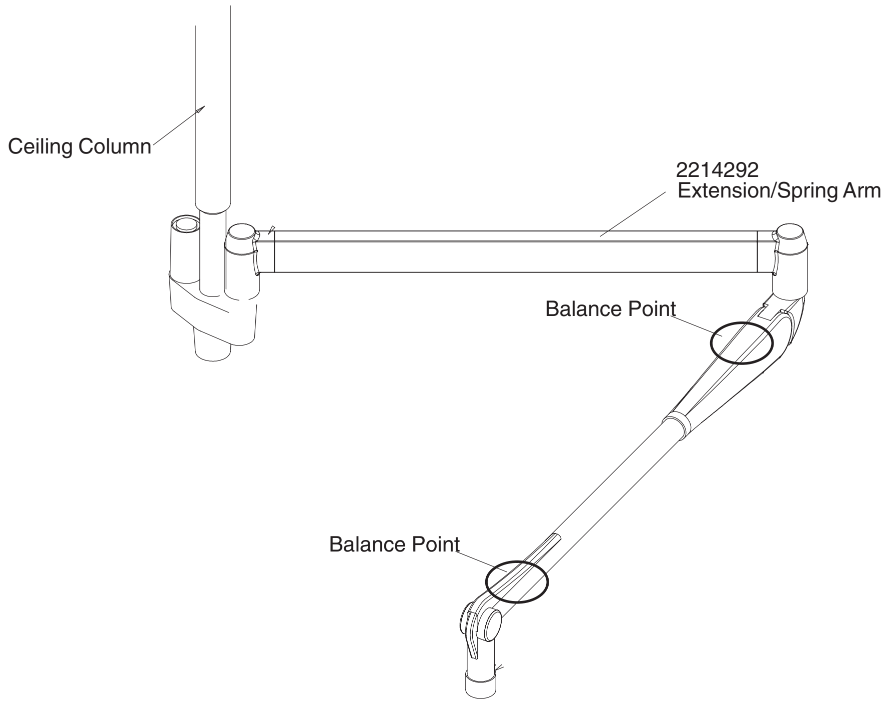


Figure 2-11 Balance the Extension/Spring Arm Assembly

2.) Locate the allen wrench adjusting tool, attached to the plastic end cover. Remove the adjusting tool from its storage clip.
3.) Slowly move the arm up and down to center the adjustment nut in the adjustment slots.
4.) Insert the tool into the adjustment nut, and rotate the nut toward the minus (-) sign.
5.) As you rotate the nut in the minus direction, the arm lowers. Continue to adjust the nut until the arm and monitor remain in position, without drifting.
6.) Raise the LCD assembly to its uppermost position. If the arm strikes the ceiling:
    a.) Use the adjusting tool to pry the cover off the height adjustment slot.
    b.) Move the arm horizontally, until the vertical adjustment nut appears in the adjustment slot.
    c.) Insert the adjusting tool, and turn the nut to adjust the tension until the arm no longer touches the ceiling.

WARNING **COLLISION WITH OTHER EQUIPMENT CAN CAUSE STRUCTURAL FAILURE OF SUSPENSION ARM, RESULTING IN OPERATOR INJURY.**

7.) Return the adjusting tool to the storage clip, and replace all the covers you removed.
8.) As a final test, move the LCD to the highest (storage) position; make sure it doesn't drift downward. Move the LCD to about eye-level; make sure it doesn't drift up or down. Repeat the adjustment process until the LCD remains in position without drifting.

Section 5.0
Restore Power to the System

5.1 Restore System Power

1.) Turn on power to the system.
2.) Turn on the following Gantry Service switches (if you turned on any of them off during installation):
   - 550V
   - 120V
   - Gantry Rotate (Axial Drive Enable)
3.) Turn on LCD monitor power, to ensure that it is working.

WARNING **PREVENT ACCIDENTAL SHOCK TO PATIENTS. CAREFULLY CHECK ALL GROUND CONNECTIONS, THEN PERFORM THE LEAKAGE/GROUND CURRENT TEST PROCEDURE IN THE FOLLOWING SECTION.**

5.2 Leakage/Ground Current Test for Exam Suite Monitor with LCD

1.) Turn both the microammeter and LCD off.
2.) Plug the LCD into the test meter, then plug the meter into a tested outlet.
3.) Set the meter switches for a current reading around 50 microamps.
4.) Turn on the meter
5.) Touch one lead to the metal part of the LCD.
6.) Touch the other lead to the metal base of the gantry, preferably to a ground stud.
7.) With power on, the meter should read less than 100 microamps.
8.) If using a meter, resistance should read less than 0.20 ohms.

Section 6.0
B7710WM Exam Suite LCD Monitor & Suspension Option Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	2214293	Ceiling Column with Flange	2	1	
2	2214292	Extension/Spring Arm	2	1	
3	2202532	CT LCD Monitor	1	1	
4	2211031	LCD Bracket — Short	2	1	
5	2216181	Video Cable for LCD	1	1	LCD to Ceiling Plate
6	2213213	Gnd. Cable for LCD	2	1	LCD to Ceiling Plate
7	2213218	Power Cable for LCD	2	1	LCD to Ceiling Plate
8	2216182	Video Cable — LCD to Console	2	1	Ceiling Plate to Console
9	2213219	Power Cable — LCD to Console	2	1	Ceiling Plate to Console
10	2237018-2	4-Way Video Splitter	2	1	
11	2154425	Splitter to Console Monitor Cable	1	1	
12	2256485	Adapter, 3 BNC to HD15	1	1	Use to connect new style monitor
13	2154427	Monitor Video Cable (Console)	2	1	

Table 2-3 B7710WM Exam Suite LCD Monitor and Suspension Option Parts List

Chapter 3
Dentascan Option Installation

Section 1.0
System Requirement

None. Works with all software versions.

Section 2.0
Install Dentascan Option Software

Software Options must be loaded/re-loaded every time the OC software is reloaded.

1.) In Service Desktop, select TOOLBOX/UTILITIES -> INSTALL-> INSTALL OPTIONS.

This program must be run once for each option MOD you are loading. For example, if you have two options MODs you must run this program twice.

An Options Window appears.

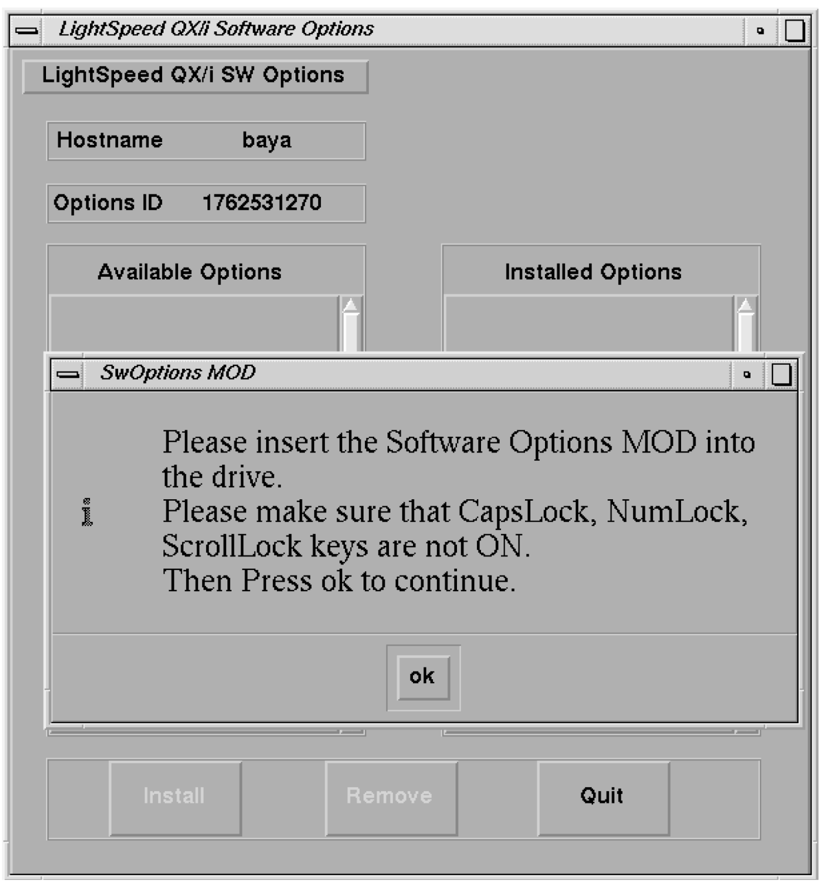


Figure 3-1 Software Options Installation Window

2.) Insert the Options MOD into the drive
3.) Click on OK

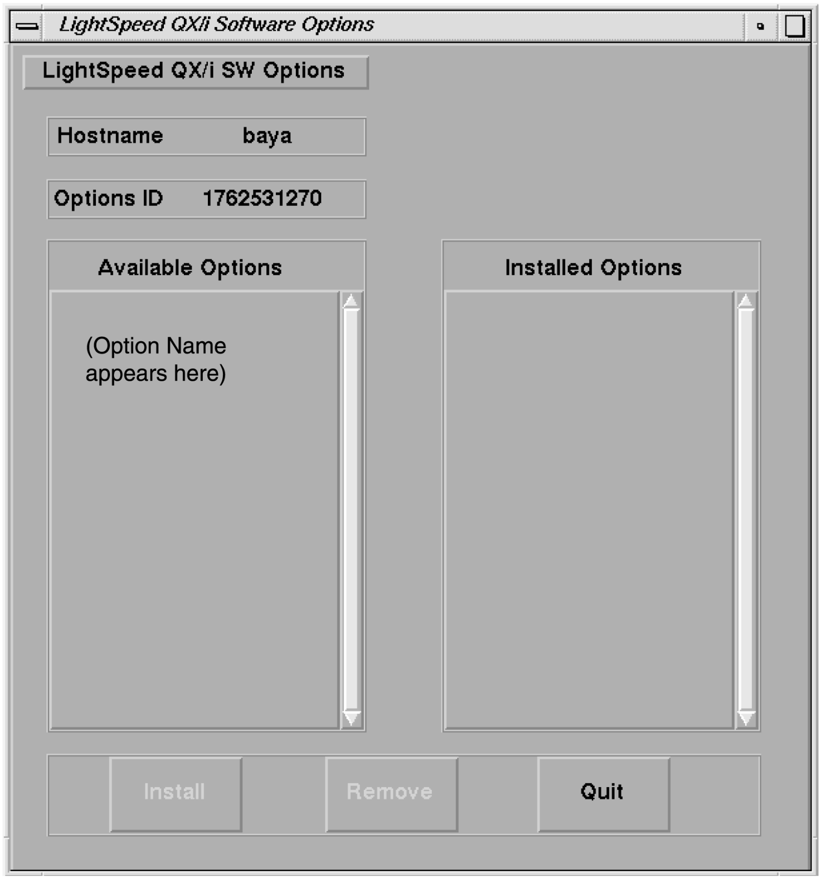


Figure 3-2 Software Options Available/Installed Screen Example

4.) The following message will appear:

Please select a printer in the list

[1] <camera1 name>
[2] <camera2 name>
[3] <camera3 name>
... (system shows complete list of available cameras)

Enter index of the camera selected and press <Enter>

Follow the on-screen instructions.

5.) The following message will appear:

Do you want to support Small jaws (about 130 mm) or Large jaws (about 165 mm)?
Type S for Small (default) or L for Large and press <Enter>

Follow the on-screen instructions.

6.) Select the Dentascan option to be installed (ex: click to highlight the option, then release).

Note: The distance that is calculated here needs to be communicated to the user, for use as the DFOV for DentaScan acquisitions.

7.) Click on the INSTALL button.

A box may appear while the options are loading.

When the option is displayed in the “Installed Options” list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half a minute (due to the fact that they are installing software).

Note: If no camera is attached, the following message will appear:

There is no camera attached. Installing DentaScan with no camera.

8.) After all the options are installed, select QUIT, then OK.

9.) Remove the MOD and write protect the side with options.

Note: The initial install requires the MOD to not be write protected; subsequent installs can be done with the MOD write protected)

10.) If you have another option MOD to load, go back to step 1.

NOTICE **Once you have installed Dentascan, you need to follow the procedures in Section 3.0, for installing Dentascan filming option, before proceeding to step 9.**

11.) Go to the SERVICE DESKTOP ->UTILITIES ->APPLICATIONS SHUTDOWN.

12.) After Applications have shut down, open the console window, or a UNIX shell window, type **st** **ENTER** to start up applications.

13.) When the system applications return, the system is ready to use.

Section 3.0 Procedures for Installing Dentascan Filming Option

1.) Open UNIX shell from the toolchest.

2.) Type **cd ~ctuser/install** **ENTER**

3.) Type **installcamera.denta** **ENTER**

4.) Message appears: Sending a film to the camera

...

Enter the distance Dist 1 in (mm):

5.) After the film is developed, use a metric scale to measure the distance between the vertical lines for “Dist 1” and “Dist 2”. Enter the value(s) in MILLIMETERS (mm).

a.) Enter the measurement of Dist 1: ***XXX.X*** **ENTER**

b.) If Zoom is supported by the type of Camera, the message will appear:

Enter the distance Dist 2 in (mm):

Enter the measurement of Dist 2: ***YYY.Y*** **ENTER**

6.) Message appears: DentaScan filming option installed.

7.) Type **exit** in the UNIX shell.

8.) Go to Section 2.0 step 11.) on page 53 for continuing installation.

Section 4.0 Finish Installation

4.1 Install Rating Plate

1.) Refer to Figure 3-3

2.) Locate the current Modification Rating plate(s) on the rear of the console.

3.) Fasten the Dentascan Option rating plate to the left of the existing rating plates.

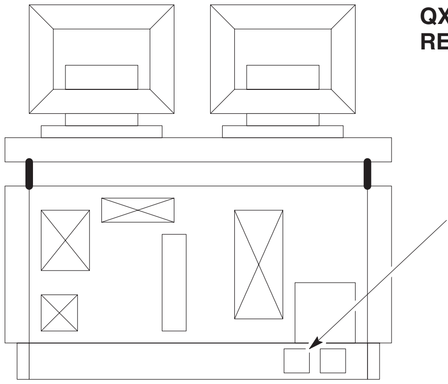


Figure 3-3 Rating Plate Location

4.2 Return Installation Card

1.) Complete and verify the site-specific information on the Red Installation Product Locator Card.
2.) Return the card to the Mailing Address that is printed on the front of the card.

Section 5.0
B7540SD Dentascan Option Kit Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	2184746	CBT CD-ROM	N	1	Dentascan On Line Operators Manual
2	2184749	Options MOD	N	1	Dentascan Option Software

Table 3-1 Dentascan Option Kit

Chapter 4
Direct-3D Option

Section 1.0
System Requirement

Version 1.2 software or later must be installed on the system before Direct-3D can be loaded.

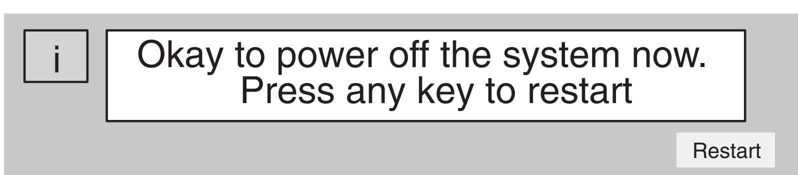
Section 2.0
CPU Memory Option Installation Preparation

The CPU Memory Option, part of the Direct-3D Option, adds an additional 1.0 GB of CPU memory to the Silicon Graphics Computer. This additional memory allows more efficient execution of system software by reducing the amount of disk swap activity and allowing larger amounts of data to be available in memory at any given moment.

2.1 Shutdown the System

1.) Select the (square red) SHUTDOWN button (shown below) on the Image Display Monitor. Select OK to shutdown Applications. This will stop both QX/i Applications and UNIX software and bring you to the boot prompt.

2.) After the software shuts down, the following message will appear.



3.) Turn off the power switch on the console.

2.2 Preparing the Workstation to Install the Optional Memory

NOTICE **THE MEMORY CIRCUIT BOARDS ARE VERY STATIC-SENSITIVE.**
Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards.

To prepare the workstation for installation of optional memory, follow these steps:

1.) Remove the front cover of the Console.
2.) Unplug the power cord from the electrical outlet and from the OCTANE workstation.
3.) Wait 5 minutes before removing the system module.

CAUTION **The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.**

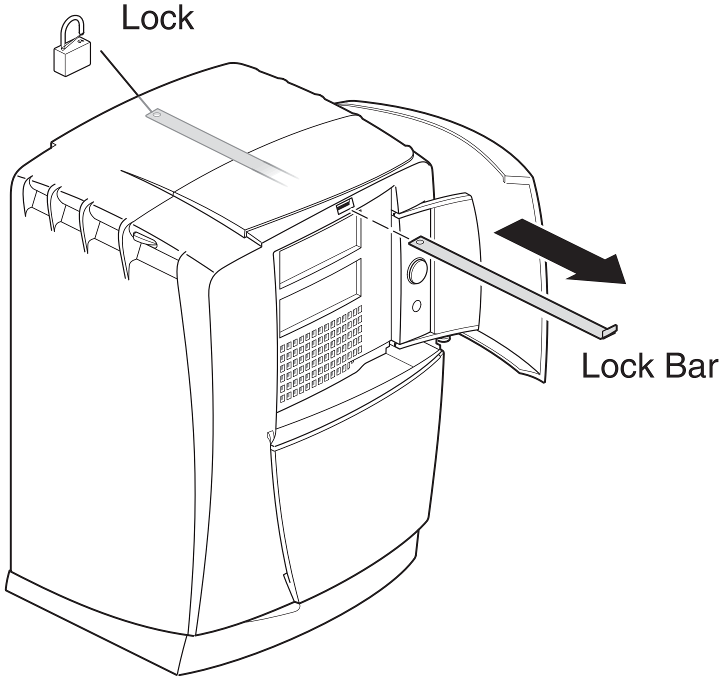


Figure 4-1 Lock Bar Removal (If Applicable)

4.) If applicable, remove the lockbar which locks the system module and bezel to the chassis.
- Unlock the lock at the back of the OCTANE workstation.
- Pull the lockbar out of the OCTANE workstation.

5.) Face the rear of the workstation.
6.) Remove the cables attached to the system module.

2.3 Removing the System Module

NOTICE **THE CIRCUIT BOARDS ARE VERY STATIC-SENSITIVE.**
Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards.

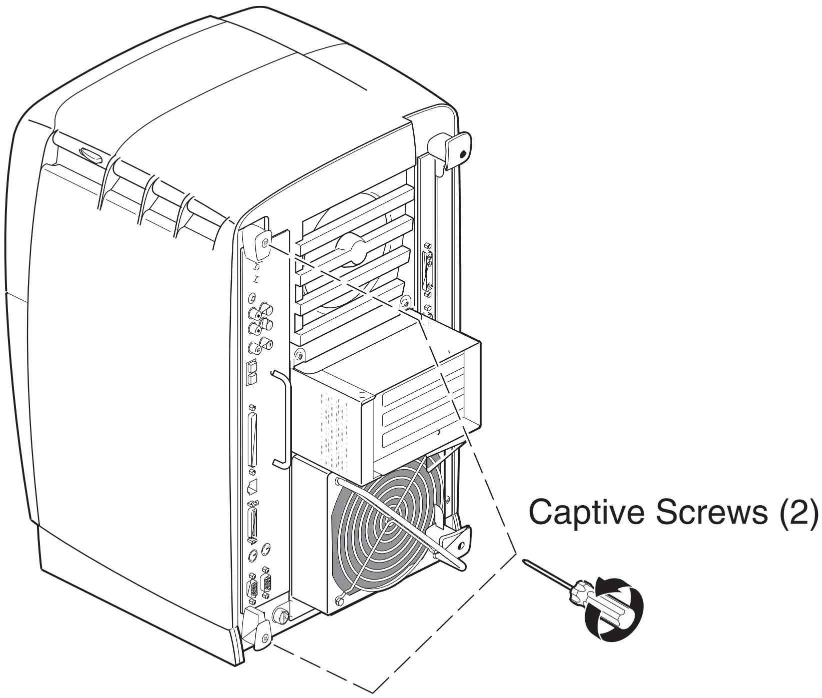


Figure 4-2 Removing the System Module Screws

CAUTION **The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.**

1.) To remove the system module, you need to power off the system, wait 5 minutes after powering off the workstation to allow the heat sinks to cool, and attach the wrist strap.
2.) Loosen the captive screws that hold the sliding handles to the OCTANE workstation.

Look for the two compression connectors when you take the system module out of the chassis, and do not touch them. They are on the back of the system module, and connect to the chassis when the system module is installed in the workstation.

NOTICE **The gold bristled pad on the compression connectors on the back of the system module is a very delicate and easily damaged bristled pad.**

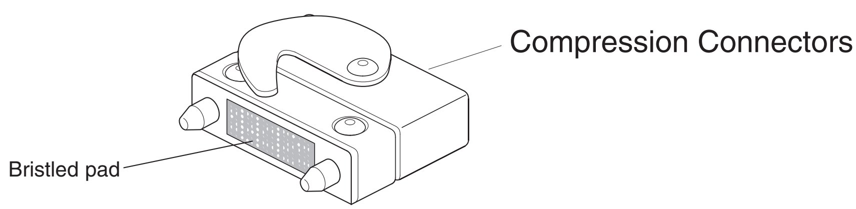


Figure 4-3 Locating the Compression Connectors

3.) Pull both sliding handles simultaneously until they are completely extended. This action releases the system module from the workstation.

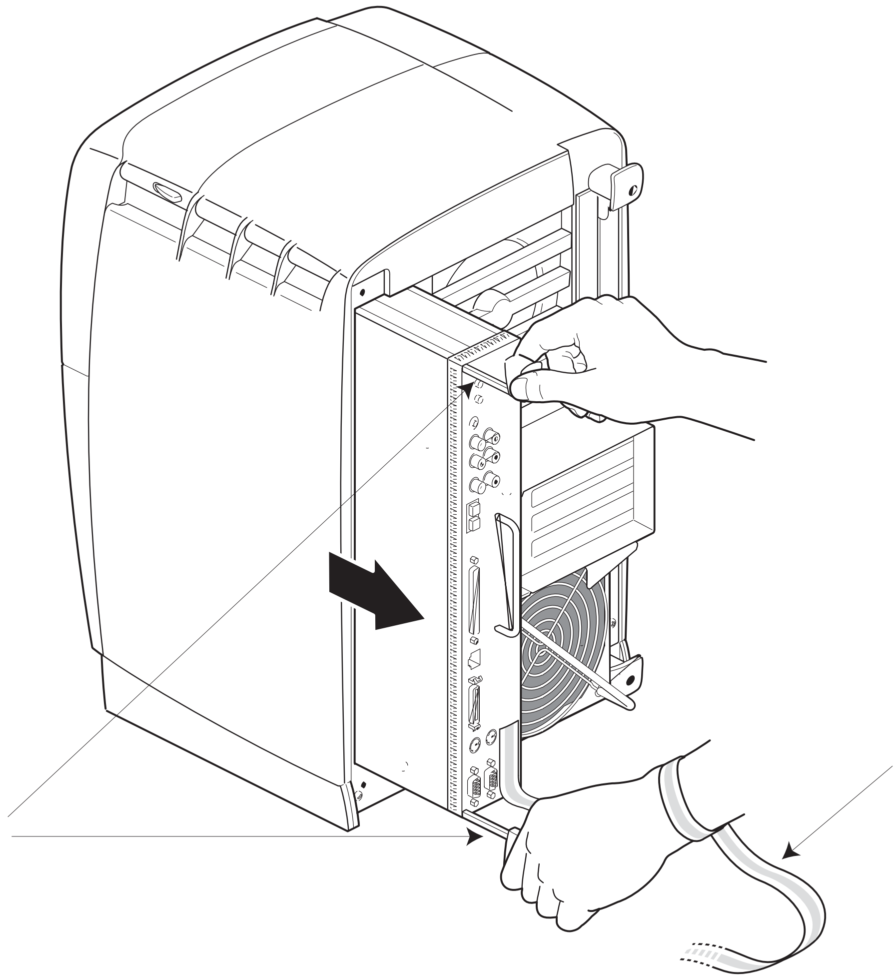


Figure 4-4 Releasing the System Module from the Front Plane

4.) Pull the system module from the chassis by grasping the module handle with your left hand and bracing your right hand against the top of the workstation. Support the bottom of the module as you pull it out.

5.) Place it on a dry, antistatic surface with the CPU and DIMMs facing up. A desktop works well.

NOTICE **Do not touch the compression connectors as you remove the system module. The side of the compression connector that connects to the workstation is extremely delicate.**

6.) Place a cap on each compression connector on the back of the system module. (Compression connector caps come with the system.)

CAUTION **Warning: The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.**

Section 3.0
Installing and Removing Memory

To install or remove memory, you must power off the workstation, wait 5 minutes after powering off the workstation to allow the heat sinks to cool, attach the wrist strap, and remove the system module.

NOTICE **DIMMS ARE VERY STATIC-SENSITIVE.**
Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards. Dual In-line Memory Modules (DIMMs) are extremely sensitive to static electricity. Handle the DIMMs carefully, and wear the wrist strap to avoid the effects of static electricity.

3.1 About Memory

Before you install DIMMs in your OCTANE workstation, read the following information.

- Bank 1 (sockets 1 and 2) must always be filled.
    - Minimum memory configuration is 64 MB in a bank. (2 x 32 MB DIMMs)
    - Maximum configuration is 512 MB in a bank. (2 x 256 MB DIMMs)
- Banks are filled sequentially; when Bank 1 is full, fill Bank 2. Do not skip banks.
- By common practice, if all of the memory installed is not of the same capacity, it is always a good idea to populate the DIMM Banks closest to the processor with the highest available capacity DIMMs. (i.e. Bank 1 contains the highest capacity DIMMs, bank 2 contains the same capacity DIMMs or the next lower capacity, and so on for banks 3 & 4.
- Each bank is either empty, or contains two DIMMs, one in each of the two sockets.
- Each bank of DIMMS must contain two DIMMs of the same capacity or density and of the same type.
    - Capacity refers to the number of megabytes in a DIMM: 32, 64, 128, 256, and so on.
    - Type refers to the construction of the DIMM.

      Because DIMM construction cannot be determined visually, it is important to know that two DIMMs are the same type before you begin installation. If you insert two DIMMs of different types, the system will not work. Banked DIMMs and stacked DIMMs are two examples of many different types of memory.
    - It is always safest to ensure that both DIMMs in a bank are from the same manufacturer.
- Minimum memory capacity in the OCTANE workstation is 64 MB. (2 x 32 MB DIMMs)
- Maximum memory capacity in the OCTANE workstation is 2 GB. (8 x 256 MB DIMMs)

3.2 Installing Memory

Note: This option kit consists of 4 x 256 MB DIMMs (1GB), which will be installed in the lowest numbered Memory Banks (Banks 1 & 2). Your system will already have at least 512 MB of memory installed. This may consist of either 2 x 256 MB DIMMs, or 4 x 128 MB DIMMs. If your system has additional memory (beyond 512 MB) installed (such as the standard 3-D Option memory), you may have to remove and discard it to install the 4 x 256 MB DIMMs included in this kit.

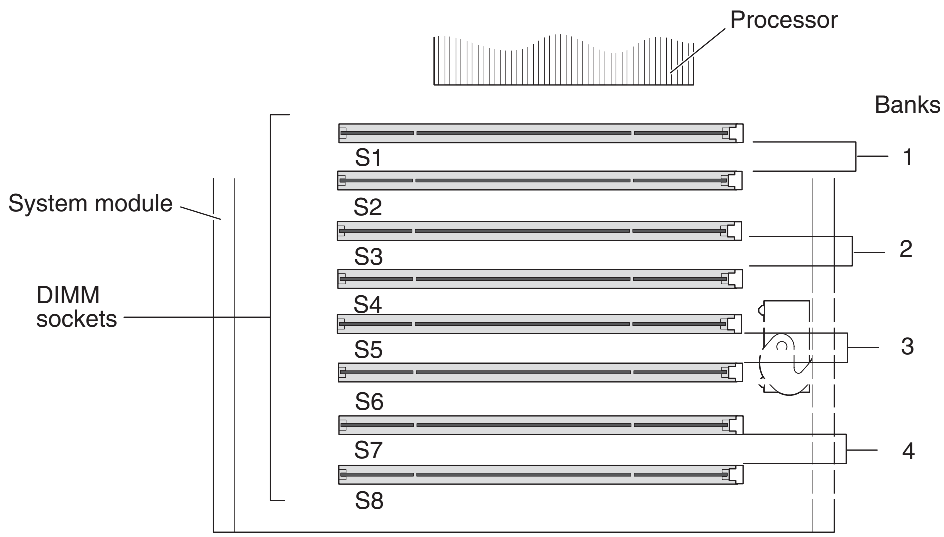


Figure 4-5 Identifying DIMM Sockets and DIMM Banks

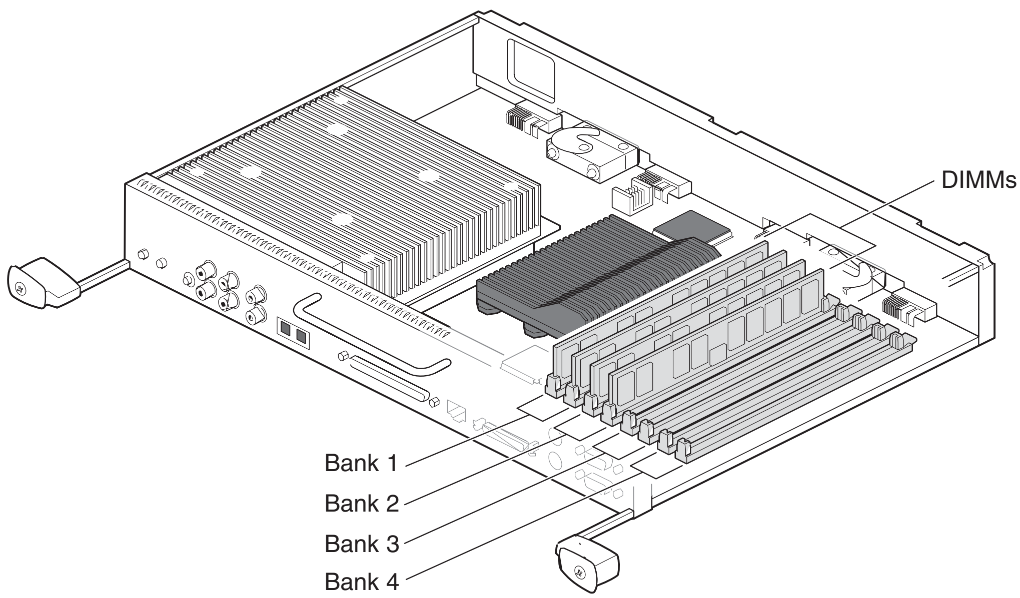


Figure 4-6 Locating DIMMs on the System Module

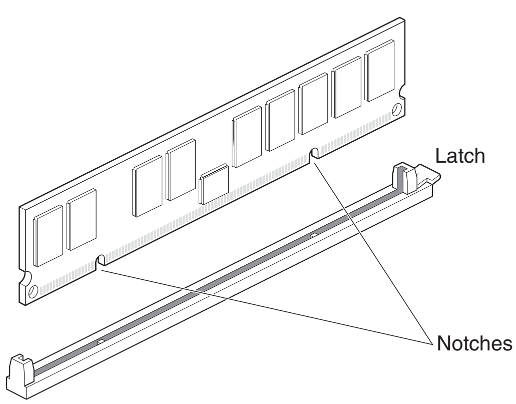


Figure 4-7 Locating the Notches on a DIMM

DIMMs are notched on the bottom so that they cannot be inserted incorrectly (see Figure 4-7).

1.) Remove all existing DIMMs from all sockets, and set them aside.
2.) Using the DIMMs provided in this kit, insert each DIMM into the socket gently but firmly, starting with slot location S1, and continuing through slot S4 (Banks 1 & 2). You hear a click as the DIMM is seated, and the latch on the end of the socket moves up.
3.) Check to be sure both sockets in each bank are full. DIMMs must be installed in pairs.
4.) Install all 4 x 256 MB DIMMs provided in this kit into Banks 1 & 2. Refer to section 3.1 on page 59.
5.) Re-install the removed DIMMs (maximum of 4) into the remaining memory sockets (S5 through S8) of Banks 3 & 4. If these DIMMs are of different capacities (or if you aren't sure), see Table 4-1, "DIMM Identification," on page 61 for information about DIMM identification, and review the information about socket location vs. DIMM capacity in section 3.1 on page 59.
6.) Once you have finished installing the memory, you are ready to replace the system module.

DIMM Manufacturer / Memory Capacity	Manufacturer Part Number	Other Coding (i.e. label color)	Comments
SGI - 32MB	9940069	Yellow	Never used on QX/i
Dataram - 32MB	60056	n/a	Never used on QX/i
SGI - 64MB	9940084 9470178	Blue Green	Used on standard 3-D Option only
Dataram - 64MB	62614	n/a	Used on standard 3-D Option only
SGI - 128MB	9470168	Brown*	Standard Memory
SGI - 128MB	9010020	Brown**	Standard Memory
Dataram - 128MB	62615	n/a	Standard Memory
SGI - 256MB	9010021	White***	Standard/Direct-3D
Dataram - 256MB	62621	High Profile	Direct-3D Option
Dataram - 256MB	62649	Low Profile	Direct-3D Option

Notes:

* 2K Refresh DIMM's, must be used as a matched pair in a bank, typically used in the original IP-30 System Module (SGI P/N 030-0887-003).

** 4K Refresh DIMMs, must be used as a matched pair in a bank, and will only function in an enhanced IP-30 System Module (SGI P/N 030-1467-001).

*** 4K Refresh DIMMs, must be used as a matched pair in a bank, and will only function in an enhanced IP-30 System Module (SGI P/N 030-1467-001).

Table 4-1 DIMM Identification

3.3 Replacing the System Module

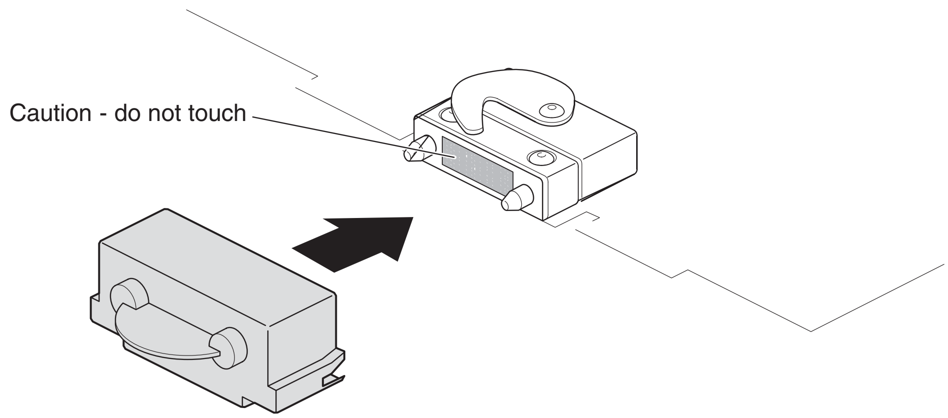


Figure 4-8 Removing the Compression Connector Cap

1.) Remove the caps on the compression connectors on the back of the system module. Do not touch the system module compression connectors as you lift the module into the workstation.

NOTICE **Do not insert the system module into the workstation unless the power cord is unplugged from the electrical socket. Inserting the system module into an OCTANE workstation with the power cord connected to an electrical socket could cause damage to the OCTANE workstation.**

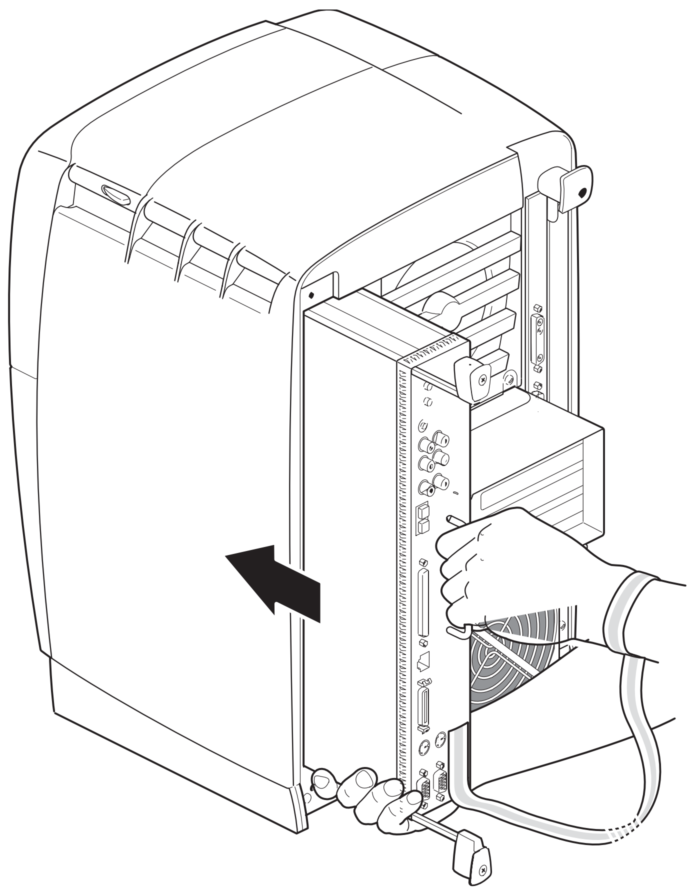


Figure 4-9 Supporting the System Module

2.) Grasp the system module by its immovable handle and support it with one hand as you slide it into the chassis.

3.) Push the module completely into the chassis using the immovable handle. (Both sliding handles protrude.) The system module stops when it is slightly out of the chassis.

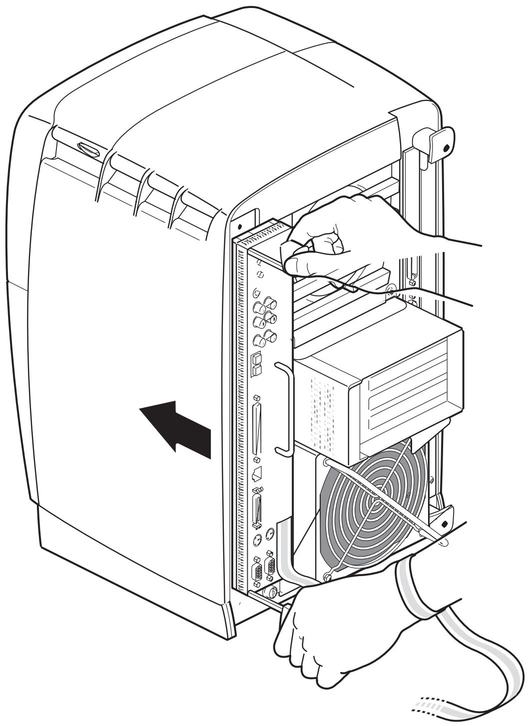


Figure 4-10 Replacing the System Module in the OCTANE Workstation

4.) Push the sliding handles simultaneously to connect the system module to the workstation.

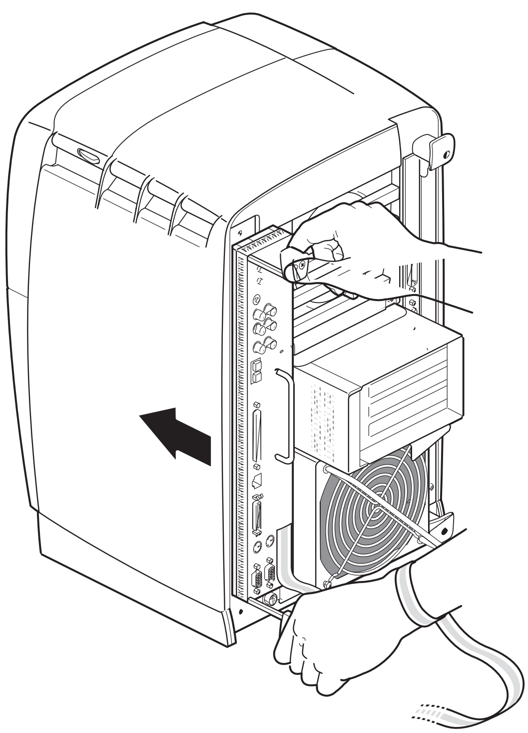


Figure 4-11 Locking the System Module to the Frontplane

5.) Push firmly on the handles to completely lock the system module to the workstation.

The handles are completely recessed and the system module is flush with the chassis when it is fully seated and locked to the workstation.

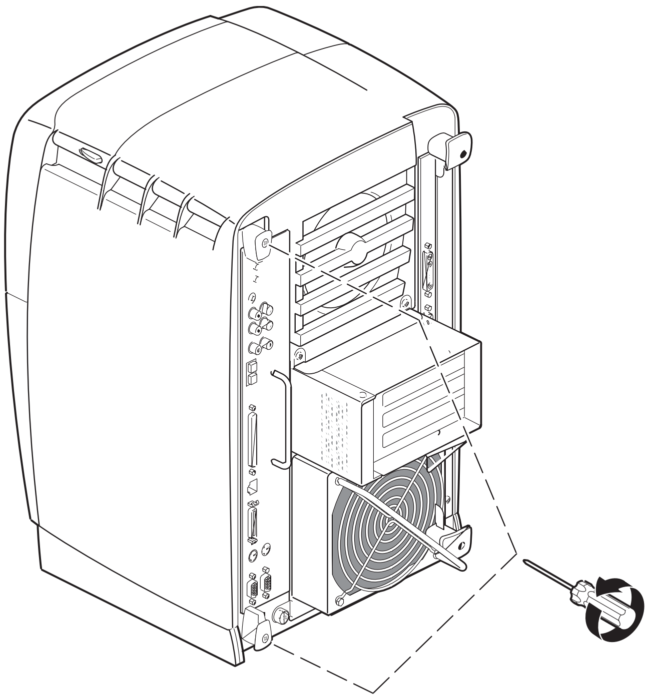


Figure 4-12 Replacing the System Module Screws

6.) Tighten the captive screws in the sliding handles until the system module is attached to the chassis.

7.) Remove the wrist strap.

3.4 Powering On the OCTANE Workstation

1.) Plug in the power cord.

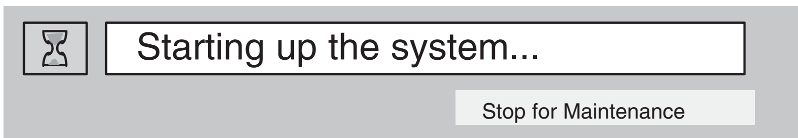
2.) Replace the lockbar (if applicable).

- Open the cover on the front of the OCTANE workstation.
- Slide the lockbar into the lockbar slot.
- Insert the lock through the lockbar at the back of the workstation.

Section 4.0
Start Up System and Verify Installed Memory

4.1 Start System

1.) Turn on the console power switch.
2.) The following message will appear. Select the STOP FOR MAINTENANCE button.



4.2 Verify Installed Memory

The system Maintenance menu appears.

1.) Select ENTER COMMAND MONITOR.
2.) At the shell prompt type: **hinv**
3.) Verify that the Main memory size is at least 1.5 GB (1512 MB).

Note: If the amount of memory is incorrect, follow the directions at the beginning of this chapter to power off your system and check the installation procedure.

- Check the angle of the DIMMs. They should be upright and completely seated.
- Check that each bank is populated with two DIMMs, and that they are of the same type. You must have an even number of DIMMs installed.

4.) Exit the shell by typing: **exit**
5.) Select START SYSTEM

The system software will start normally.

6.) After the system is started, open a shell tool and re-check the hardware environment:

{ctuser@somesystem} [1] **hinv**

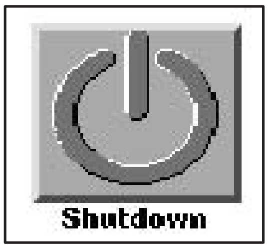
```
1 XXX MHZ IP30 Processor
CPU: MIPS RXXXXX Processor Chip Revision: 2.7
FPU: MIPS RXXXXX Floating Point Chip Revision: 0.0
Main Memory size: 1512 Mbytes                  <- This is the minimum memory you should see.
Instruction cache size: 32 Kbytes
Data cache size: 32 Kbytes
Secondary unified instruction/data cache size: 1 Mbytes
Intergal SCSI controller 0: Version QL1040B (rev. 2)
 Disk drive: unit 1 on SCSI controller 0
Integral SCSI controller 1: Version QL1040B (rev. 2)
 Optical disk: unit 4 on SCSI controller 1
 Comm device: unit 4 on SCSI controller 1
 Comm device: unit 4, lin 1 on SCSI controller 1
 CDROM: unit 6 on SCSI controller 1
IOC3 serial port: tty1
IOC3 serial port: tty2
IOC3 parallel port: plp1
Graphics board: ESI with texture option
Graphics board: ESI
Integral Fast Ethernet: ef0, version 1, pci 1
Fast Ethernet: ef1, version 1, pci 3
Iris Audio Processor: version RAD revision 12.0, number 1
```

Section 5.0
Octane Dual Processor Option Installation Preparation

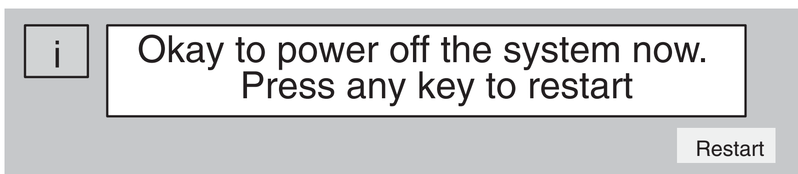
The Octane Dual Processor Option, part of the Direct-3D Option, replaces the single CPU Module with a dual CPU Module in the Silicon Graphics Computer.

5.1 Shutdown the System

1.) Select the (square red) SHUTDOWN button (shown below) on the Image Display Monitor. Select OK to shutdown Applications. This will stop both QX/i Applications and UNIX software and bring you to the boot prompt.



2.) After the software shuts down, the following message will appear.



3.) Turn off the power switch on the console.

4.) Unplug the power cord from the electrical outlet and from the OCTANE workstation.

5.) Wait 5 minutes before removing the system module.

CAUTION **The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.**

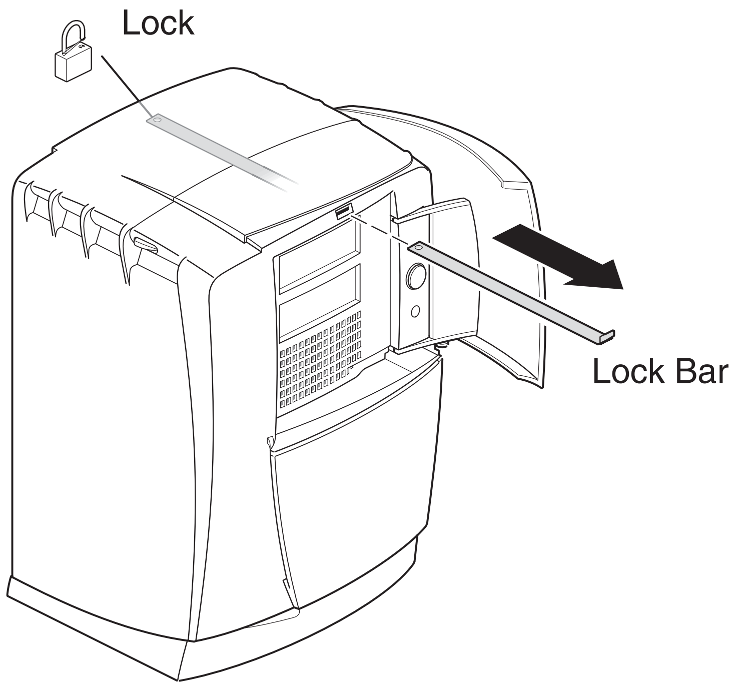


Figure 4-13 Lock Bar Removal (If Applicable)

6.) If applicable, remove the lockbar that locks the system module and bezel to the chassis.
   - Unlock the lock at the back of the OCTANE workstation.
   - Pull the lockbar out of the OCTANE workstation.

- Face the rear of the workstation.
- Remove the cables attached to the system module.

5.2 Removing the System Module

NOTICE **THE CIRCUIT BOARDS ARE VERY STATIC-SENSITIVE.**
Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards.

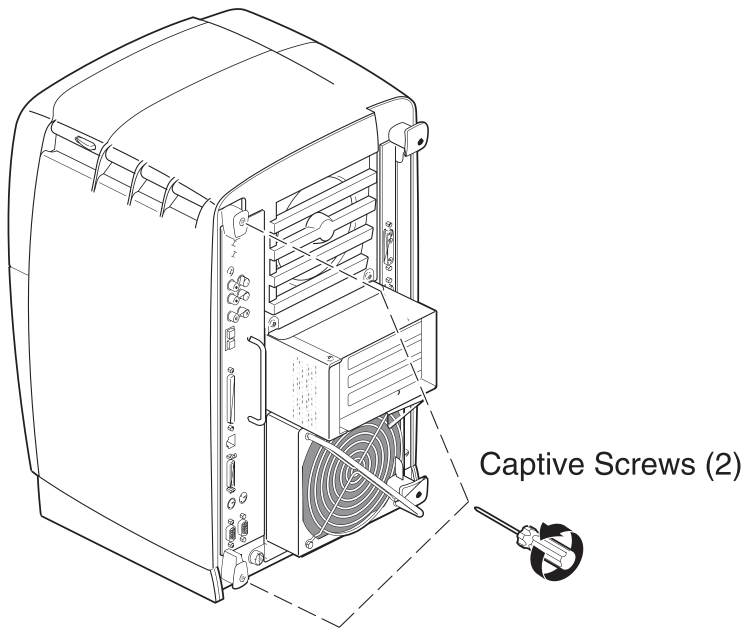


Figure 4-14 Removing the System Module Screws

CAUTION **The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.**

1.) To remove the system module, you need to power off the system, wait 5 minutes after powering off the workstation to allow the heat sinks to cool, and attach the wrist strap.
2.) Loosen the captive screws that hold the sliding handles to the OCTANE workstation.

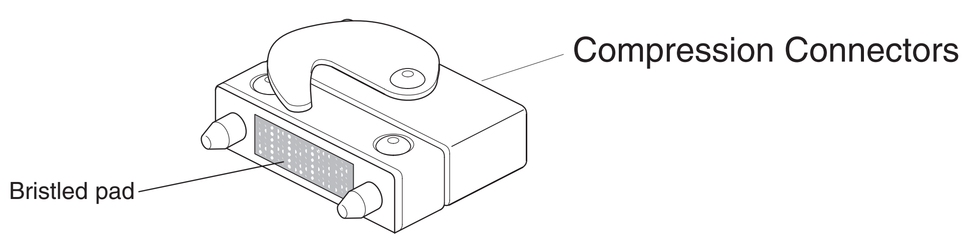
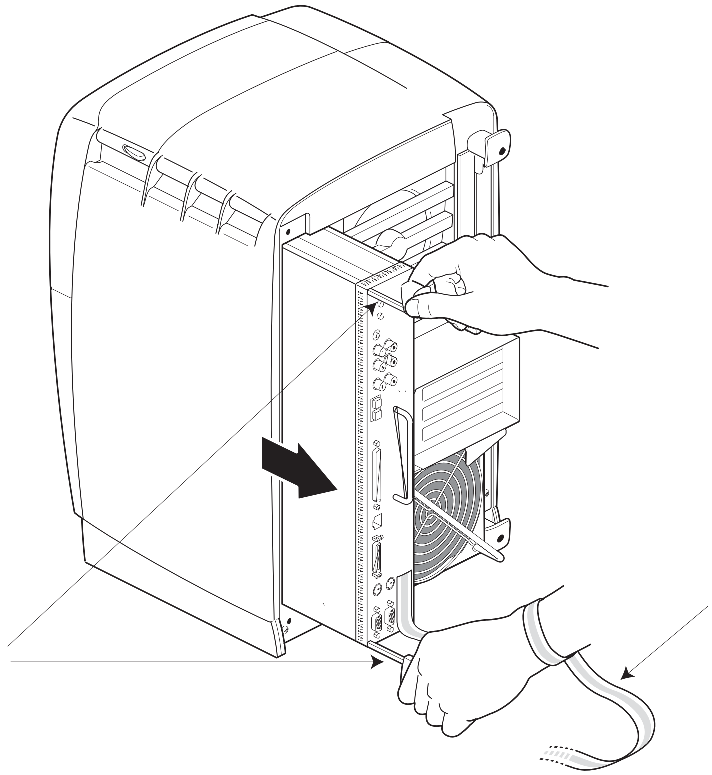


Figure 4-15 Locating the Compression Connectors

Look for the two compression connectors when you take the system module out of the chassis, and do not touch them. They are on the back of the system module, and connect to the chassis when the system module is installed in the workstation.

NOTICE **The gold bristled pad on the compression connectors on the back of the system module is a very delicate and easily damaged bristled pad.**

3.) Pull both sliding handles simultaneously until they are completely extended. This action releases the system module from the workstation.



4.) Pull the system module from the chassis by grasping the module handle with your left hand and bracing your right hand against the top of the workstation.

Support the bottom of the module as you pull it out.

5.) Place it on a dry, antistatic surface with the CPU and DIMMs facing up. A desktop works well.

NOTICE **Do not touch the compression connectors as you remove the system module. The side of the compression connector that connects to the workstation is extremely delicate.**

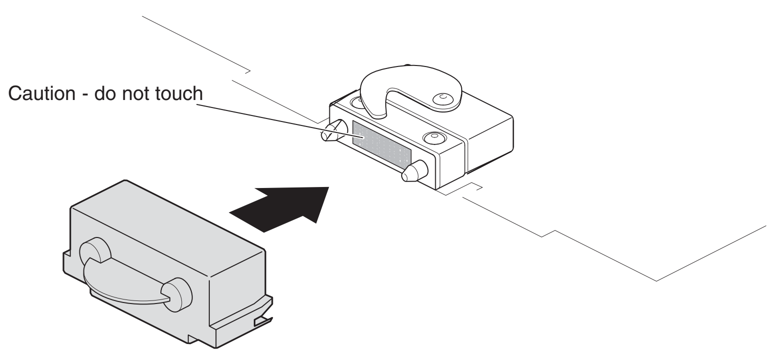


Figure 4-16 Placing the Compression Cap On the Compression Connector

6.) Place a cap on each compression connector on the back of the system module. (Compression connector caps come with the system.)

CAUTION **The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.**

Section 6.0
Installing the Optional Dual Processor

6.1 Removing the Single Processor

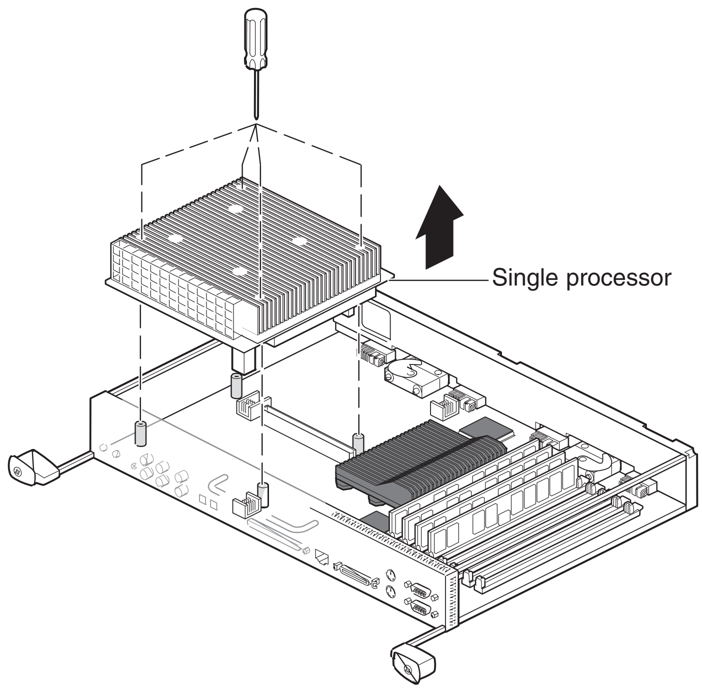


Figure 4-17 Removing the Single Processor

1.) Loosen the four captive Phillips screws holding the single processor CPU in place. (The dual processor has six screws.)

NOTICE **Do not remove the four inner screws (not Phillips) holding the heat sink to the CPU.**

2.) Slide your fingers under the edge of the single processor closest to the back of the module, and push up to release it. You may need to use two hands to lift it out.

6.2 Installing The Optional Dual Processor

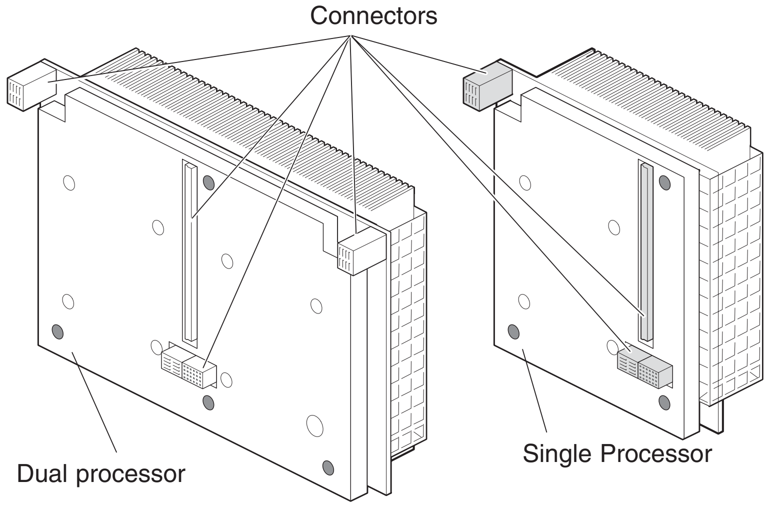


Figure 4-18 Viewing the Connectors On the Underside Of the CPUs

1.) Turn over the CPU to determine where the connectors are located. Align the connectors on the base of the CPU with the connectors on the system board.
2.) Place the dual processor on the side of the system module closest to the panel of connectors.

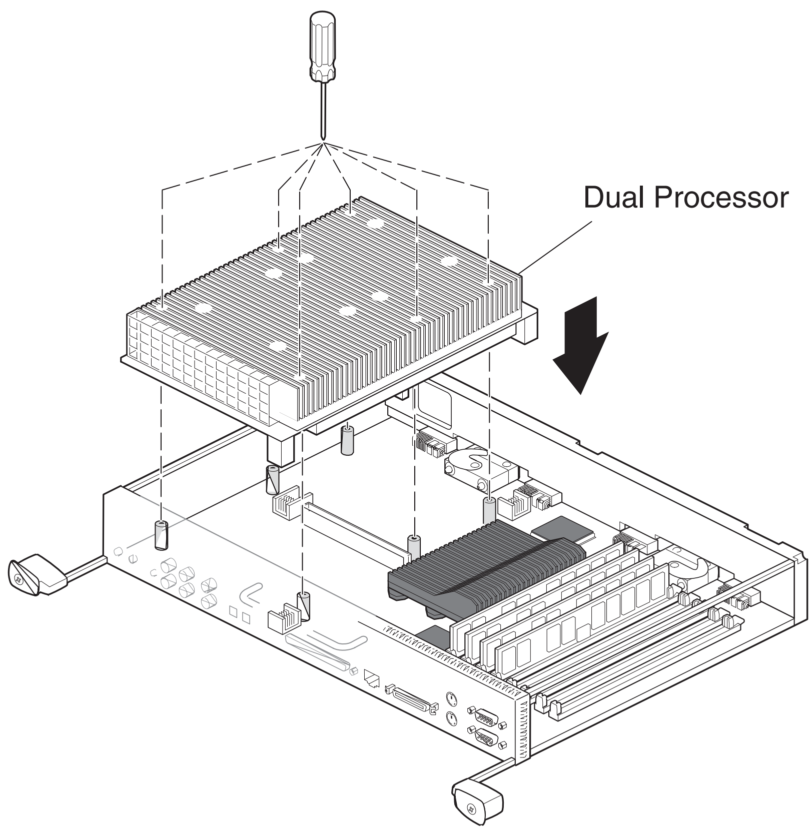


Figure 4-19 Installing the Dual Processor

3.) Lower the CPU onto the standoffs and connectors as shown.
4.) Tighten the six captive screws to the standoffs.

6.3 Replacing the System Module

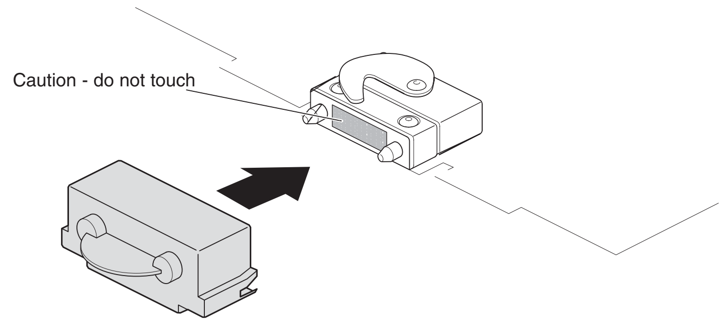


Figure 4-20 Removing the Compression Connector Cap

1.) Remove the caps on the compression connectors on the back of the system module. Do not touch the system module compression connectors as you lift the module into the workstation.

NOTICE **Do not insert the system module into the workstation unless the power cord is unplugged from the electrical socket. Inserting the system module into an OCTANE workstation with the power cord connected to an electrical socket could cause damage to the OCTANE workstation.**

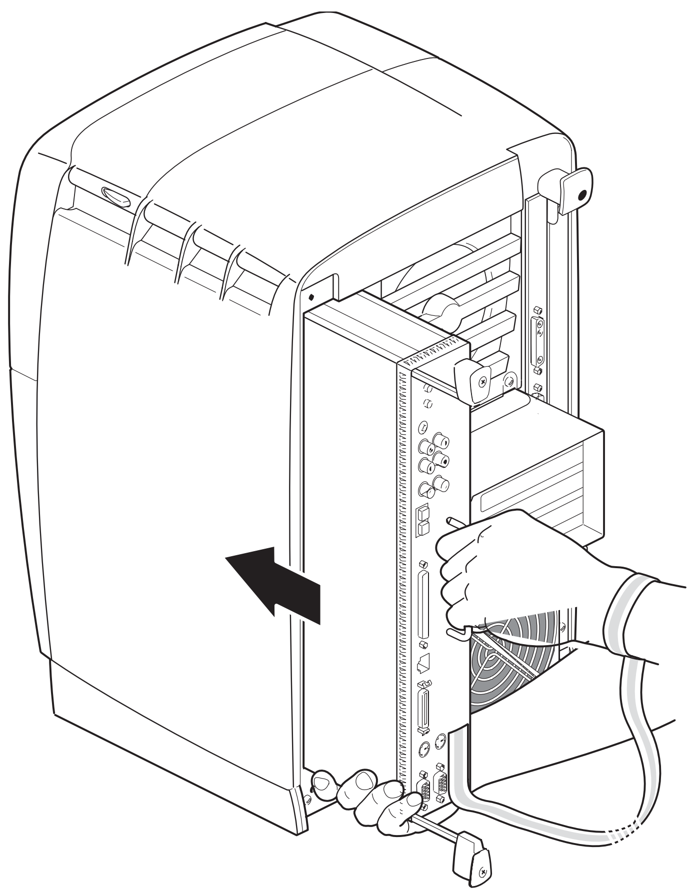


Figure 4-21 Supporting the System Module

2.) Grasp the system module by its immovable handle and support it with one hand as you slide it into the chassis.

3.) Push the module completely into the chassis using the immovable handle. (Both sliding handles protrude.) The system module stops when it is slightly out of the chassis.

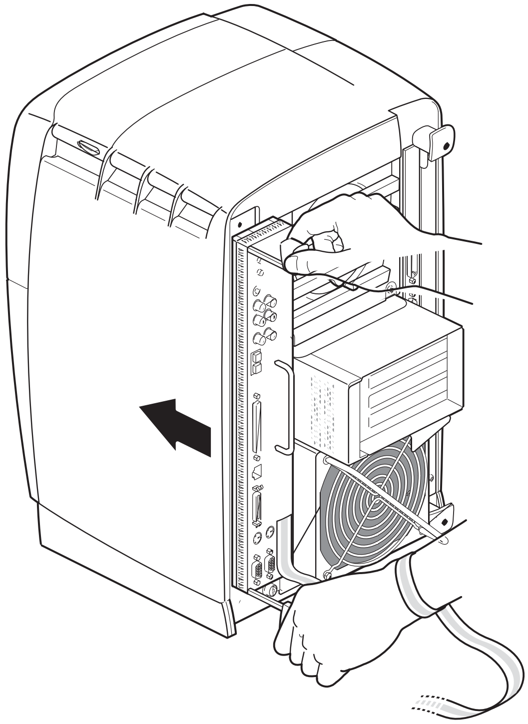


Figure 4-22 Replacing the System Module in the OCTANE Workstation

4.) Push the sliding handles simultaneously to connect the system module to the workstation.

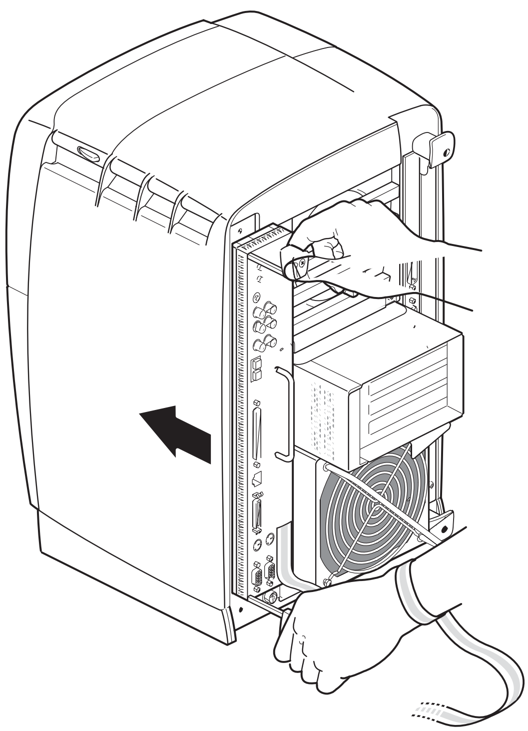


Figure 4-23 Locking the System Module to the Frontplane

5.) Push firmly on the handles to completely lock the system module to the workstation.

The handles are completely recessed and the system module is flush with the chassis when it is fully seated and locked to the workstation.

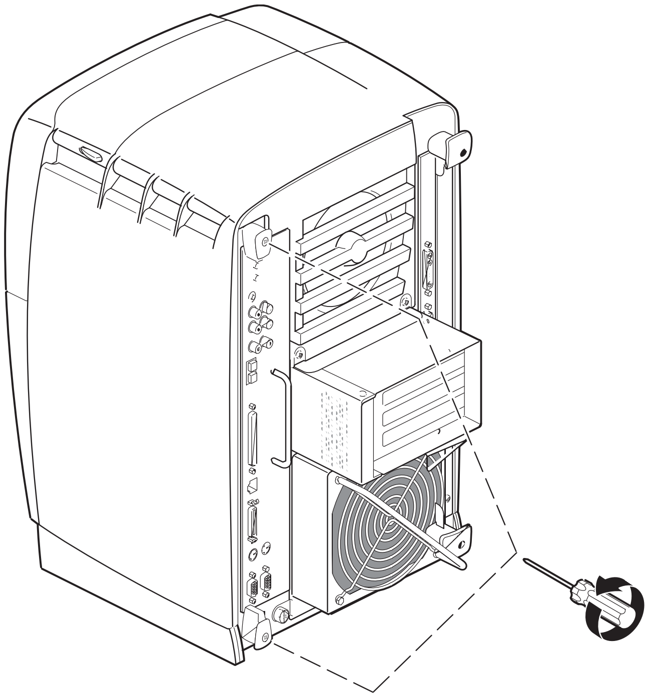


Figure 4-24 Replacing the System Module Screws

6.) Tighten the captive screws in the sliding handles until the system module is attached to the chassis.

7.) Remove the wrist strap.

6.4 Powering On the OCTANE Workstation

1.) Plug in the power cord.

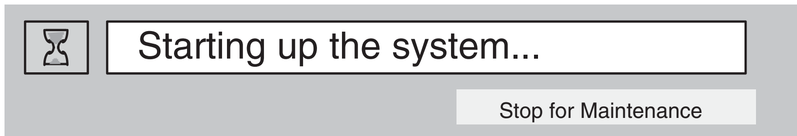
2.) Replace the lockbar (if applicable).

- Open the cover on the front of the OCTANE workstation.
- Slide the lockbar into the lockbar slot.
- Insert the lock through the lockbar at the back of the workstation.

Section 7.0
Start Up System and Verify Installed Dual Processor

7.1 Start System

1.) Turn on the console power switch.
2.) The following message will appear. Select the STOP FOR MAINTENANCE button.



7.2 Verify Installed Dual Processor

The system Maintenance menu appears.

1.) Select ENTER COMMAND MONITOR.
2.) At the shell prompt type: **hinv**
3.) Verify that the processor is: "2 xxx mhz IP30 Processors"
4.) Exit the shell by typing: **exit**
5.) Select START SYSTEM
The system software will start normally.
6.) After the system is started, open a shell tool and re-check the hardware environment:
{ctuser@somesystem} [1] **hinv**
2 XXX MHZ IP30 Processor <- This is the line that identifies # of processors.
CPU: MIPS RXXXXX Processor Chip Revision: 2.7
FPU: MIPS RXXXXX Floating Point Chip Revision: 0.0
Main Memory size: 1512 Mbytes
Instruction cache size: 32 Kbytes
Data cache size: 32 Kbytes
Secondary unified instruction/data cache size: 1 Mbytes
Integral SCSI controller 0: Version QL1040B (rev. 2)
 Disk drive: unit 1 on SCSI controller 0
Integral SCSI controller 1: Version QL1040B (rev. 2)
 Optical disk: unit 4 on SCSI controller 1
 Comm device: unit 4 on SCSI controller 1
 Comm deveice: unit 4, lin 1 on SCSI controller 1
 CDROM: unit 6 on SCSI controller 1
IOC3 serial port: tty1
IOC3 serial port: tty2
IOC3 parallel port: plp1
Graphics board: SI with texture option
Graphics board: SI
Integral Fast Ethernet: ef0, version 1
Integral Ethernet: et0, IO0
Iris Audio Processor: version RAD revision 12.0, number 1
PCI card, bus 0, slot 221, Vendor 0x0, Device 0x0

Section 8.0
Install Direct-3D Option Software

Software Options must be loaded/re-loaded every time the OC software is reloaded.

1.) In Service Desktop, select TOOLBOX/UTILITIES -> INSTALL-> INSTALL OPTIONS.

This program must be run once for each option MOD you are loading. For example, if you have two options MODs you must run this program twice.

An Options Window appears.

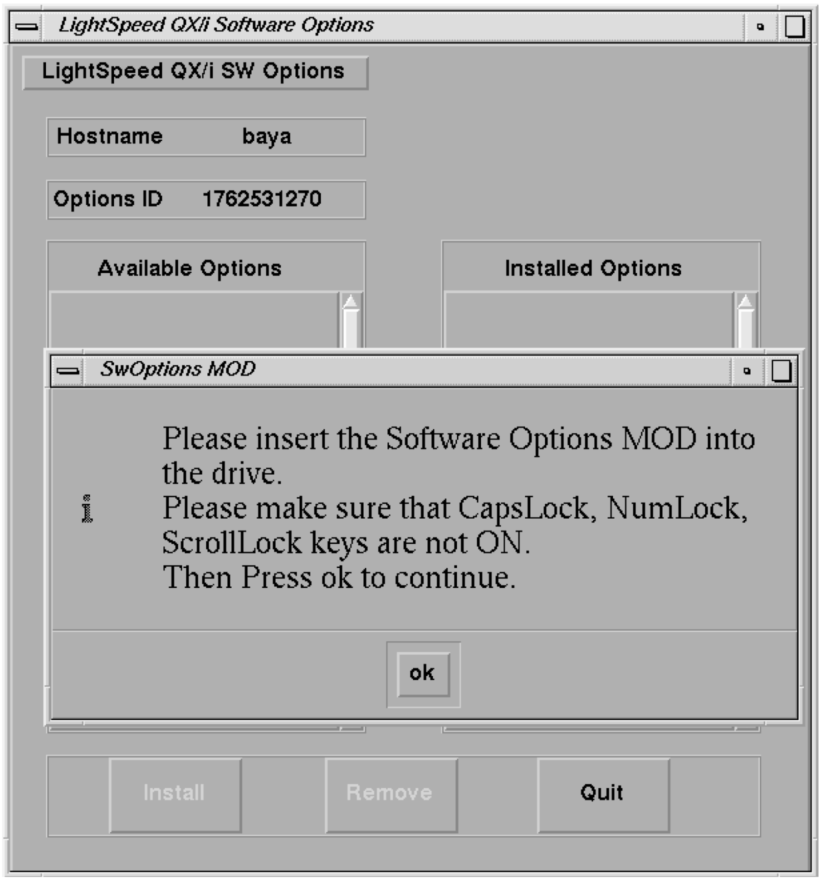


Figure 4-25 Software Options Installation Window

2.) Insert the Options MOD into the drive

3.) Click on OK

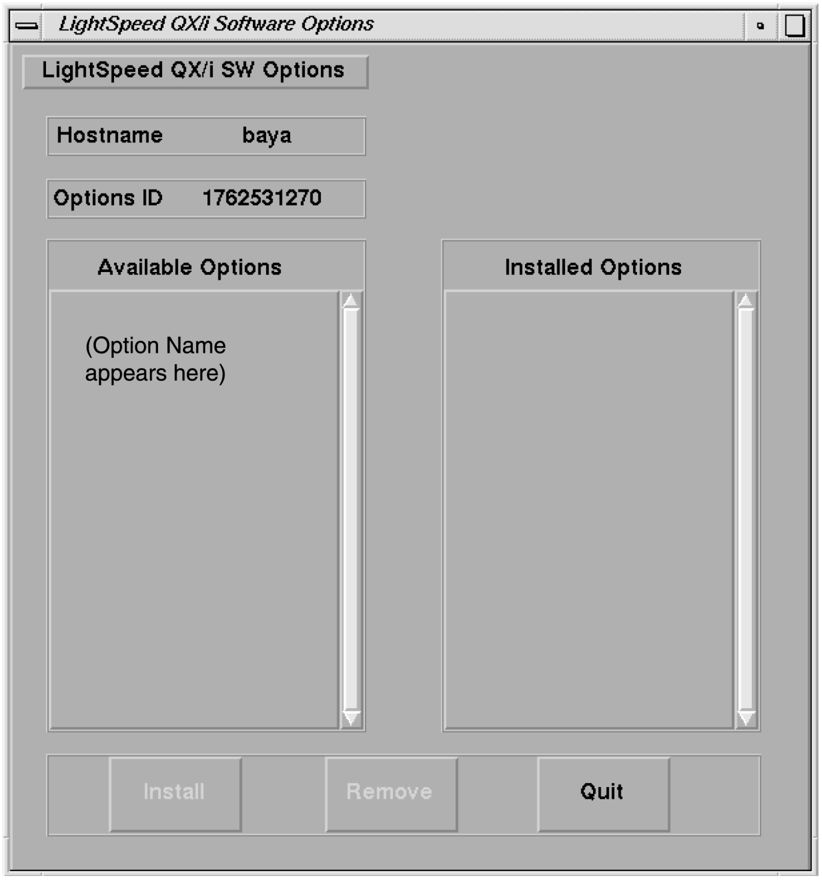


Figure 4-26 Software Options Available/Installed Screen Example

4.) Select the Direct-3D option to be installed (ex: click to highlight the option).

5.) Click on the INSTALL button

A box may appear while the option is loading

When the option is displayed in the "Installed Options" list, then installation of that option is complete. After the Direct-3D option is installed, select QUIT, then OK.

6.) Remove the MOD and write protect the side with options.

Note: The initial install requires the MOD to not be write protected; subsequent installs can be done with the MOD write protected)

7.) If you have another option MOD to load, go back to step 1.

8.) Go to the SERVICE DESKTOP ->UTILITIES ->APPLICATIONS SHUTDOWN.

9.) After Applications have shut down, open the console window, or a UNIX shell window, type **st** **ENTER** to start up applications.

10.) When the system applications return, the system is ready to use.

Section 9.0
Finish Installation

9.1 Install Rating Plate

Locate the Modification Rating plate, left of the Rating Plates already on the rear of the cabinet.

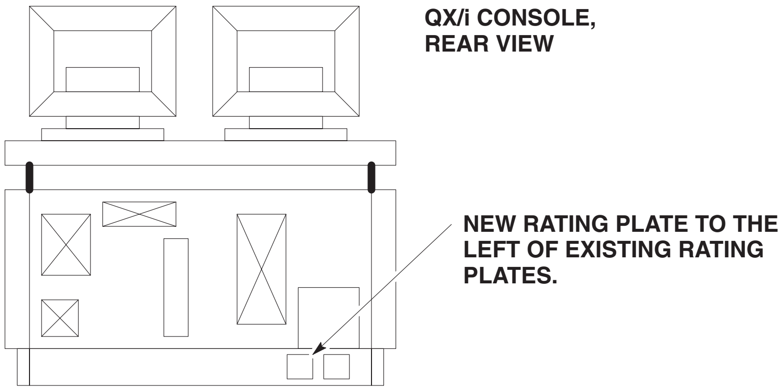


Figure 4-27 Rating Plate Location

9.2 Return Installation Card

Complete/Verify the site specific information on the Red Product Locator Installation Card and return to the appropriate location printed on the front of the card.

Section 10.0
Field Replaceable Parts

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	2169940-48	Dual 270 Mhz Processor	1	1	Option Dual Processor Module
or					
2	2169940-49	Dual 300 Mhz Processor	1	1	Option Dual Processor Module
3	2199806-7	2 x 256 MB DIMMs	1	2	2 x 256 MB SDRAM DIMMs (pair of "High Profile" DIMMs)
or					
4	2199806-8	2 x 256 MB DIMMs	1	2	2 x 256 MB SDRAM DIMMs (pair of "Low Profile" DIMMs)
5	2268937	Option Key	N	1	Direct-3D Option Key MOD

Table 4-2 Field Replaceable Parts

Chapter 5
3-D Analysis Option Installation

Section 1.0
System Requirement

None. Works with all software versions.

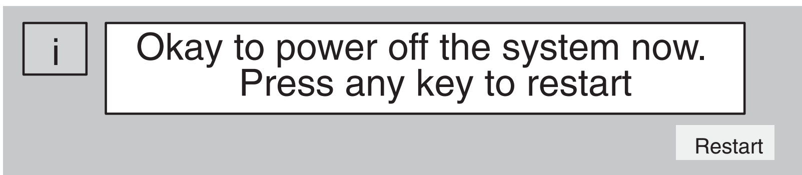
Section 2.0
CPU Memory Option Installation Preparation

The CPU Memory Option, B7710M provides an additional 128 MB of CPU memory for the Silicon Graphics Computer. This memory allows more efficient execution of system software by reducing the amount of disk swap activity and allowing larger amounts of data to be available in memory at any given moment.

2.1 Shutdown the System:

1.) Select the (square red) SHUTDOWN button (shown below) on the Image Display Monitor. Select OK to shutdown Applications. Shutdown stops the QX/i Applications and the UNIX software, then displays the boot prompt.

2.) After the software shuts down, the following message appears:



3.) Turn off the power switch on the console.

2.2 Preparing the Workstation to Install the Optional Memory

NOTICE **THE CIRCUIT BOARDS ARE VERY STATIC-SENSITIVE.**
Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards.

To prepare the workstation for installation of optional memory, follow these steps:

1.) Remove the front cover of the Console.
2.) Unplug the power cord from the electrical outlet and from the OCTANE workstation.
3.) Wait 5 minutes before removing the system module.

CAUTION **The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the workstation before you remove the system module. Test before touching the CPU or heat sinks.**

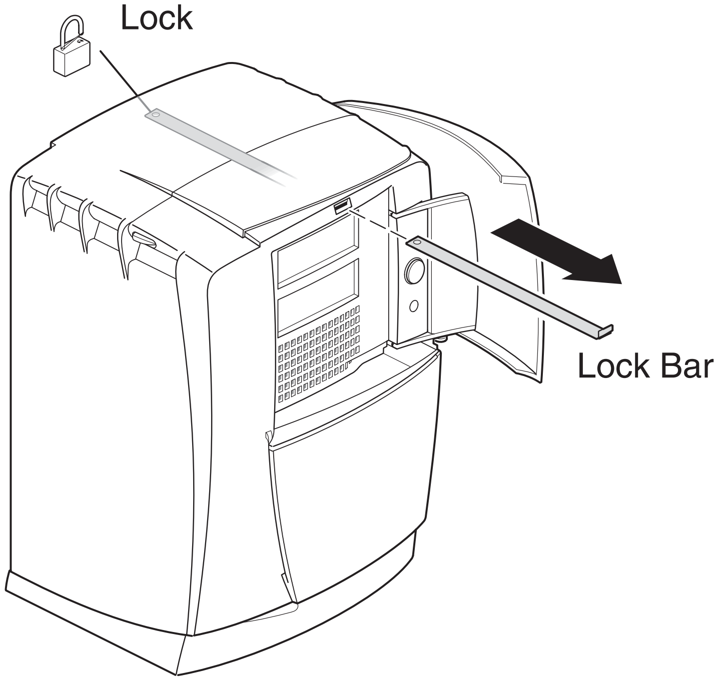


Figure 5-1 Lock Bar Removal (If Applicable)

4.) If applicable, remove the lockbar which locks the system module and bezel to the chassis.
- Unlock the lock at the back of the OCTANE workstation
- Pull the lockbar out of the OCTANE workstation.

5.) Face the rear of the workstation.
6.) Remove the cables attached to the system module.

2.3 Removing the System Module

NOTICE **THE CIRCUIT BOARDS ARE VERY STATIC-SENSITIVE.**
Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards.

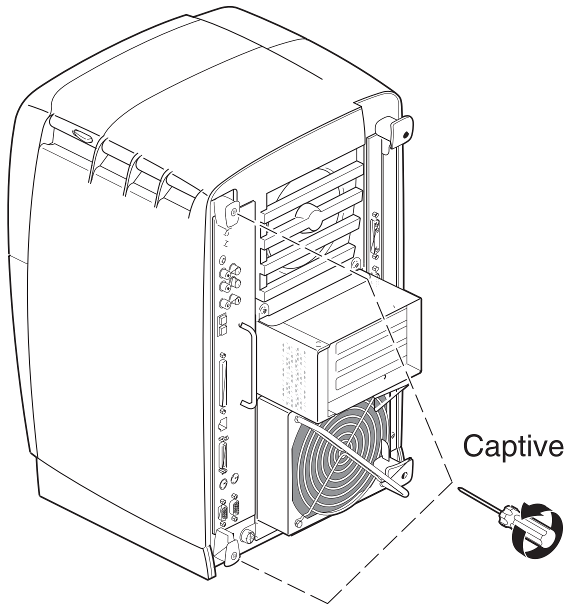


Figure 5-2 Removing the System Module Screws

CAUTION **The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the workstation before you remove the system module. Test before touching the CPU or heat sinks.**

1.) To remove the system module, you need to power off the system, wait 5 minutes after powering off the workstation to allow the heat sinks to cool, and attach the wrist strap.
2.) Loosen the captive screws that hold the sliding handles to the workstation.

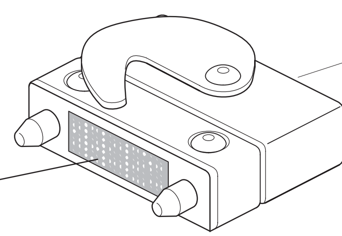


Figure 5-3 Locating the Compression Connectors

Look for the two compression connectors when you take the system module out of the chassis, and do not touch them. They are on the back of the system module, and connect to the chassis when the system module is installed in the workstation.

NOTICE **The gold bristled pad on the compression connectors on the back of the system module is a very delicate and easily damaged bristled pad.**

3.) Pull both sliding handles simultaneously until they are completely extended. This action releases the system module from the workstation.

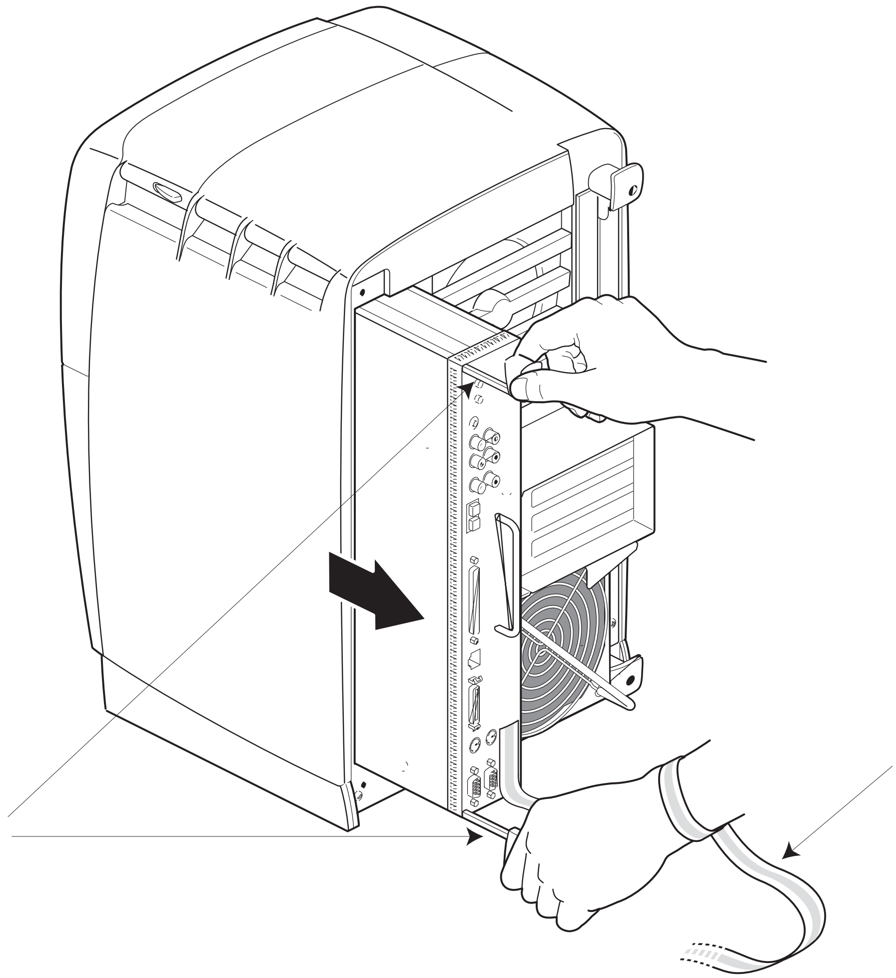


Figure 5-4 Releasing the System Module from the Front Plane

4.) Pull the system module from the chassis by grasping the module handle with your left hand and bracing your right hand against the top of the workstation.

Support the bottom of the module as you pull it out.

5.) Place it on a dry, antistatic surface with the CPU and DIMMs facing up. A desktop works well.

NOTICE **Do not touch the compression connectors as you remove the system module. The side of the compression connector that connects to the workstation is extremely delicate.**

6.) Place a cap on each compression connector on the back of the system module. (Compression connector caps come with the system.)

CAUTION **Warning: The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.**

Section 3.0
Installing and Removing Memory

To install or remove memory, you must power off the workstation, wait 5 minutes after powering off the workstation to allow the heat sinks to cool, attach the wrist strap, and remove the system module.

NOTICE **DIMMS ARE VERY STATIC-SENSITIVE.**
Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards. Dual in line memory modules (DIMMs) are extremely sensitive to static electricity. Handle the DIMMs carefully and wear the wrist strap to avoid the flow of static electricity.

3.1 About Memory

Before you install DIMMs in your OCTANE workstation, read the following information.

- Bank 1 (sockets 1 and 2) must always be filled.
  - Minimum memory configuration is 64 MB in a bank. (2 x 32 MB DIMMs)
  - Maximum configuration is 512 MB in a bank. (2 x 256 MB DIMMs)
- Banks are filled sequentially; when Bank 1 is full, fill Bank 2. Do not skip banks.
- By common practice, if all of the memory installed is not of the same capacity, it is always a good idea to populate the DIMM Banks closest to the processor with the highest available capacity DIMMs. (i.e. Bank 1 contains the highest capacity DIMMs, bank 2 contains the same capacity DIMMs or the next lower capacity, and so on for banks 3 & 4.
- Each bank is either empty, or contains two DIMMs, one in each of the two sockets.
- Each bank of DIMMS must contain two DIMMs of the same capacity or density and of the same type.
  - Capacity refers to the number of megabytes in a DIMM: 32, 64, 128, 256, and so on.
  - Type refers to the construction of the DIMM.

    Because DIMM construction cannot be determined visually, it is important to know that two DIMMs are the same type before you begin installation. If you insert two DIMMs of different types, the system will not work. Banked DIMMs and stacked DIMMs are two examples of many different types of memory.
  - It is always safest to ensure that both DIMMs in a bank are from the same manufacturer.
- Minimum memory capacity in the OCTANE workstation is 64 MB. (2 x 32 MB DIMMs)
- Maximum memory capacity in the OCTANE workstation is 2 GB. (8 x 256 MB DIMMs)

3.2 Installing Memory

Note: This option kit, to support 3-D Analysis, consists of 2 x 64 MB DIMMs (128 MB). Your system should already have 512 MB of memory installed. This may consist of either 2 x 256 MB DIMMs, or 4 x 128 MB DIMMs. You will need to install the new memory DIMMs in the next available bank (see Figure 5-5 and Figure 5-6).

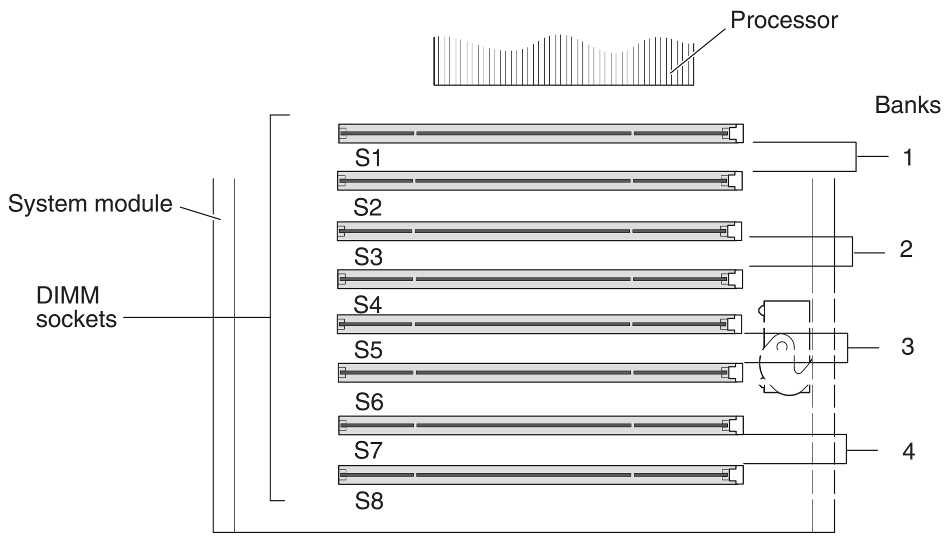


Figure 5-5 Identifying DIMM Sockets and DIMM Banks

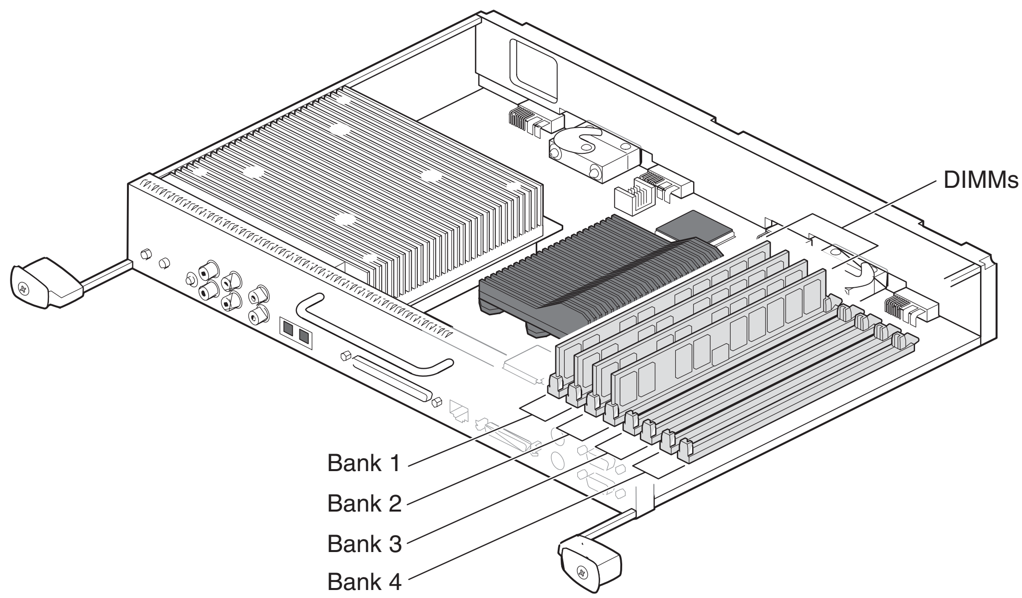


Figure 5-6 Locating DIMMs on the System Module

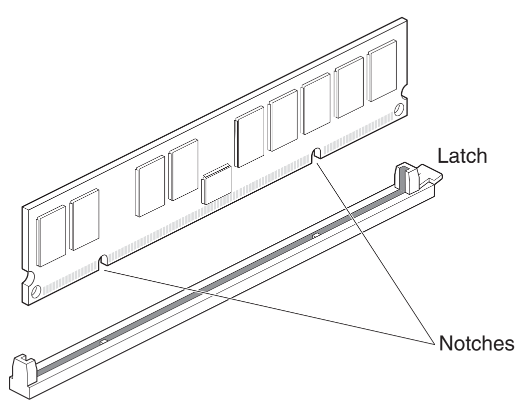


Figure 5-7 Locating the Notches on a DIMM

DIMMs are notched on the bottom so that they cannot be inserted incorrectly.

1.) Using the DIMMs provided in this kit, insert each DIMM into the socket gently but firmly, starting with the first open DIMM slot location (typically slot S3 or S5), and fill the bank with the two DIMMs from this kit. You hear a click as the DIMM is seated, and the latch on the end of the socket moves up.
2.) Check to be sure both sockets in the bank are full. DIMMs must be installed in pairs.
3.) Install both (2X64 MB) DIMMs provided in this kit into the selected bank. If you need additional information about the DIMMs you are installing, or about the DIMMs already installed, see Table 5-1 for information about DIMM identification, and review the information about socket location vs. DIMM capacity in section 3.1 on page 83.
4.) Once you have finished installing the memory you are ready to replace the system module.

DIMM Manufacturer / Memory Capacity	Manufacturer Part Number	Other Coding (i.e. label color)	Comments
SGI - 32MB	9940069	Yellow	Never used on QX/i
Dataram - 32MB	60056	n/a	Never used on QX/i
SGI - 64MB	9940084 9470178	Blue Green	Used on standard 3-D Option only
Dataram - 64MB	62614	n/a	Used on standard 3-D Option only
SGI - 128MB	9470168	Brown*	Standard Memory
SGI - 128MB	9010020	Brown**	Standard Memory
Dataram - 128MB	62615	n/a	Standard Memory
SGI - 256MB	9010021	White***	Standard/Direct-3D
Dataram - 256MB	62621	High Profile	Direct-3D Option
Dataram - 256MB	62649	Low Profile	Direct-3D Option

Notes:

* 2K Refresh DIMM's, must be used as a matched pair in a bank, typically used in the original IP-30 System Module (SGI P/N 030-0887-003).

** 4K Refresh DIMMs, must be used as a matched pair in a bank, and will only function in an enhanced IP-30 System Module (SGI P/N 030-1467-001).

*** 4K Refresh DIMMs, must be used as a matched pair in a bank, and will only function in an enhanced IP-30 System Module (SGI P/N 030-1467-001).

Table 5-1 DIMM Identification

3.3 Replacing the System Module

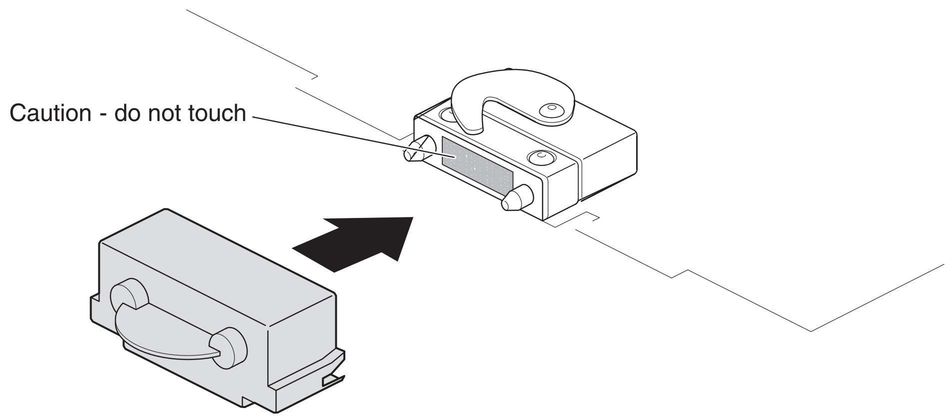


Figure 5-8 Removing the Compression Connector Cap

1.) Remove the caps on the compression connectors on the back of the system module. Do not touch the system module compression connectors as you lift the module into the workstation.

NOTICE **Do not insert the system module into the workstation unless the power cord is unplugged from the electrical socket. Inserting the system module into an OCTANE workstation with the power cord connected to an electrical socket could cause damage to the OCTANE workstation.**

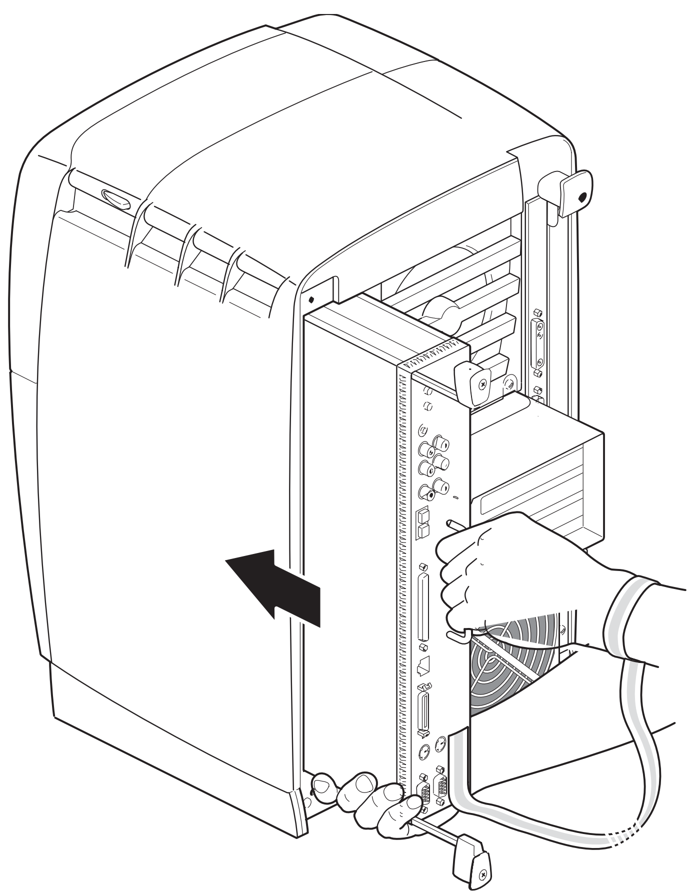


Figure 5-9 Supporting the System Module

2.) Grasp the system module by its immovable handle and support it with one hand as you slide it into the chassis.

3.) Push the module completely into the chassis using the immovable handle. (Both sliding handles protrude.) The system module stops when it is slightly out of the chassis.

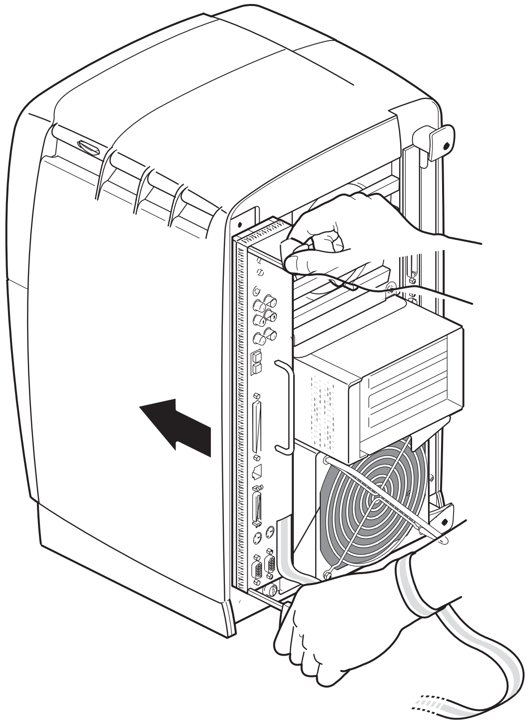


Figure 5-10 Replacing the System Module in the OCTANE Workstation

4.) Push the sliding handles simultaneously to connect the system module to the workstation.

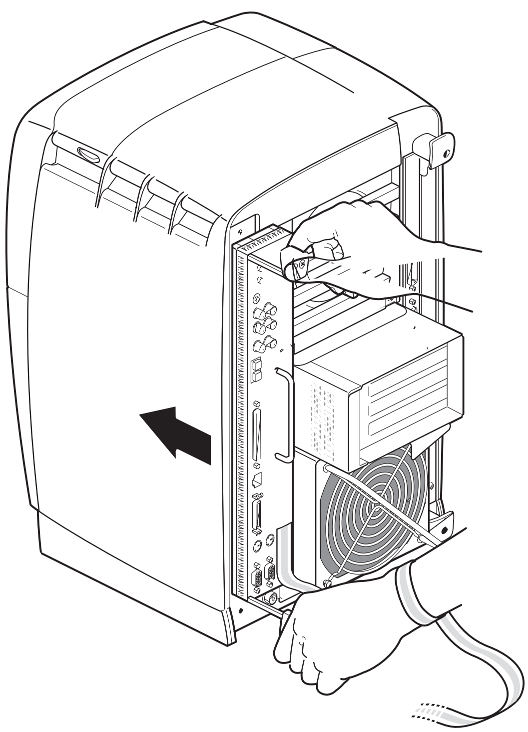


Figure 5-11 Locking the System Module to the Front Plane

5.) Push firmly on the handles to completely lock the system module to the workstation.

The handles are completely recessed and the system module is flush with the chassis when it is fully seated and locked to the workstation.

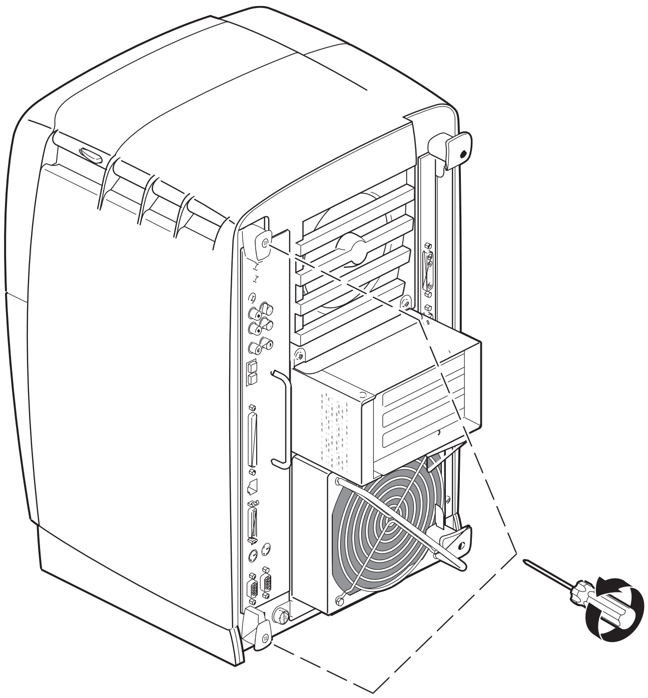


Figure 5-12 Replacing the System Module Screws

6.) Tighten the captive screws in the sliding handles until the system module is attached to the chassis.

7.) Remove the wrist strap.

3.4 Powering On the OCTANE Workstation

1.) Plug in the power cord.

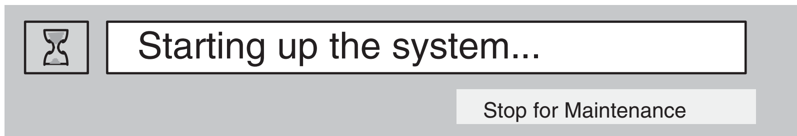
2.) Replace the lockbar (if applicable).

- Open the cover on the front of the OCTANE workstation.
- Slide the lockbar into the lockbar slot.
- Insert the lock through the lockbar at the back of the workstation.

Section 4.0
Start Up System and Verify Installed Memory

4.1 Start System

1.) Turn on the console power switch.
2.) The following message will appear. Select the STOP FOR MAINTENANCE button.



4.2 Verify Installed Memory

The system Maintenance menu appears.

1.) Select ENTER COMMAND MONITOR.
2.) At the shell prompt type: **hinv**
3.) Verify that the Main memory size: is 640 MB.

Note: If the amount of memory is incorrect, follow the directions at the beginning of this chapter to power off your system and check the installation procedure.

- Check the angle of the DIMMs. They should be upright and completely inserted.
- Check that each bank is populated with two DIMMs, and that they are of the same type. You must have an even number of DIMMs installed.

4.) Exit the shell by typing: **exit**
5.) Select START SYSTEM

The system software will start normally.

6.) After the system is started, open a shell tool and re-check the hardware environment:

{ctuser@somesystem} [1] **hinv**

1 XXX MHZ IP30 Processor
CPU: MIPS RXXXXX Processor Chip Revision: 2.7
FPU: MIPS RXXXXX Floating Point Chip Revision: 0.0
Main Memory size: 640 Mbytes <- This is the minimum memory you should see.
Instruction cache size: 32 Kbytes
Data cache size: 32 Kbytes
Secondary unified instruction/data cache size: 1 Mbytes
Intergal SCSI controller 0: Version QL1040B (rev. 2)
 Disk drive: unit 1 on SCSI controller 0
Integral SCSI controller 1: Version QL1040B (rev. 2)
 Optical disk: unit 4 on SCSI controller 1
 Comm device: unit 4 on SCSI controller 1
 Comm device: unit 4, lin 1 on SCSI controller 1
 CDROM: unit 6 on SCSI controller 1
IOC3 serial port: tty1
IOC3 serial port: tty2
IOC3 parallel port: plp1
Graphics board: ESI with texture option
Graphics board: ESI
Integral Fast Ethernet: ef0, version 1, pci 1
Fast Ethernet: ef1, version 1, pci 3
Iris Audio Processor: version RAD revision 12.0, number 1

Section 5.0
Install 3-D Analysis Option Software

Software Options must be loaded/re-loaded every time the OC software is reloaded.

1.) In Service Desktop, select TOOLBOX/UTILITIES -> INSTALL-> INSTALL OPTIONS.

This program must be run once for each option MOD you are loading. For example, if you have two options MODs you must run this program twice.

An Options Window appears.

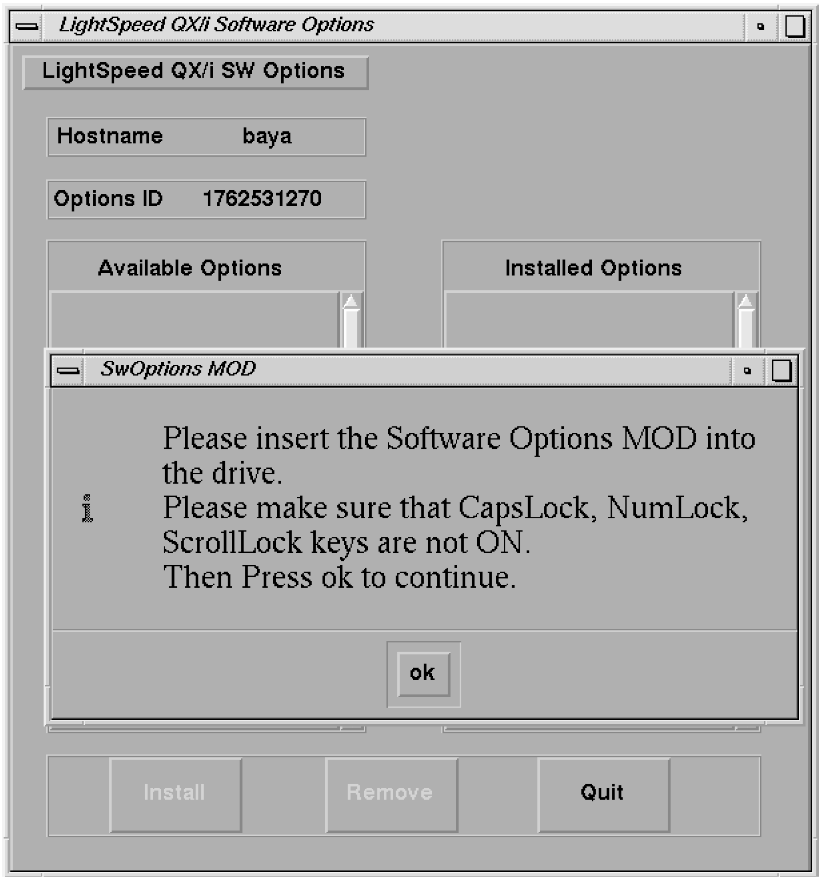


Figure 5-13 Software Options Installation Window

2.) Insert the Options MOD into the drive

3.) Click on OK

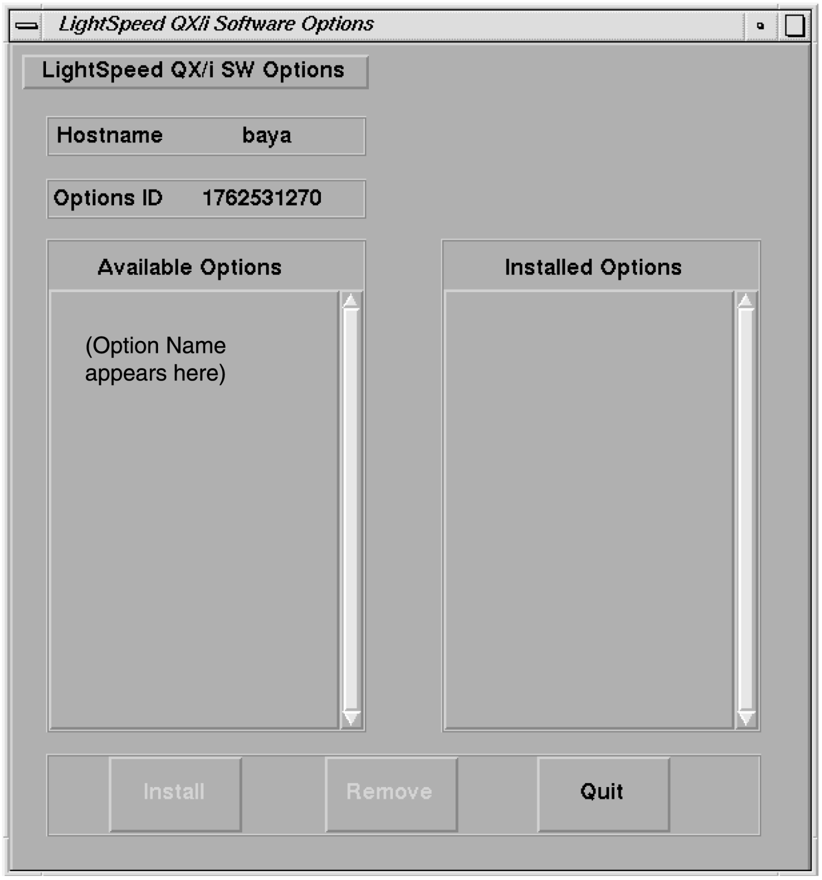


Figure 5-14 Software Options Available/Installed Screen Example

4.) Select the 3-D option to be installed (ex: click to highlight the option, then release).

5.) Click on the INSTALL button

A box may appear while the options are loading

When the option is displayed in the “Installed Options” list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half a minute (due to the fact that they are installing software).

6.) After the option is installed, select QUIT, then OK.

7.) Remove the MOD and write protect the side with options.

Note: The initial install requires the MOD to not be write protected; subsequent installs can be done with the MOD write protected)

8.) If you have another option MOD to load, go back to step 1.

9.) Go to the SERVICE DESKTOP ->UTILITIES ->APPLICATIONS SHUTDOWN.

10.) After Applications have shut down, open the console window, or a UNIX shell window, type **st** **ENTER** to start up applications.

11.) When the system applications return, the system is ready to use.

Section 6.0
Finish Installation

6.1 Install Rating Plate

Locate the Modification Rating plate, left of the Rating Plates already on the rear of the cabinet.

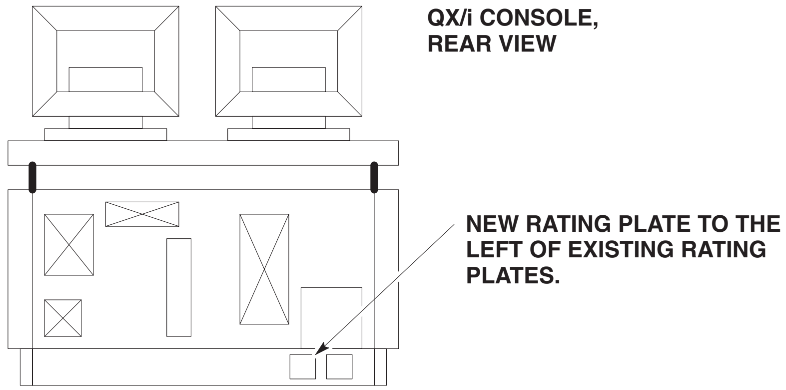


Figure 5-15 Rating Plate Location

6.2 Return Installation Card

Complete/Verify the site specific information on the Red Product Locator Installation Card and return to the appropriate Location printed on the front of the card.

Section 7.0
Parts

7.1 B7710MD 3-D Analysis Option Kit Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	B7710M	Memory Opt. Kit	N	1	128 MB Memory Opt. Kit
2	2149344	3-D Analysis Opt. Mod	N	1	3-D Option MOD

Table 5-2 3-D Analysis Option Kit

7.2 B7710M Memory Option Kit Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	2199806-2	128 MB Memory	1	1	2 x 64 MB DIMMs

Table 5-3 Memory Option Kit

Chapter 6
ConnectPRO & Bar Code Reader Option

Section 1.0
System Requirement

None. Works with all software versions.

Section 2.0
(Octane) ConnectPRO Option Installation Preparation

Two versions of the ConnectPRO Option exist:

B7540RB BAR CODE READER ONLY

Use the hand-held bar code scanner to directly input information to the Exam Rx patient information screen. The Bar Code Reader hardware option reads information from the paper schedule or patient wrist strap bar codes.

B7500PM HIS/RIS INTERFACE SOFTWARE WITH BAR CODE READER

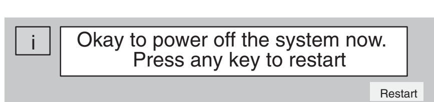
Use the HIS/RIS software option to network patient information directly to the QX/i Exam Rx desktop from the Hospital Information System (HIS) and Radiology Information Systems (RIS) database. This option kit also contains the hand-held bar code reader.

Before you start the option installation, take an inventory of the supplied kit. Refer to Section 5.0 for the corresponding list(s).

2.1 Bar Code Reader Kit Hardware Installation

1.) Shutdown the System:
   - a.) Select the (square red) SHUTDOWN button (shown below) on the Image Display Monitor.
   - b.) Select OK to shutdown Applications. Shutdown stops the QX/i Applications and the UNIX software, then displays the boot prompt.

2.) After the software shuts down, the following message appears:



3.) Turn off the power switch on the console.

2.2 Console Hardware Installation

1.) Remove the lower front cover of the console.
2.) Route the bar code scanner cable through the opening in the rear of the console (Figure 6-1).

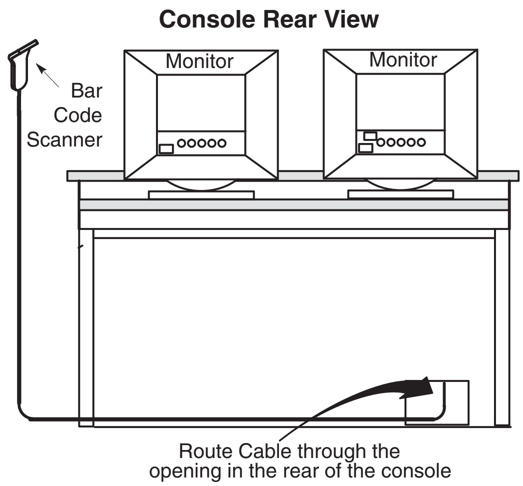


Figure 6-1 Bar Code Reader Installation

3.) Attach the Y-cable assembly (2180690-2) to the bar code scanner cord.
(Refer to Figure 6-2 for details)
4.) Unplug the existing keyboard connector from rear of Octane Computer.
(Refer to Figure 6-2 for details)
5.) Attach keyboard connector to the short length female connector on the Y-Cable assembly.
(Refer to Figure 6-2 for details)
6.) Connect the long length male connector from the Y-Cable assembly to keyboard connection in the back of the Octane Computer. (Refer to Figure 6-2 for details)
7.) Optional: Mount the bar code scanner's holder to the left side of the Exam Rx monitor, or to a location selected by the site Technologist. Use the double-sided tape (supplied with kit) to attach the holder.

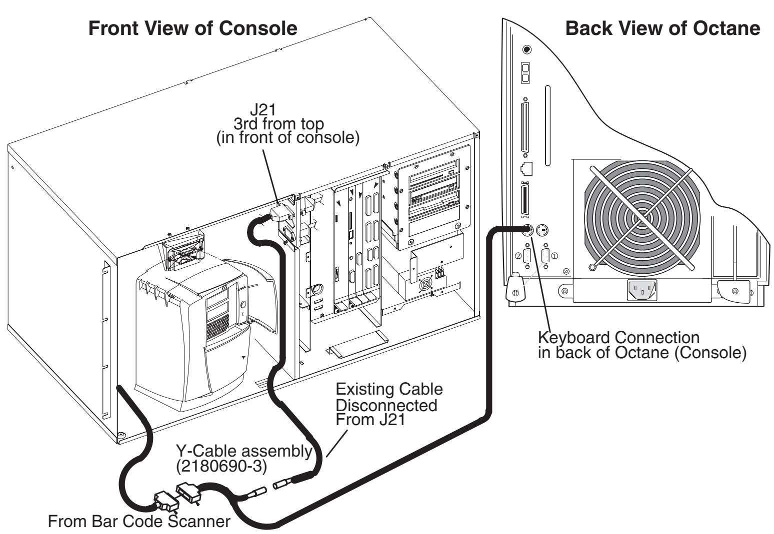


Figure 6-2 Bar Code Reader Installation (Detail)

Note: Do not replace the console front cover until you verify the bar code scanner operation.

2.3 Verify Bar Code Scanner Operation

1.) Restore power to the console and boot up the system to the scan user level.

2.) Select PATIENT SCHEDULE and ADD PATIENT on the Exam Rx desktop.

3.) Use the hand-held bar code scanner to scan the PATIENT ID# bar code on one of the Bar code examples in Figure 6-3. Make sure the screen updates with the corresponding number, either 555 666 777 or 777 888 999.

The bar code scanner beeps when it successfully reads a bar code.

Comment: *Practice with the bar code scanner and bar code samples first. Delete screen entries and re-read bar codes until you feel comfortable with the bar code scanner operation.*

Recommended: New users, take some time to practice with the bar code scanner and bar code samples. Delete the screen entry and reread the bar codes until you feel comfortable with the bar code scanner operation.

A User's Manual comes with the kit. It contains detailed usage information, as well as additional sample bar codes. If you encounter difficulties, refer to the Operation User Maintenance and Troubleshooting section of this manual. If you still cannot successfully scan the sample bar codes, select the DIAGNOSTICS menu from the service desktop, OPERATOR I/O from the Selection menu, and run the KEYBOARD CONFIDENCE TEST diagnostic to identify a bad cable connection or failed FRU.

4.) Select the REQUISITION # data field.

5.) Use the hand held bar code scanner to scan the REQUISITION# bar code on one of the Bar code examples in Figure 6-3. Make sure the screen updates with the corresponding number, either 508 or 510.

6.) Manually type/enter the *Patient Name*, and select ACCEPT.

7.) Select VIEW SCHEDULE

8.) Make sure the Schedule screen displays the test patient entry.

9.) Select DELETE.

10.) Select QUIT.

11.) This completes the bar code scanner and keyboard function test.

12.) Reinstall the console front cover.

Note: For customer option **#B7500PM** ONLY, proceed to Section 3.0.

Note: If you are using a foreign language (non-English) keyboard, you may see other numbers/characters displayed in the Patient ID# field when you scan the codes in Figure 6-3.

REQUISITION #
508
PATIENT ID #
555 666 777
REQUISITION #
510
PATIENT ID #
777 888 999

Figure 6-3 Bar Code Samples to Scan

Section 3.0
ConnectPRO Software Option Installation & Checkout

Note: Before you install the ConnectPRO option software, obtain the following information from the hospital's network administrator:

HIS Server IP Address

HIS Server AE Title

HIS Server AE Port#

CT Server AE Title

3.1 Install and Setup ConnectPRO Option from MOD

1.) Open the Service desktop
2.) Select the TOOLBOX/UTILITIES menu.
3.) Select INSTALL.
4.) Select INSTALLOPTIONS to display the options window shown in Figure 6-4.

This program is run for each software option you plan to load from the MOD. For example, if you have to load two option MODs, run this program twice.

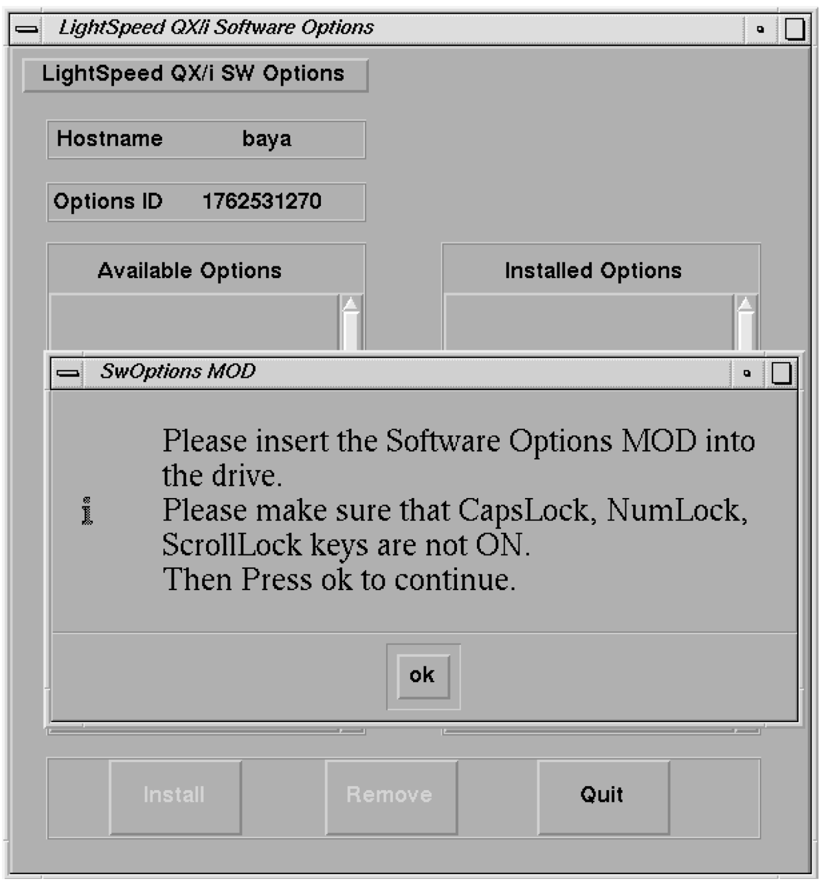


Figure 6-4 Options Window

5.) Insert the ConnectPRO option MOD into the drive.

6.) Select OK to display a software options screen similar to the one shown in Figure 6-5.

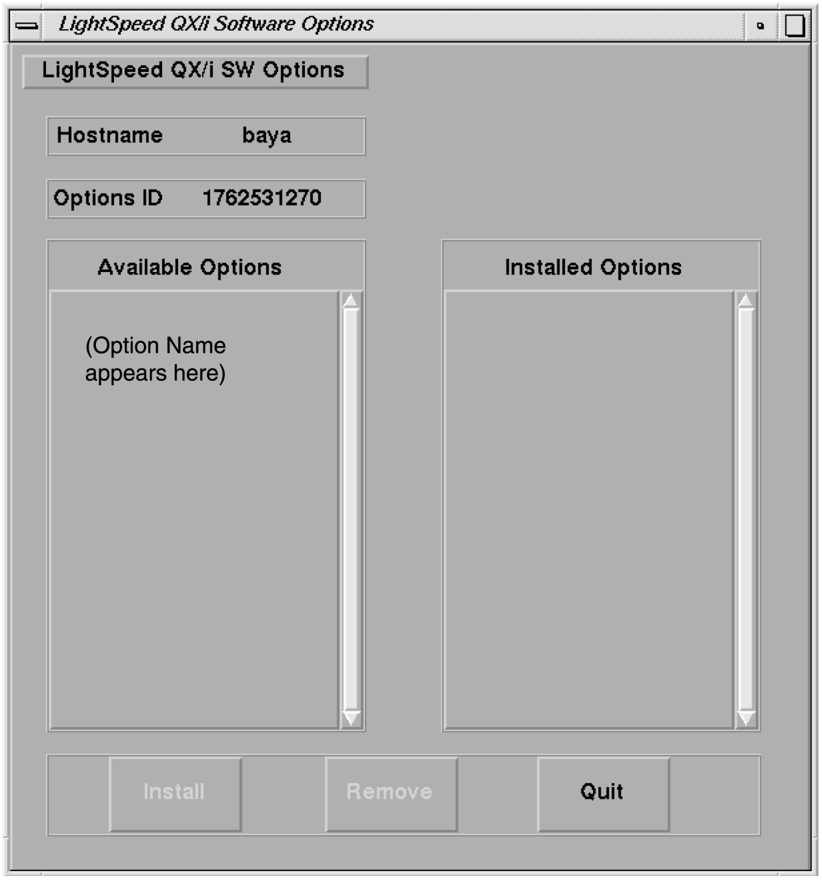


Figure 6-5 Sample Software Options Screen

7.) Select CONNECTPRO from the Available Options list box.

8.) Select INSTALL.A message box may appear while the option loads. The option label appears in the Installed Options list, when installation completes. Be patient, as some options take a fraction of a second to install. Some may take minutes.

9.) When ConnectPRO appears in the Installed Options list, the ConnectPRO Setup screen appears similar to the one in Figure 6-6.

10.) Enter the HIS information you recorded on page 96 into the corresponding data fields on the ConnectPRO Setup screen.

a.) If you need additional information, select HELP to display a HELP window explaining each data field.

b.) Select DISMISS to exit the ConnectPRO Setup screen without any changes.

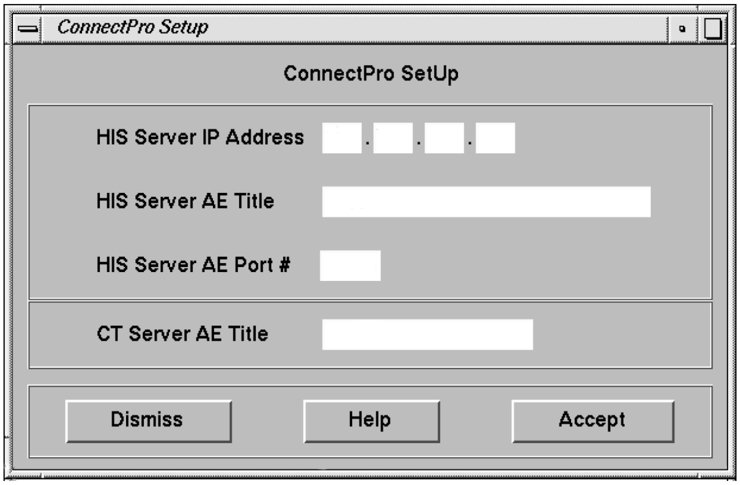


Figure 6-6 ConnectPRO Setup Screen

11.) Click on ACCEPT.

12.) If you have PPS information, enter the data you recorded on page 96 into the PPS Setup screen data fields. Then click ACCEPT. This returns you to the Software Options screen again.

Note: PPS Data Optional — If you don have a PPS network information, leave the data fields blank and press ACCEPT.

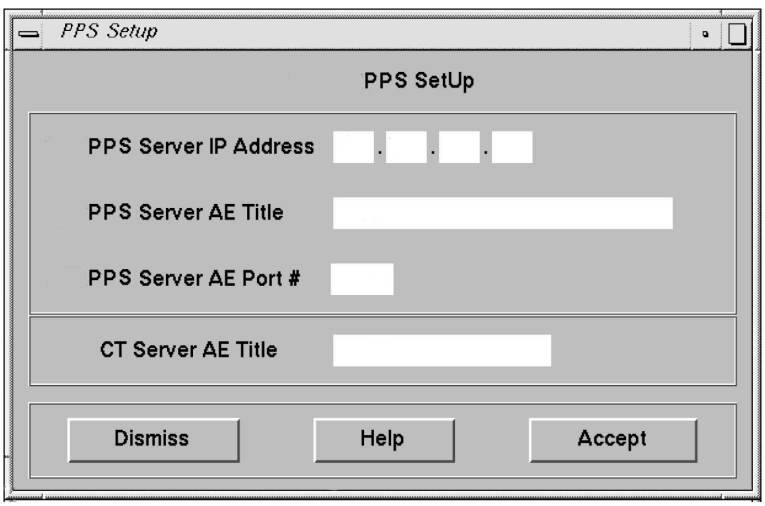


Figure 6-7 PPS Setup Screen

13.) Refer to Figure 6-5

a.) Select QUIT to exit the installation program.

b.) Select OK.

14.) Remove the MOD, and write protect the side containing the ConnectPRO option.

The initial install requires a write enabled MOD; but you may subsequently install the option(s) with a write protected MOD.

15.) If you have another option MOD to load, go back to step 1.

16.) Go to the SERVICE DESKTOP ->UTILITIES ->APPLICATIONS SHUTDOWN.

17.) After Applications have shut down, open the console window, or a UNIX shell window, type **st** **ENTER** to start up applications.

18.) When the system applications return, the system is ready to use.

3.2 ConnectPRO Feature Checkout

1.) Start up the Exam Rx applications.
2.) Query the HIS/RIS system to verify operation. (Refer to the User Manual/CBT for details.)

3.3 "Use Study ID" Problem Check on LightSpeed QXI Systems

Due to a problem with LightSpeed QXI systems (Application Software Release 10.5_2.8.2I_H1.3M4), systems using the "Use Study ID" feature of the Connect Pro HIS/RIS option may encounter DICOM image creation problems. To ensure that the system can create images using the Hospital HIS/RIS Study UID format, perform the following functional check prior to customer turnover.

3.3.1 Description

This problem occurs when the Hospital HIS/RIS system sends a Study ID that is longer than 50 characters. The result is that "LandMark UID" ends with a "." (dot) and this is unacceptable by DICOM standards. This is due to a bug in the host software that truncates the landmark uid and results in a "." at the end of the string. Landmark UID is created on the system by concatenating "Study Instance UID" and an "internal algorithm". Investigation showed that this problem occurs when the Preference setup has set the option "Use MWL study instance UID" to "Yes" under Patient schedule. This option helps to put the "Study instance UID" from HIS/RIS system into the DICOM image header.

3.3.2 Preferred Check of "Use Study ID" with Radiology Department

To simply check whether the Radiology Department may use the "Use Study ID" feature where the system accepts an Study ID from the HIS/RIS System, request a list of Study ID's from the Radiology Department (typical day, typical week, day of option install, etc.). Be sure to review as large a list as possible to avoid any future problems. Review the list of ID's and look for the following:

If any of the Study ID's generated by the HIS/RIS System are greater than 50 characters, the Connect Pro Patient Schedule Preference "Use Study UID?" parameter should be set to "No" to avoid image creation problems.

If none of the Study ID's generated by the HIS/RIS System are greater than 50 characters, the customer may have the Connect Pro Patient Schedule Preference "Use Study UID?" set to "Yes".

3.3.3 Manual Check of "Use Study ID" Procedure

Note: Because Study UID's from the Radiology Department HIS/RIS System are created dynamically, it may be necessary to repeat this "single patient exam" procedure multiple times to create a Patient Study with an image creation problem on the system.

If the system demonstrates failures due to performing this procedure, be sure to clean the recon queue of any images that are corrupted due to this error after each failed Patient Study prior to returning the system to the customer.

Prepare the system to perform a HIS/RIS scheduled exam:

1.) Select ExamRx Desktop
2.) Select New Patient
3.) Select Patient Schedule
4.) Setup on the Patient Schedule Display:
    - Select Preferences from the Patient Schedule display.
    - Select "Yes" for "Use Study UID".
    - Select "OK" to close Patient Schedule Preferences display.

- Select “Update” from the Patient Schedule display.
- Select “All CT Systems” on the Update Parameters display.
- Select “Continue Update” on the Update Parameters display.
- Click on the first “N” Status Patient display on the patient list form the Patient Schedule display.
- Select “Select Patient” on the Patient Schedule Display (system returns to New Patient Display with the selected patient information filled in from the Patient Schedule list).

5.) Setup on the “New Patient” Display:
- Under the Anatomical Selector, select “Service”
- From the Service List of Anatomical Selections, select “Manufacturing”.
- Select Protocol # 45.10 (Image Series QA).

6.) Position the QA Phantom for a QA Phantom Image Series:
- Mount the Phantom Holder on the head-end of the table.
- Mount the 20cm QA Phantom on the Phantom Holder.
- Align, level, and center the 20cm QA Phantom:

  Align black line on phantom using the internal laser lights

  Level phantom using bubble level and the Z Axis knob on the Phantom Holder.

  Center phantom using the Center Phantom procedure in the left head Scanner Utilities selection and the X and Y Axis knobs of the Phantom Holder

7.) Set internal Landmark

3.3.4 Patient Imaging Check

Acquire the first 20cm QA Phantom Series using the Patient selected from the Patient Schedule Display. Verify that the system completes the 1st series without any image creation problems (bad UID syntax). Look for the following in the GESYslog:

Mon Jun 4 14:44:43 2001 Error: 200002712

Host: hmerct3 Process: imagecreate

File: ic_mfm_main.c Line: 425

Image : 6320/1/1 Recon Sequence Id : 1841

MFM error : Format failure

Group Number : 20 Element : 52 Error status : -102

Error String : **[bad UID syntax]**

Mon Jun 4 14:45:07 2001 Error: 200002714

Host: hmerct3 Process: imagecreate

File: ic_imagecreate_req.c Line: 399

Image : 6320/1/1 Recon Sequence Id : 1845

MFM Error : Format failure DICOM image not created

If there are no image create problems using the HIS/RIS System Study UID, the customer can continue to use this setting in the Patient Schedule Preferences setting.

If the image create problem appears (Error Code 200002712 with a “bad UID syntax” as the Error String) in the GESYslog, the customer cannot use the “Use Study UID” feature. Return to the Patient Schedule Preferences display and set this option to “No”.

3.3.5 “Study UID” Image Corruption Recovery Procedure

Use the following procedure to remove corrupted images from the Recon Queue.

QXI systems have 6 recon queues on the SBC, (Priority_Job_ 0 through 4 and Prospective queues), one or more of the queues may be corrupt, identify the corrupt queue directory then remove all of it's contents. See example below; all EMPTY Queues have a file size of 9 bytes.

Bring application down via service desktop.

rsh sbc
su - root

ct01_sbc 3# cd /usr/g/queue
ct01_sbc 4# ls -al

drwxrwxr-x 9 ctuser ctuser 4096 May 7 1999 .
drwxrwxr-x 19 ctuser ctuser 4096 Jan 2 06:18 ..
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 16:57 priority0_jobs <-file size 9 = empty
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 16:31 priority1_jobs <-file size 9 = empty
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 13:41 priority2_jobs
drwxrwxr-x 2 ctuser ctuser 61 Mar 22 17:53 priority3_jobs ***<--Problem Possibly corrupt***
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 17:02 priority4_jobs
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 17:52 prospective
drwxrwxr-x 2 ctuser ctuser 9 May 7 1999 recovery

ct01_sbc 8# cd priority3_jobs ***<-- Change Directory to the Directory with the problem.***

ct01_sbc 9# ls -al

drwxrwxr-x 2 ctuser ctuser 61 Mar 22 17:53 .
drwxrwxr-x 9 ctuser ctuser 4096 May 7 1999 ..
-rw-r--r-- 1 ctuser ctuser 736 Mar 22 18:27 2684.3.21.2684.953733818.518923.pro_helical ***<--problem***

ct01_sbc 10# pwd

/usr/g/queue/priority3_jobs

ct01_sbc 12# rm -i * ***<---remove all images from the queue***

Answers y to remove

ct01_sbc 14# cd /usr/g/queue

ct01_sbc 15# ls -al

drwxrwxr-x 9 ctuser ctuser 4096 May 7 1999 .
drwxrwxr-x 19 ctuser ctuser 4096 Jan 2 06:18 ..
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 16:57 priority0_jobs
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 16:31 priority1_jobs
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 13:41 priority2_jobs
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 18:29 priority3_jobs ***<-- All removed***
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 17:02 priority4_jobs
drwxrwxr-x 2 ctuser ctuser 9 Mar 22 17:52 prospective
drwxrwxr-x 2 ctuser ctuser 9 May 7 1999 recovery

Shutdown and restart applications to flush the queue manager process.

3.4 Troubleshooting

To troubleshoot the network hardware: Open a Shell Tool, and type:

more /usr/g/service/menus/networkHelp.txt ENTER

to display the network diagnostics help menu, and follow the instructions displayed on the screen.

3.5 Save System State

The following procedure saves Characterization Data, Calibration Data, Protocols, and reconfig information including Connect-Pro setup information), to your System State MOD. You must save this information to the site's System State MOD in order to prevent having to track down the information and manually enter it every time a Load-From-Cold is performed.

NOTICE for Data Loss **Do Not display any images in Image Works when you save the System State. Exit to the Browser level before you Save System State.**

1.) Insert the Save System State MOD into the MOD drive.
2.) Open the Service desktop.
3.) Select the PM option from the list.
4.) Select SYSTEM STATE.
5.) Select ALL.
6.) Select SAVE.
Review the output for errors or missing files. You may use the scroll bar on the right after the tool completes its tasks.
7.) When Save completes, select DISMISS.

Section 4.0
Finish Installation

4.1 Install Rating Plate

1.) Refer to Figure 6-8.
2.) Locate the current Modification Rating plate(s) on the rear of the console.
3.) Fasten the Option Kit rating plate to the left of the existing rating plates.

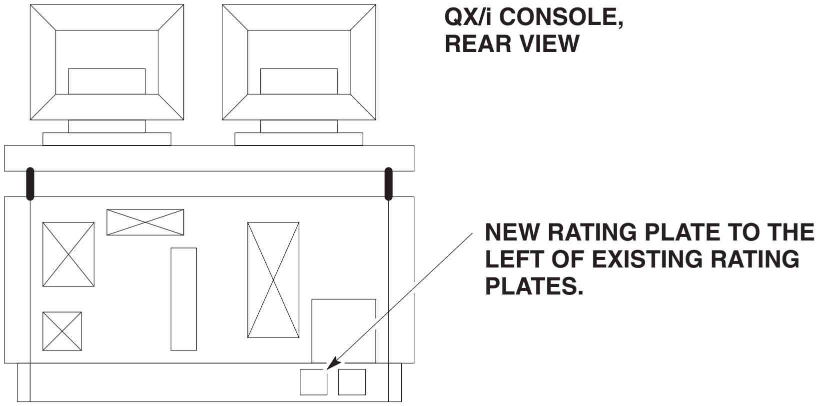


Figure 6-8 Rating Plate Location

4.2 Return Installation Card

1.) Complete and verify the site-specific information on the Red Installation Product Locator Card.
2.) Return the card to the Mailing Address that is printed on the front of the card.

Section 5.0
Parts

5.1 B7500PM HIS/RIS Interface w/Bar Code Reader Option Kit Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	B7540RB	Bar Code Reader Kit	N	1	Bar Code Reader Kit including Y-Cable and rubber mount (B7540RB).
2	2182070	Option MOD	N	1	HIS/RIS Option MOD

Table 6-1 HIS/RIS Interface w/Bar Code Reader Option Kit

5.2 B7540RB Bar Code Reader Only Option Kit Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	2180690-2	Bar Code Scanner	1	1	Hand Held Bar Code Scanner ONLY
2	2180690-3	Y-cable assembly.	1	1	Console/Keyboard/Scanner interface cable ONLY
3	2180690-4	Scanner Holder	2	1	Plastic holder ONLY (to store bar code scanner when not in use)

Table 6-2 Bar Code Reader Only Option Kit

Chapter 7
Navigator Option Installation

Section 1.0
System Requirement

None. Works with all software versions.

Section 2.0
Install Navigator Option Software

Navigator Option provides Navigator software for LightSpeed. This option allows the creation of 3-D surface rendered images of hollow structures and allows the user to navigate or "fly through" them.

Note: This option requires the 3-D analysis option (B7710MD) to also be installed.

ADD SOFTWARE OPTIONS

Software Options must be loaded/re-loaded every time the OC software is reloaded.

1.) In Service Desktop, select TOOLBOX/UTILITIES -> INSTALL-> INSTALL OPTIONS.

This program must be run once for each option MOD you are loading. For example, if you have two options MODs you must run this program twice.

An Options Window appears.

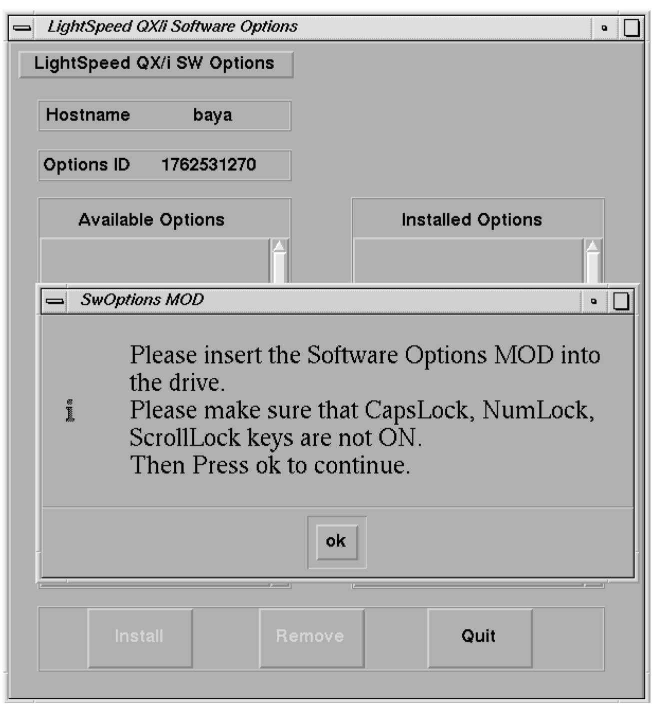


Figure 7-1 Software Options Installation Window

2.) Insert the Options MOD into the drive

3.) Click on OK

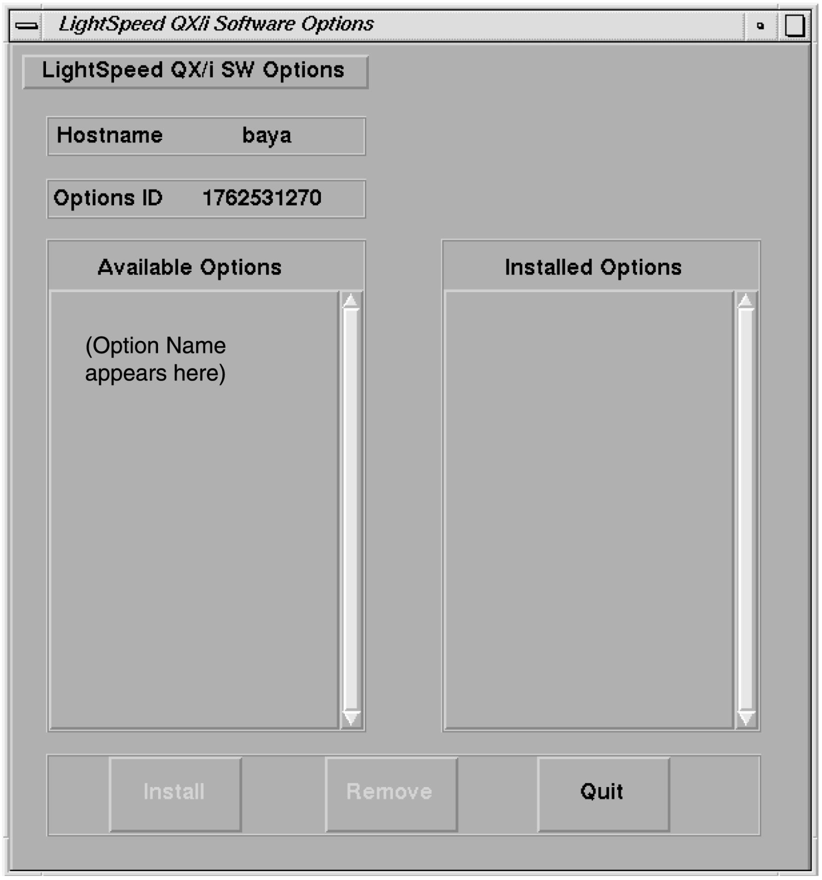


Figure 7-2 Software Options Available/Installed Screen Example

4.) Select the Navigator option to be installed (ex: click to highlight the option, then release).

5.) Click on the INSTALL button

A box may appear while the options are loading

When the option is displayed in the "Installed Options" list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half a minute (due to the fact that they are installing software).

6.) After the option is installed, select QUIT, then OK.

7.) Remove the MOD and write protect the side with options.

Note: The initial install requires the MOD to not be write protected; subsequent installs can be done with the MOD write protected)

8.) If you have another option MOD to load, go back to step 1.

9.) Go to the SERVICE DESKTOP ->UTILITIES ->APPLICATIONS SHUTDOWN.

10.) After Applications have shut down, open the console window, or a UNIX shell window, type **st** **ENTER** to start up applications.

11.) When the system applications return, the system is ready to use.

Section 3.0
Finish Installation

3.1 Install Rating Plate

1.) Refer to Figure 7-3
2.) Locate the current Modification Rating plate(s) on the rear of the console.
3.) Fasten the Navigator Option rating plate to the left of the existing rating plates.

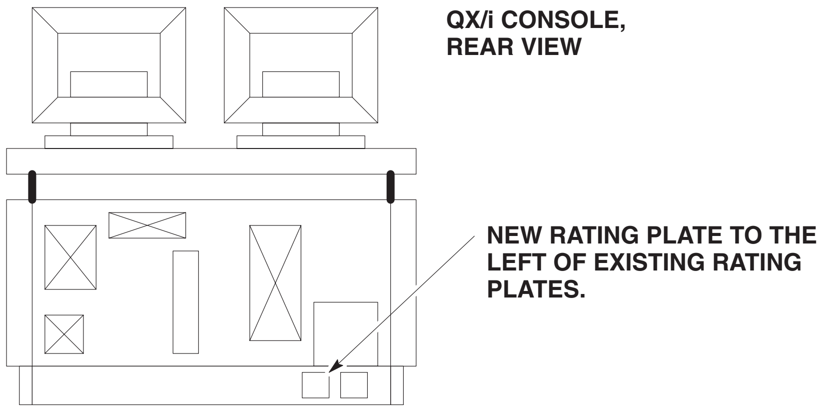


Figure 7-3 Rating Plate Location

3.2 Return Installation Card

1.) Complete and verify the site-specific information on the Red Product Locator Installation Card.
2.) Return the card to the Mailing Address that is printed on the front of the card.

Section 4.0
B7540N Navigator Option Kit Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	2184748	CBT CD-ROM	N	1	Navigator On Line Operators Manual
2	2184750	Options MOD	N	1	Navigator Option Software

Table 7-1 Navigator Option Kit

This page intentionally blank.

Chapter 8
SmartPrep Option Installation

The SmartPrep Option provides SmartPrep contrast monitoring.

Section 1.0
System Requirement

None. Works with all software versions.

Section 2.0
Install SmartPrep Option Software

Software Options must be loaded/re-loaded every time the OC software is reloaded.

1.) In Service Desktop, select TOOLBOX/UTILITIES -> INSTALL-> INSTALL OPTIONS.
This program must be run once for each option MOD you are loading. For example, if you have two options MODs you must run this program twice.
An Options Window appears.

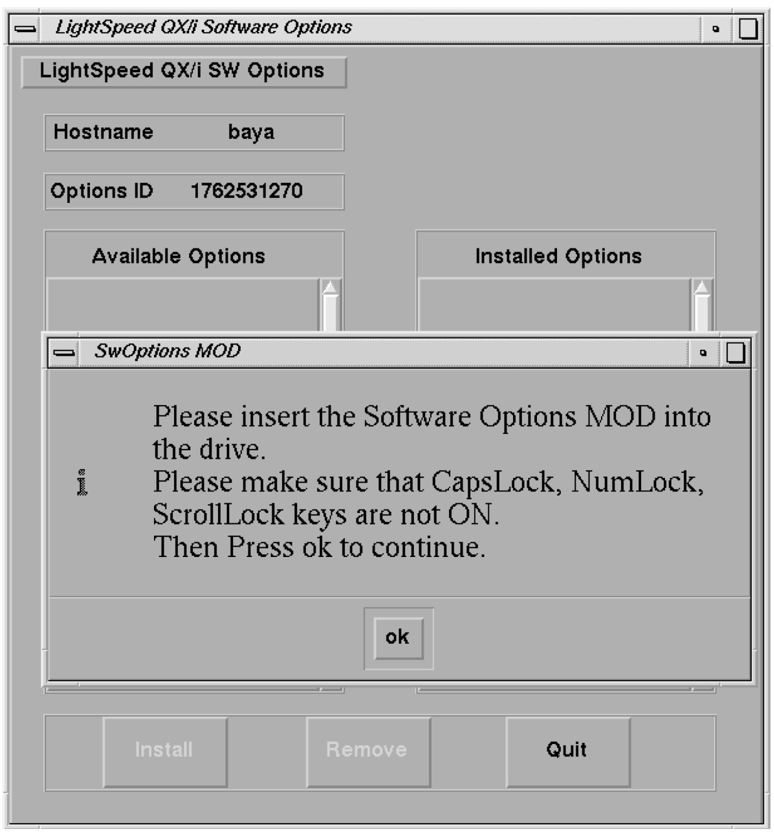


Figure 8-1 Software Options Installation Window

2.) Insert the Options MOD into the drive

3.) Click on OK

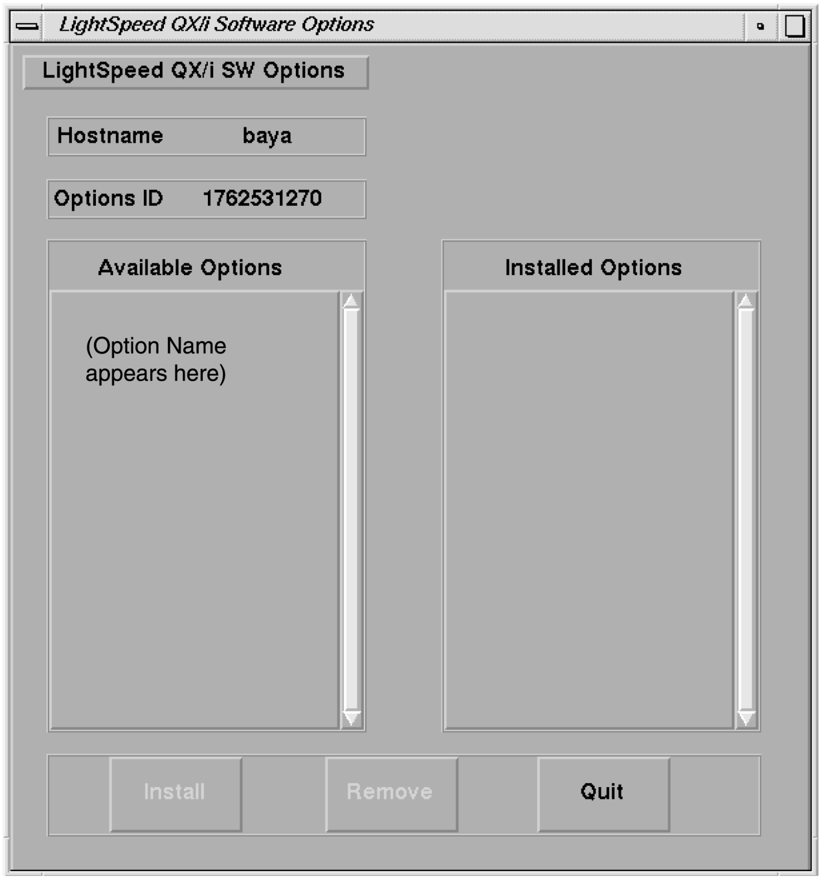


Figure 8-2 Software Options Available/Installed Screen Example

4.) Select the SmartPrep option to be installed (ex: click to highlight the option, then release).

5.) Click on the INSTALL button

A box may appear while the options are loading

When the option is displayed in the “Installed Options” list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half a minute (due to the fact that they are installing software).

6.) After the option is installed, select QUIT, then OK.

7.) Remove the MOD and write protect the side with options.

Note: The initial install requires the MOD to not be write protected; subsequent installs can be done with the MOD write protected)

8.) If you have another option MOD to load, go back to step 1.

9.) Go to the SERVICE DESKTOP ->UTILITIES ->APPLICATIONS SHUTDOWN.

10.) After Applications have shut down, open the console window, or a UNIX shell window, type **st** **ENTER** to start up applications.

11.) When the system applications return, the system is ready to use.

Section 3.0
Finish Installation

3.1 Install Rating Plate

1.) Refer to Figure 8-3
2.) Locate the current Modification Rating plate(s) on the rear of the console.
3.) Fasten the SmartPrep Option rating plate to the left of the existing rating plates.

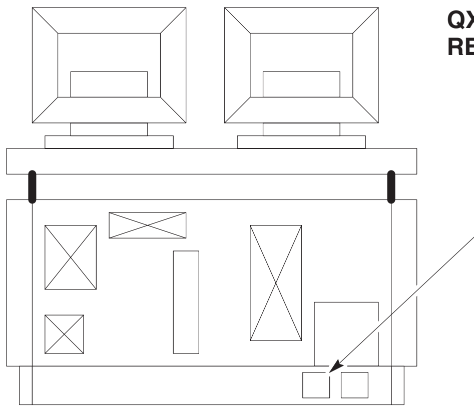


Figure 8-3 Rating Plate Location

3.2 Return Installation Card

1.) Complete and verify the site-specific information on the Red Installation Product Locator Card.
2.) Return the card to the Mailing Address that is printed on the front of the card.

Section 4.0
B7830SP SmartPrep Option Kit Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1	2148905	Options MOD	N	1	SmartPrep Option Software

Table 8-1 SmartPrep Option Kit

This page intentionally blank.

Chapter 9
CT Perfusion Option Installation

Section 1.0
System Requirement

Version 1.2 software or later must be installed on the system before CT Perfusion can be loaded.

Section 2.0
Install CT Perfusion Option Software

Software Options must be loaded/re-loaded every time the OC software is reloaded.

1.) In Service Desktop, select TOOLBOX/UTILITIES -> INSTALL-> INSTALL OPTIONS.

Note: This program must be run once for each option MOD you are loading. For example, if you have two options MODs you must run this program twice.

An Options Window appears.

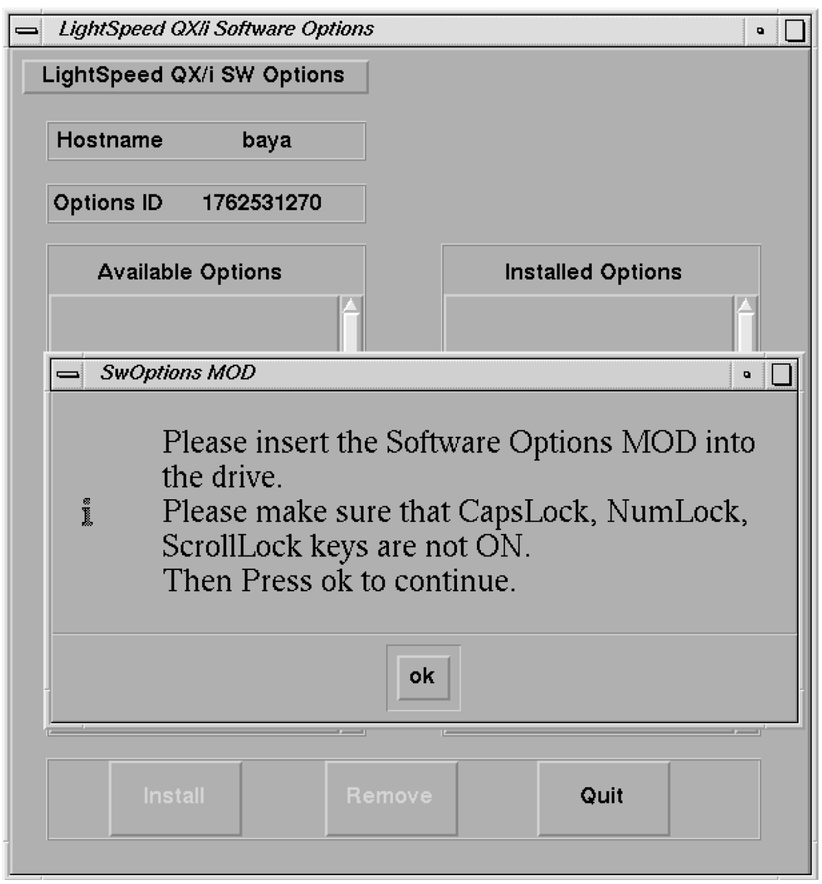


Figure 9-1 Software Options Installation Window

2.) Insert the Options MOD into the drive

3.) Click on OK

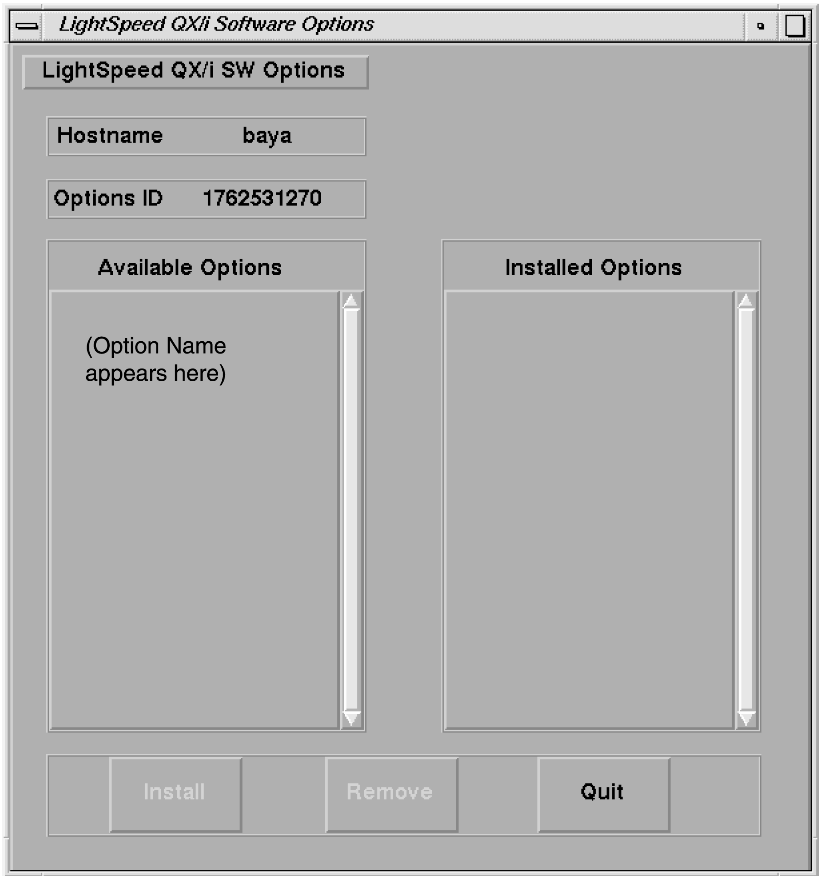


Figure 9-2 Software Options Available/Installed Screen Example

4.) Select the CT Perfusion option to be installed (ex: click to highlight the option, then release).

5.) Click on the INSTALL button

A box may appear while the options are loading

When the option is displayed in the “Installed Options” list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half a minute (due to the fact that they are installing software).

6.) After the option is installed, select QUIT, then OK.

7.) Remove the MOD and write protect the side with options.

Note: The initial install requires the MOD to not be write protected; subsequent installs can be done with the MOD write protected)

8.) If you have another option MOD to load, go back to step 1.

9.) Go to the SERVICE DESKTOP ->UTILITIES ->APPLICATIONS SHUTDOWN.

10.) After Applications have shut down, open the console window, or a UNIX shell window, and type **st** **ENTER** to start up applications.

11.) When the system applications return, the system is ready to use.

Section 3.0
Finish Installation

3.1 Install Rating Plate

1.) Refer to Figure 9-3
2.) Locate the current Modification Rating plate(s) on the rear of the console.
3.) Fasten the CT Perfusion Option rating plate to the left of the existing rating plates.

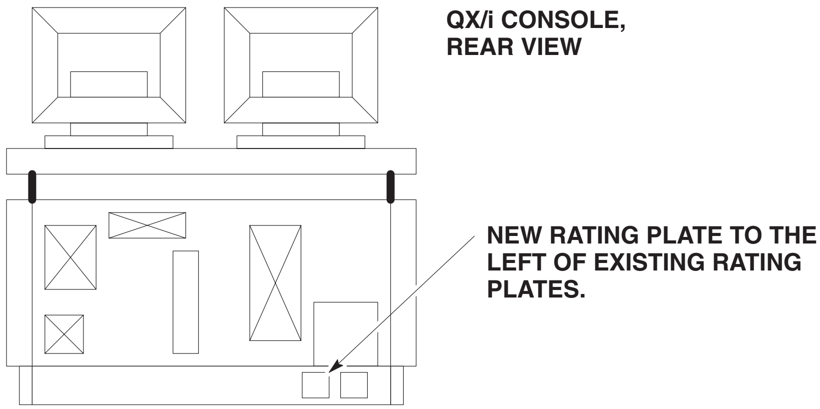


Figure 9-3 Rating Plate Location

3.2 Return Installation Card

1.) Complete and verify the site-specific information on the Red Installation Product Locator Card.
2.) Return the card to the Mailing Address that is printed on the front of the card.

Section 4.0
CT Perfusion Option Kit Parts List

ITEM	P/N	NAME	FRU	QTY	DESCRIPTION
1		Options MOD	N	1	CT Perfusion Option Software

Table 9-1 Perfusion Kit

This page intentionally blank.

Chapter 10
DASM/Camera Installation

Section 1.0
Hardware Installation

1.1 Console w/Serial Expander

DASMs are available in two types, digital (refer to Figure 10-1) and analog (refer to Figure 10-2). If DASMs are used at your site, they may already be set at the factory. Complete the following steps to connect a DASM to the Octane computer.

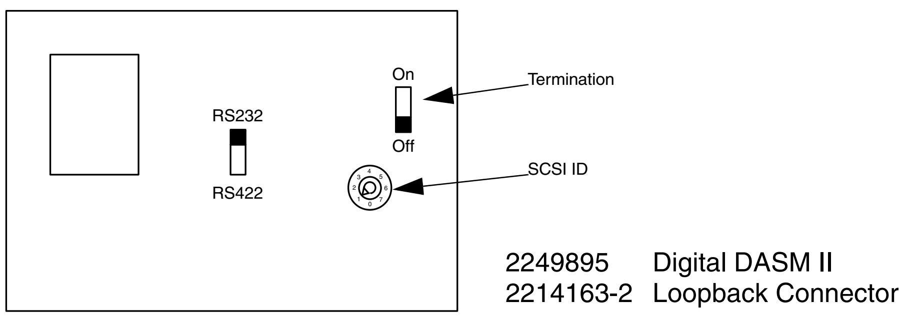


Figure 10-1 Digital DASM Bottom

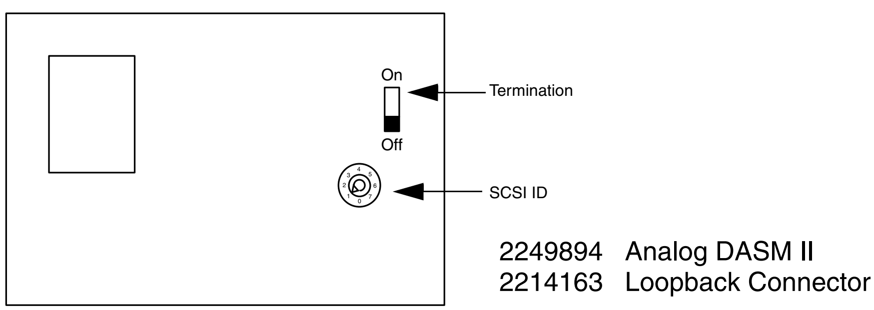


Figure 10-2 Analog DASM Bottom

1.) Turn the DASM over so you are looking at the bottom.
2.) Set the DASM SCSI ID to 1.
3.) Set the DASM termination to off.
4.) Set the RS232/RS422 switch to the RS232 position if you have a digital DASM.

5.) Disconnect the Central Data Box end of SCSI cable A (refer to Figure 10-3).

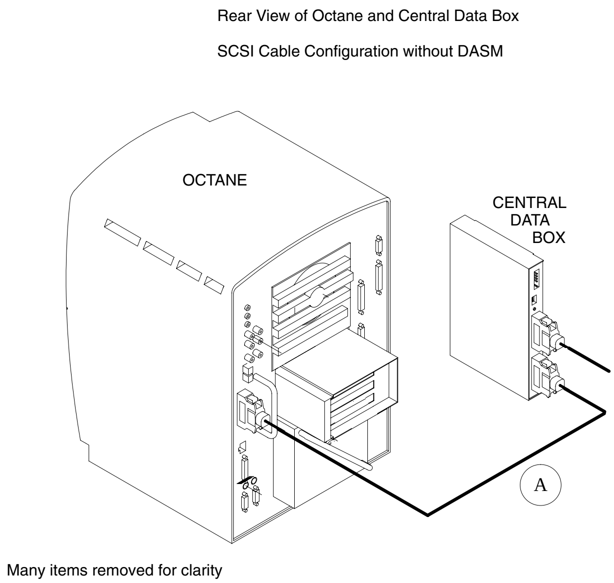


Figure 10-3 SCSI Cable Configuration without DASM

6.) Refer to Figure 10-4. Install SCSI 3 to 1 Adapter to the back of the DASM.

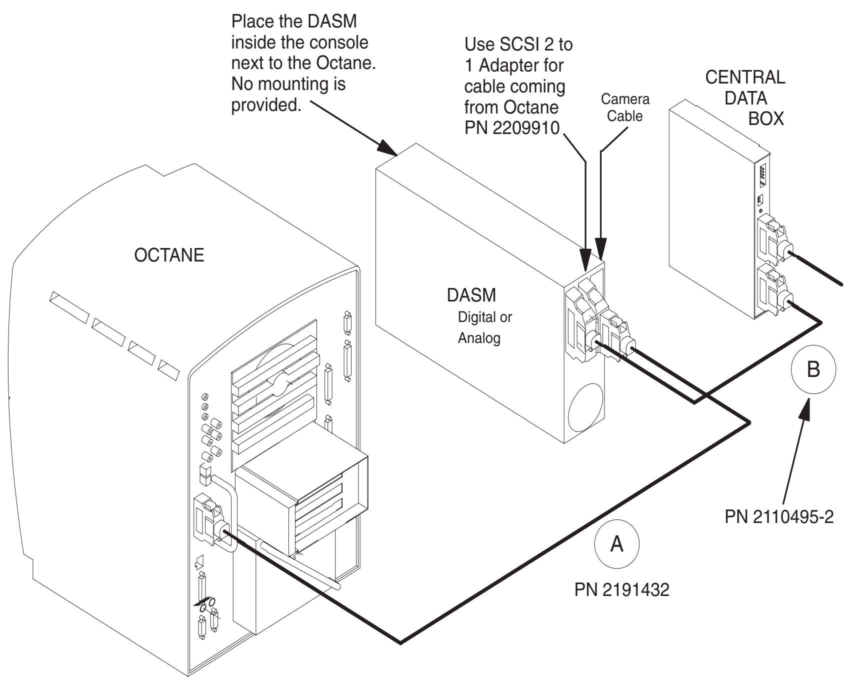


Figure 10-4 SCSI Cable Configuration with DASM

7.) Attach the free end of cable A to adapter installed in step (6).

8.) Attach cable B between the Central Data Box and DASM as shown.

9.) Plug DASM power cord into J10.

10.) Re-attach remaining cable(s) from camera to DASM and install the DASM next to the computer.

1.2 Console w/o Serial Expander

Although DASMs are available in two types, digital and analog, setup is identical. Complete the following steps to connect a DASM to the Octane computer.

1.) Turn the DASM over so you are looking at the bottom.
2.) Set the DASM SCSI ID to 1.
3.) Set the DASM termination to ON.

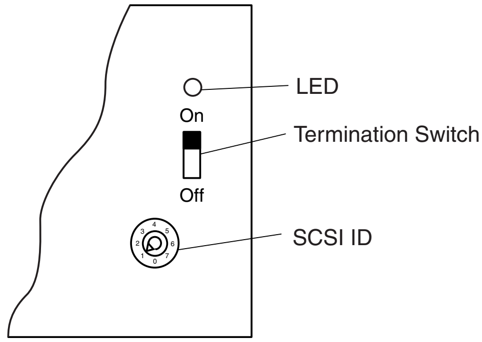


Figure 10-5 DASM Bottom

4.) Set the RS232/RS422 switch to the RS232 position, if you have a digital DASM.
5.) Refer to Figure 10-6. Attach the SCSI cable supplied with DASM kit to the back of the DASM.

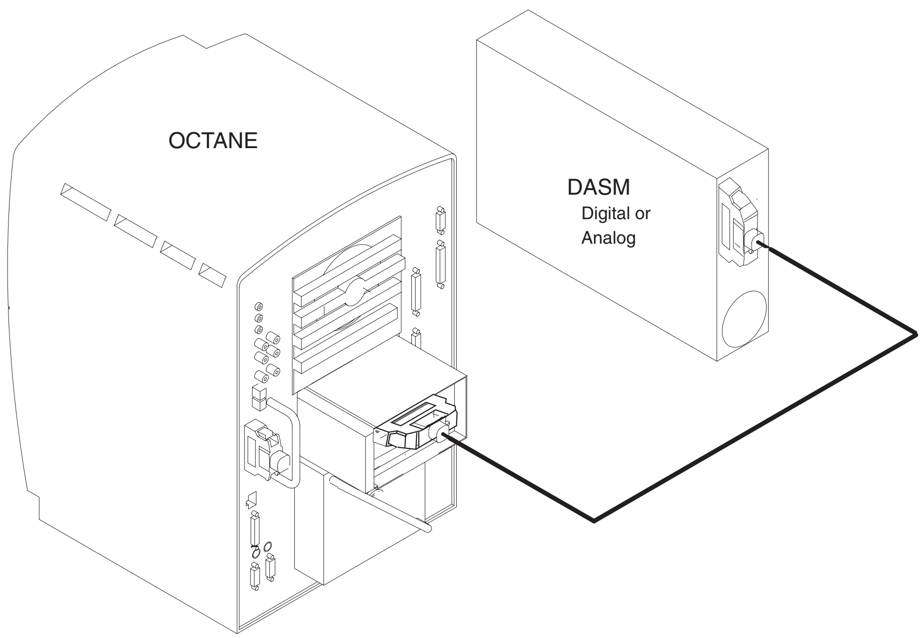


Figure 10-6 SCSI Cable Configuration with DASM

6.) Attach the free end of the SCSI cable to the PCI module's SCSI connector (slot 1 — top slot).
7.) Plug DASM power cord into J10.
8.) Attach remaining cable(s) from camera to DASM and install the DASM next to the computer.

Section 2.0
Camera Configuration

See your Software Load Procedures manual for instructions on configuring cameras.

Chapter 11
FX/Xtream Workflow Options

Section 1.0
Options

These options are productivity features for the customer.

- B7864KH: Exam Split FX

For HP8000-based consoles:

- B7866PA: WorkFlow FX, with host memory
- B7866PE: WorkFlow FX, without host memory
- B7866PC: Xtream II Platform FX, with host memory
- B7866PG: Xtream FX WorkFlow Package, with host memory

For HP8200-based consoles:

- B7866PB or B7866PF: WorkFlow FX
- B7866PD: Xtream II Platform FX
- B7866PH: Xtream FX WorkFlow Package

Section 2.0
Prerequisite

Requires 05MW14.5 software release, labeled as “Q2 2005 Software Set,” (p/n 5134305), or more recent release.

Section 3.0
Install Options

Install per FX/Xtream service document 5134798-800.

Section 4.0
Finish Installation

Confirm the software option has been installed.

This page intentionally blank.

Chapter 12
Ivy Model 3150 ECG Monitor

Section 1.0
Hardware Installation

This hardware is Ethernet-based.

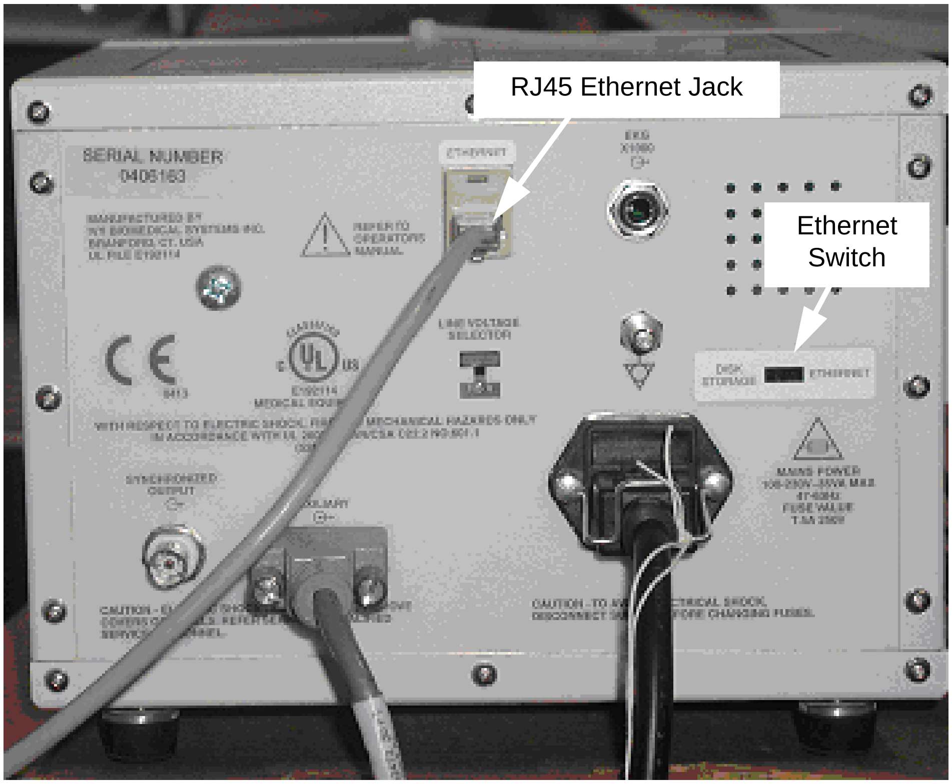


Figure 12-1 ECG Monitor Cabling and Switch Setting

1.) At the back of the ECG monitor, connect the long Ethernet RJ45 cable to the Ethernet jack as shown in Figure 12-1.
2.) Connect the other end of this Ethernet cable to the AW-labeled RJ45 jack (near the a.c. power plug) on the back of the Xtream console.
3.) On the ECG monitor, set the Ethernet switch to “Ethernet.”
4.) Toggle power to the ECG monitor so that it will operate in Ethernet mode.

Section 2.0 Installation Test

Check the operation of the ECG monitor by prescribing a patient cardiac scan as follows.

1.) Start applications.
2.) Click NEW PATIENT.
3.) For patient ID, type: **ECG Test ENTER**
4.) Click on the GE tab.
5.) Click PROTOCOL 25.1 ROUTINE CHEST.
6.) Click the GATING button.
    The system should respond with "System waiting for signal….".
7.) At the ECG Monitor, connect ECG leads to the left-side test points on the monitor.
    - White lead to white test point
    - Red lead to red test point
    - Black lead to black test point
8.) Turn on the ECG monitor (front button).
9.) Press **TEST MODE** (front button).
10.) Press **SIM RATE OFF** (front button).
    Press the **SIM RATE OFF** button to toggle to the desired beats-per-minute setting.
    The monitor should now be outputting a test waveform at the beats per minute selected.
11.) At the Xtream console, view this waveform.
    If the waveform is not displayed at the console, check for proper Ethernet communications between the console and ECG monitor by executing the following:
    a.) Open a Unix shell and log in as ctuser.
    b.) Type: **ping 10.44.22.21** ENTER

Note: If the connection is correct, the ECG monitor should repeatedly respond with a "ping" time and packet information.

    c.) If the ECG monitor did not respond, attempt to communicate to the host port (eth4) for the ECG monitor. To do so, type: **ping 10.44.22.20** ENTER

Note: If the connection is correct, the host port (eth4) should repeatedly respond with a "ping" time and packet information.

    d.) If the host port (eth4) did not respond, the Ethernet port is faulty.

© 2000-2006, General Electric Company.
GE Medical Systems, a General Electric Company, going to market as GE Healthcare.
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com